EXPONENTIAL SUMS OVER THE RIEMANN ZEROS VERSUS THE RIEMANN–VON MANGOLDT CLOCK, WITH FINITE MÖBIUS INTERPOLATION AND PRIME-POWER SPECTRAL IDENTITIES

VÍCTOR CASAÑA-GINER

ABSTRACT. The manuscript studies the exponential zero walk $S_N(\omega) = \sum_{k \le N} e^{i\omega\gamma_k}$ through its exact comparison with the inverse Riemann–von Mangoldt clock. The indexed phase discrepancy is decomposed into a linear displacement sum and a quadratic remainder. Global and short-window bounds for the displacement square sum are derived from classical estimates for the argument of zeta; these bounds do not make the quadratic remainder negligible and therefore isolate a specific open local-control problem.

For admissible regularized test functions, the Guinand–Weil explicit formula gives two exact prime-power branches with frequencies $\omega \pm \log n$. Removing the regularizing weights, passing to a sharp counting window, and treating the resulting unweighted branches as a complete spectral model are explicitly separated as nonstandard conjectural steps. At the coefficient level, the manuscript proves the elliptic geometry of each mode, the positivity of $d_n = |c_n^-|^2 - |c_n^+|^2$, and the asymptotic law $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$. For $\omega = 1$, the $n = 3$ mode is proved to be the unique closest prime-power resonance and the unique largest mode under several exact coefficient norms. This proves a genuine analytic influence of $\log 3$, but not by itself the bimodality of the full radial histogram. A deterministic matched-amplitude frequency scan on the archived zero data is therefore included as finite numerical support.

A separate finite construction starts from the symmetric affine interpolation of the terminal prime coefficients of the Möbius Dirichlet polynomial. It yields the exact identity $\Phi_N(s) = M(N) + h_N R_N(s)/P_N(s)$, together with $\Phi_N(0) = -2$ and $\Phi_N(\sigma) \to M(N)$, an explicit rounding criterion, and a rigorous recovery formula for the first nonzero Möbius coefficient after N.

Finally, the complete-zero phase measure and the ordinate-only phase measure are shown to differ by a positive atomic defect whose vanishing for one fixed $q > 1$ is equivalent to the Riemann Hypothesis. Exact finite-sample formulas are proved for the raw and jointly fitted carrier coefficients and for the projected residual variance. Seven conditional routes toward the Riemann Hypothesis are stated with their precise hypotheses, their rigorous logical implications, and the local or uniform estimates that remain open. The numerical chapter precedes the conclusion; the complete MATLAB program, console transcript, and matched-frequency scan are annexed after the bibliography.

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Status taxonomy

Every substantive statement is assigned one of the following statuses at the point of use.

Status	Meaning
Classical result – unconditional	A standard theorem from the cited literature; the normalization used here is stated explicitly.
Exact result proved here – unconditional	An algebraic, analytic, or finite sample identity proved in the manuscript. This status does not by itself assert worldwide novelty.
Conditional route	The implication is proved, but one or more analytic hypotheses required to activate it are open.
Heuristic or conjecture	A proposed asymptotic or formal interpretation; it is not used as a proved premise.
Numerical observation	A finite computation, isolated in the numerical part and never promoted to an asymptotic theorem.

1. Introduction and conceptual map

The investigation starts from one comparison. If $0 < \gamma_1 \le \gamma_2 \le \cdots$ are the positive ordinates of the nontrivial zeros, repeated with multiplicity, and $\omega > 0$, define

$$S_N(\omega) = \sum_{k \le N} e^{i\omega\gamma_k}. \tag{1.1}$$

The benchmark is obtained by inverting the principal Riemann–von Mangoldt term; see [9, 14] for the classical zero-counting framework

$$N_0(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{7}{8} \tag{1.2}$$

and setting $N_0(\widetilde{\gamma}_k) = k - 1/2$. The original object is the phase discrepancy

$$\mathcal{E}_N(\omega) = \sum_{k \le N} \left(e^{i\omega\gamma_k} - e^{i\omega\widetilde{\gamma}_k}\right). \tag{1.3}$$

The first analytic task is therefore to separate the smooth zero density from the oscillatory remainder without discarding the endpoint atom.

The explicit formula supplies the arithmetic interface. For admissible regularized test functions, its prime side is supported on prime powers and contains two displaced frequencies $\omega - \log n$ and $\omega + \log n$ [5, 6, 7, 15]. This origin is classical and exact: for each admissible test function the prime-power terms are not auxiliary fitted modes, but the two prime-side Fourier branches of the Guinand–Weil formula. A different statement, used later as a finite computational ansatz, is that after removing the test-function weights these unweighted modes form a complete spectral description of the sharp residual. That extrapolation is nonstandard and is always marked as heuristic or conjectural.

The finite Möbius interpolation is a second, algebraically independent construction. Its smallest instance already gives the intended meaning. For $N = 10$,

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \tag{1.4}$$

$$P_{10}(s) = 7^{-s}, \qquad B_{10} = A_{10}(0) = 0. \tag{1.5}$$

The first interpolation, with numerical parameter m, is

$$\mathcal{I}_{10}(s; m) = A_{10}(s) + m7^{-s}. \tag{1.6}$$

Solving $\mathcal{I}_{10}(s; m) = \zeta(s)^{-1}$ gives

$$\Phi_{10}(s) = 7^s\left(\zeta(s)^{-1} - A_{10}(s)\right). \tag{1.7}$$

The symbolic construction replaces the single prime 7 by all primes in $(N/2, N]$ and produces exact finite identities for Φ_N.

The manuscript has four logical layers. Classical input consists of the Riemann–von Mangoldt formula, Guinand–Weil, prime-number asymptotics, and Landau–Gonek type information. Exact results proved here include the finite Möbius interpolation, the coefficient identities, the positive atomic defect, the signed tail, and the finite sampled carrier and variance formulas. Conjectural statements concern collective asymptotics of the sharp residual. All finite computations are placed in the numerical part, while the complete MATLAB program and console transcript are annexed after the bibliography.

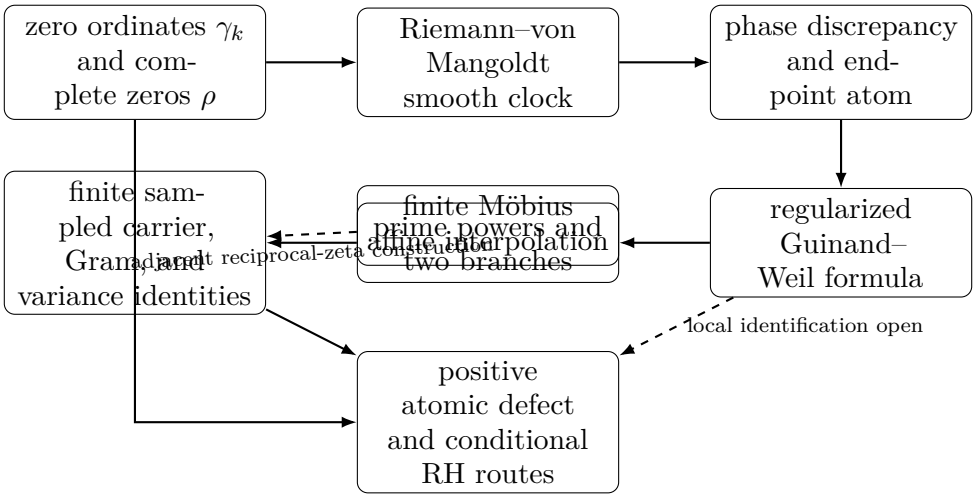


FIGURE 1. Logical flow of the manuscript. Solid arrows indicate proved constructions or exact reductions. Dashed arrows indicate an adjacent construction or an open identification. In particular, the use of unweighted prime-power modes as a complete sharp-residual model is not a proved arrow.

2. Main results and status map

The following table is a navigation device, not a list of novelty claims. "Proved here" means that a complete proof is supplied in this manuscript.

Statement	Status	Precise scope
Riemann–von Mangoldt counting formula and Guinand–Weil explicit formula	Classical, unconditional	Used with the normalization and admissible test class stated in the text.
Exact RVM phase-displacement decomposition	Proved here, unconditional	Exact for the indexed walk; the quadratic remainder is not proved small.
Two prime-power branches	Classical origin plus exact specialization	Exact with regularizing weights; completeness of the unweighted sharp model is conjectural.
Finite Möbius interpolation	Proved here, unconditional	Includes the canonical identity, endpoint value, real-axis limit, rounding, and tail extraction.
Positive ordinate-local defect	Proved here, unconditional	For one fixed $q > 1$, vanishing of the atomic defect is equivalent to RH.
Signed tail $\Delta_\omega(Q)$	Proved here, unconditional	$\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$ in the nonresonant case.
Finite carrier and projected variance identities	Proved here, unconditional	Exact identities for the chosen finite model and the actual joint projection used numerically.
Dominance of the $n = 3$ coefficient system at $\omega = 1$	Proved here, unconditional	Does not by itself imply bimodality of the full radial distribution.
Finite spectral alignment from equality on an interval	Classical linear independence, proved here	Applies to finite exponential polynomials after grouping repeated frequencies.
Seven RH routes	Conditional implications	Each implication is rigorous; the activating local or uniform estimate is open.
Global asymptotics for the sharp residual	Conjectural	No completeness, transport, shell localization, or energy-exhaustion theorem is claimed.
Full-data tables and frequency controls	Numerical	Finite support only; no asymptotic conclusion is inferred.

3. Notation and logical dependencies

Symbol	Meaning
γ_k	Positive zero ordinates repeated with multiplicity.
τ_j, m_j	Distinct positive ordinate and its total multiplicity.
$N(T), N^{\#}(T)$	Right-continuous zero count and midpoint count.
$\widetilde{\gamma}_k$	Inverse RVM clock value satisfying $N_0(\widetilde{\gamma}_k) = k - 1/2$.
ε_k	Ordinate-clock displacement $\gamma_k - \widetilde{\gamma}_k$. It is not a horizontal displacement and does not vanish under RH.
α_ρ	Horizontal displacement $\beta - 1/2$ of a zero $\rho = \beta + i\gamma$.
$q = e^\omega$	Phase base and angular frequency.
$M_H(x)$	Finite zero kernel $N(H)^{-1} \sum_{\gamma \le H} x^{i\gamma}$.
$R_N(s)$	Möbius Dirichlet tail $\zeta(s)^{-1} - \sum_{n \le N} \mu(n) n^{-s}$.
$\mathcal{Q}_N(\omega)$	Quadratic remainder in the indexed phase linearization; distinct from $R_N(s)$.

Symbol	Meaning
$\Delta_\omega(Q)$	Convergent signed prime-power tail.
$\widehat{c}_{Q,H}, c_{Q,H}$	Raw carrier coefficient and jointly fitted constant–carrier coefficient.

Part 1. The exponential walk and the Riemann–von Mangoldt benchmark

4. The zero-counting function and its smooth inverse

Definition 4.1 (Midpoint zero count)**.** Let $N(T)$ be the number of nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity. At a zero ordinate $T = \tau$, let

$$N^{\#}(\tau) = \frac{1}{2}\left(N(\tau^-) + N(\tau^+)\right). \tag{4.1}$$

For a simple ordered ordinate γ_k, this equals $k - 1/2$.

Definition 4.2 (Midpoint argument function and exact smooth remainder)**.** For T that is not a zero ordinate, let

$$S(T) = \frac{1}{\pi}\arg\zeta\left(\frac{1}{2} + iT\right), \tag{4.2}$$

with the standard path from 2 to $2 + iT$ and then to $1/2 + iT$. At a zero ordinate τ, define $S^{\#}(\tau)$ as the average of the two one-sided limits. Define

$$R_\vartheta(T) = 1 + \frac{\vartheta(T)}{\pi} - N_0(T), \tag{4.3}$$

where

$$\vartheta(T) = \Im \log\Gamma\left(\frac{1}{4} + \frac{iT}{2}\right) - \frac{T}{2}\log\pi. \tag{4.4}$$

Status. Classical result – unconditional.

Theorem 4.3 (Riemann–von Mangoldt formula)**.** *For every $T > 0$, with midpoint interpretation at zero ordinates,*

$$\boxed{N^{\#}(T) = N_0(T) + S^{\#}(T) + R_\vartheta(T).} \tag{4.5}$$

Moreover,

$$R_\vartheta(T) = O(T^{-1}) \qquad (T \to \infty). \tag{4.6}$$

The exact counting identity and the asymptotic expansion are standard; see [14, Chapter 9]*.*

Proof. For T that is not a zero ordinate, the argument principle applied to the completed zeta function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{4.7}$$

gives the exact identity

$$N(T) = 1 + \frac{\vartheta(T)}{\pi} + S(T). \tag{4.8}$$

At a zero ordinate, the average of the two one-sided identities gives the same formula with $N^{\#}$ and $S^{\#}$. Subtracting $N_0(T)$ and using the definition (4.3) proves (4.5) exactly.

It remains to estimate R_ϑ. Stirling's formula in a fixed sector gives

$$\log\Gamma(z) = \left(z - \frac{1}{2}\right)\log z - z + \frac{1}{2}\log(2\pi) + O(|z|^{-1}). \tag{4.9}$$

With $z = 1/4 + iT/2$, taking imaginary parts yields

$$\vartheta(T) = \frac{T}{2}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2} - \frac{\pi}{8} + O(T^{-1}). \tag{4.10}$$

Consequently,

$$1 + \frac{\vartheta(T)}{\pi} = \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{7}{8} + O(T^{-1}) \tag{4.11}$$

$$= N_0(T) + O(T^{-1}), \tag{4.12}$$

which is exactly (4.6). □

Status. Exact result proved in this manuscript – unconditional.

Theorem 4.4 (Principal Lambert–W inverse of the smooth clock)**.** *Let $x > -1/8$. The equation $N_0(T) = x$ has a unique solution $T > 2\pi$, namely*

$$\boxed{T = \frac{2\pi(x - 7/8)}{W_0((x - 7/8)/e)}.} \tag{4.13}$$

Here W_0 is the principal real branch of the Lambert function [1]*, and the value at $x = 7/8$ is understood by continuity as $T = 2\pi e$. In particular, for every $k \geq 1$,*

$$\widetilde{\gamma}_k = \frac{2\pi(k - 11/8)}{W_0((k - 11/8)/e)}. \tag{4.14}$$

Proof. Put

$$y = x - \frac{7}{8}, \qquad u = \frac{T}{2\pi e}. \tag{4.15}$$

The equation $N_0(T) = x$ is

$$y = e\,u\log u. \tag{4.16}$$

Writing $v = \log u$ gives

$$ve^v = \frac{y}{e}. \tag{4.17}$$

Since $x > -1/8$, one has $y/e > -1/e$, so the principal branch $W_0(y/e)$ is real and greater than -1. Thus

$$v = W_0(y/e), \qquad u = e^v = \frac{y/e}{W_0(y/e)}, \tag{4.18}$$

and hence (4.13). The inequality $W_0(y/e) > -1$ implies $u > e^{-1}$, hence $T > 2\pi$. Conversely, $N_0'(T) = (2\pi)^{-1}\log(T/2\pi) > 0$ on $(2\pi, \infty)$, so this solution is unique in that interval. At $y = 0$, $W_0(y/e) \sim y/e$, giving the continuous value $T = 2\pi e$. Finally, $x = k - 1/2 > -1/8$ for every $k \geq 1$, and substitution gives (4.14). □

5. Stieltjes representation and the smooth RVM walk

Definition 5.1 (Symmetric phase sum)**.** Define

$$S_\omega^{\#}(T) = \sum_{0<\gamma<T} m(\gamma)e^{i\omega\gamma} + \frac{1}{2}\sum_{\gamma=T} m(\gamma)e^{i\omega\gamma}. \tag{5.1}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 5.2 (Atomic representation and integration by parts)**.** *Let*

$$\mathrm{d}N = \sum_{\gamma>0} m(\gamma)\delta_\gamma. \tag{5.2}$$

Then

$$\boxed{S_\omega^{\#}(T) = \int_{(0,T)} e^{i\omega t}\ \mathrm{d}N(t) + \frac{1}{2}e^{i\omega T}\Delta N(T).} \tag{5.3}$$

If $0 < T_0 < T$ and neither endpoint is a zero ordinate, then

$$\int_{(T_0,T]} e^{i\omega t}\ \mathrm{d}N(t) = e^{i\omega T}N(T) - e^{i\omega T_0}N(T_0) \tag{5.4}$$

$$- i\omega \int_{T_0}^{T} e^{i\omega t}N(t)\ \mathrm{d}t. \tag{5.5}$$

The midpoint formula at a zero endpoint is obtained by averaging the two one-sided identities.

Proof. The first identity is the direct evaluation of the atomic measure $\mathrm{d}N$: atoms strictly below T receive full weight and the atom at T receives half weight. Merely changing the point value of the counting function would not change its Stieltjes measure, which is why the explicit endpoint term is required.

For endpoints away from atoms, N is a right-continuous bounded-variation function and $f(t) = e^{i\omega t}$ is continuously differentiable. The Riemann–Stieltjes integration-by-parts formula gives

$$\int_{(T_0,T]} f\ \mathrm{d}N = f(T)N(T) - f(T_0)N(T_0) - \int_{T_0}^{T} N(t)f'(t)\ \mathrm{d}t. \tag{5.6}$$

Substitution of $f'(t) = i\omega e^{i\omega t}$ proves the displayed formula. Averaging the limits from the two sides supplies half of any endpoint atom. □

Warning 5.3 (The endpoint term is an atom, not an error term)*.* At a zero ordinate $T = \tau$, the boundary contribution in (5.3) is exactly

$$\frac{m(\tau)}{2}e^{i\omega\tau}. \tag{5.7}$$

Its modulus is $m(\tau)/2$; it is not naturally small as T grows. The right-endpoint convention differs from the midpoint convention by the same half atom. Every sharp-window reconstruction in the prime-power part must retain this term explicitly rather than absorb it into an asymptotic error.

Status. Exact result proved in this manuscript – unconditional.

Theorem 5.4 (Smooth Riemann–von Mangoldt contribution)**.** *The Stieltjes integral of the smooth density*

$$N_0'(t) = \frac{1}{2\pi}\log\!\left(\frac{t}{2\pi}\right) \tag{5.8}$$

has the antiderivative

$$\boxed{A_{\mathrm{smooth},\omega}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega} + C_\omega.} \tag{5.9}$$

Consequently, for any $T_0 > 0$,

$$\int_{T_0}^{T} e^{i\omega t}N_0'(t)\ \mathrm{d}t = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0). \tag{5.10}$$

Proof. Differentiate the numerator in (5.9). The derivative of the first term is

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\log(T/2\pi)\right] = i\omega e^{i\omega T}\log(T/2\pi) + \frac{e^{i\omega T}}{T}. \tag{5.11}$$

Since $\frac{\mathrm{d}}{\mathrm{d}z}\operatorname{Ei}(z) = e^z/z$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}T}\operatorname{Ei}(i\omega T) = \frac{e^{i\omega T}}{T}. \tag{5.12}$$

The two T^{-1} terms cancel. Division by $2\pi i\omega$ leaves exactly $e^{i\omega T}N_0'(T)$. The definite identity follows from the fundamental theorem of calculus on any interval avoiding the chosen branch cut; a change of branch alters only the constant C_ω. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 5.5 (Exact RVM remainder decomposition)**.** *Let $0 < T_0 < T$ and suppose first that neither endpoint is a zero ordinate. Then*

$$\int_{(T_0,T]} e^{i\omega t}\,\mathrm{d}N(t) = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0) \tag{5.13}$$

$$+ \int_{(T_0,T]} e^{i\omega t}\,\mathrm{d}S^{\#}(t) + \int_{T_0}^{T} e^{i\omega t}R_\vartheta'(t)\,\mathrm{d}t. \tag{5.14}$$

At a zero endpoint, the corresponding symmetric identity is obtained by averaging the one-sided limits.

Proof. The exact identity (4.5) is an equality of locally bounded-variation functions:

$$N^{\#}(t) = N_0(t) + S^{\#}(t) + R_\vartheta(t). \tag{5.15}$$

Away from endpoint atoms, taking Stieltjes differentials gives

$$\mathrm{d}N = N_0'(t)\,\mathrm{d}t + \mathrm{d}S^{\#}(t) + R_\vartheta'(t)\,\mathrm{d}t. \tag{5.16}$$

Integrating against $e^{i\omega t}$ over $(T_0, T]$ and invoking Theorem 5.4 for the N_0'-term proves the formula. Midpoint endpoint weights follow from the same one-sided averaging used in Proposition 5.2. □

6. Exact comparison of the indexed zero walk with the smooth clock

Definition 6.1. For each k, define the ordinate–clock displacement

$$\varepsilon_k = \gamma_k - \widetilde{\gamma}_k. \tag{6.1}$$

Status. Exact result proved in this manuscript – unconditional.

Lemma 6.2 (Exact phase perturbation identity)**.** *For real x and h,*

$$e^{i(x+h)} - e^{ix} = ih\int_0^1 e^{i(x+th)}\,\mathrm{d}t. \tag{6.2}$$

In particular,

$$|e^{i(x+h)} - e^{ix}| \le |h|, \tag{6.3}$$

$$\left|e^{i(x+h)} - e^{ix} - ihe^{ix}\right| \le \frac{h^2}{2}. \tag{6.4}$$

Proof. Define $F(t) = e^{i(x+th)}$ on $[0,1]$. Since $F'(t) = ihe^{i(x+th)}$, the fundamental theorem of calculus gives

$$e^{i(x+h)} - e^{ix} = F(1) - F(0) = ih \int_0^1 e^{i(x+th)} \, \mathrm{d}t. \tag{6.5}$$

The integrand has modulus one, so taking absolute values yields the Lipschitz estimate. For the quadratic estimate, subtract ihe^{ix}:

$$e^{i(x+h)} - e^{ix} - ihe^{ix} = ihe^{ix} \int_0^1 (e^{ith} - 1) \, \mathrm{d}t. \tag{6.6}$$

Apply the already proved Lipschitz bound to $e^{ith} - 1$, obtaining $|e^{ith} - 1| \le t|h|$. Integration of t from zero to one gives $h^2/2$. □

Status. Exact result proved in this manuscript – unconditional.

Theorem 6.3 (Exact and first-order decompositions of the discrepancy)**.** *For every N,*

$$\mathcal{E}_N(\omega) = i\omega \sum_{k=1}^{N} \varepsilon_k \int_0^1 e^{i\omega(\widetilde{\gamma}_k + t\varepsilon_k)} \, \mathrm{d}t. \tag{6.7}$$

Moreover,

$$\mathcal{E}_N(\omega) = i\omega \sum_{k=1}^{N} \varepsilon_k e^{i\omega\widetilde{\gamma}_k} + \mathcal{Q}_N(\omega), \tag{6.8}$$

where

$$|\mathcal{Q}_N(\omega)| \le \frac{\omega^2}{2} \sum_{k=1}^{N} \varepsilon_k^2. \tag{6.9}$$

Proof. For each k, apply Lemma 6.2 to

$$x = \omega\widetilde{\gamma}_k, \qquad h = \omega(\gamma_k - \widetilde{\gamma}_k) = \omega\varepsilon_k. \tag{6.10}$$

The resulting identity is

$$e^{i\omega\gamma_k} - e^{i\omega\widetilde{\gamma}_k} = i\omega\varepsilon_k \int_0^1 e^{i\omega(\widetilde{\gamma}_k + t\varepsilon_k)} \, \mathrm{d}t. \tag{6.11}$$

Summing from 1 to N gives the exact formula. For the first-order form, the quadratic estimate in the lemma gives a remainder r_k with

$$|r_k| \le \frac{\omega^2\varepsilon_k^2}{2}. \tag{6.12}$$

The triangle inequality then gives the stated bound for the sum of all remainders. No restriction on the signs or dependence of the ordinate–clock displacements is needed. □

Definition 6.4 (Midpoint rank of a distinct ordinate)**.** Let

$$0 < \tau_1 < \tau_2 < \cdots. \tag{6.13}$$

be the distinct positive zero ordinates, and let m_j be the total multiplicity at τ_j. Define the midpoint rank

$$\kappa_j = N(\tau_j^-) + \frac{m_j}{2} = N^{\#}(\tau_j), \tag{6.14}$$

the smooth comparison ordinate

$$\widetilde{\tau}_j = N_0^{-1}(\kappa_j), \tag{6.15}$$

and the displacement

$$\varepsilon_j^{\#} = \tau_j - \widetilde{\tau}_j. \tag{6.16}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 6.5 (Displacement in terms of the Riemann–von Mangoldt remainder)**.** *For every distinct positive zero ordinate τ_j, there is a point ξ_j between τ_j and $\widetilde{\tau}_j$ such that*

$$\boxed{\varepsilon_j^{\#} = -\frac{S^{\#}(\tau_j) + R_\vartheta(\tau_j)}{N_0'(\xi_j)}.} \tag{6.17}$$

Equivalently,

$$\varepsilon_j^{\#} = -\frac{2\pi\,[S^{\#}(\tau_j) + R_\vartheta(\tau_j)]}{\log(\xi_j/2\pi)}. \tag{6.18}$$

If τ_j is simple and is the j-th ordinate when ordinates are listed with multiplicity, then $\kappa_j = j - \frac{1}{2}$, $\widetilde{\tau}_j = \widetilde{\gamma}_j$, and $\varepsilon_j^{\#} = \varepsilon_j$.

Proof. By Definitions 4.1 and 6.4,

$$N_0(\widetilde{\tau}_j) = \kappa_j = N^{\#}(\tau_j). \tag{6.19}$$

The exact Riemann–von Mangoldt identity (4.5) therefore gives

$$N_0(\widetilde{\tau}_j) - N_0(\tau_j) = S^{\#}(\tau_j) + R_\vartheta(\tau_j). \tag{6.20}$$

Both τ_j and $\widetilde{\tau}_j$ exceed 2π, so N_0 is differentiable on the closed interval joining them and $N_0'(t) = (2\pi)^{-1}\log(t/2\pi) > 0$ there. The mean-value theorem provides ξ_j between the two ordinates such that

$$N_0(\widetilde{\tau}_j) - N_0(\tau_j) = N_0'(\xi_j)(\widetilde{\tau}_j - \tau_j) \tag{6.21}$$

$$= -N_0'(\xi_j)\varepsilon_j^{\#}. \tag{6.22}$$

Solving for $\varepsilon_j^{\#}$ gives (6.17); substitution of the explicit derivative gives (6.18). In the simple case $m_j = 1$ and $N(\tau_j^-) = j - 1$, so $\kappa_j = j - \frac{1}{2}$, proving the final assertion. □

6.1. **Magnitude of the quadratic displacement sum.** The indexed phase remainder in (6.9) contains the displacements ε_k attached to the list repeated with multiplicity. The exact Riemann–von Mangoldt formula at a multiple ordinate naturally gives instead the midpoint displacement $\varepsilon_j^{\#}$. The following identity keeps those two objects separate.

Status. Exact result proved in this manuscript – unconditional.

Proposition 6.6 (Multiplicity correction between indexed and atomic clocks)**.** *Let τ_j have multiplicity m_j, let $a_j = N(\tau_j^-)$, and write the corresponding repeated indices as $k = a_j + r$, $1 \le r \le m_j$. Put*

$$\theta_{j,r} = r - \frac{m_j + 1}{2}. \tag{6.23}$$

There exists a point $\xi_{j,r}$ between $\widetilde{\tau}_j$ and $\widetilde{\gamma}_{a_j+r}$ such that

$$\boxed{\varepsilon_{a_j+r} = \varepsilon_j^{\#} - \frac{\theta_{j,r}}{N_0'(\xi_{j,r})}.} \tag{6.24}$$

Consequently,

$$\sum_{r=1}^{m_j} |\varepsilon_{a_j+r}|^2 \le 2m_j|\varepsilon_j^{\#}|^2 + \frac{m_j(m_j^2-1)}{6\,d_j^2}, \tag{6.25}$$

where

$$d_j = \min_{t \text{ between } \widetilde{\tau}_j \text{ and all } \widetilde{\gamma}_{a_j+r}} N_0'(t) > 0. \tag{6.26}$$

For a simple zero, $m_j = 1$ and the correction term vanishes.

Proof. The midpoint rank is $\kappa_j = a_j + m_j/2$, whereas the repeated index $a_j + r$ uses rank $a_j + r - 1/2$. Their difference is exactly

$$(a_j + r - 1/2) - \kappa_j = \theta_{j,r}. \tag{6.27}$$

Apply the mean-value theorem to the inverse relation $N_0(\widetilde{\gamma}_k) = k - 1/2$:

$$\widetilde{\gamma}_{a_j+r} - \widetilde{\tau}_j = \frac{\theta_{j,r}}{N_0'(\xi_{j,r})}. \tag{6.28}$$

Subtract this identity from $\tau_j - \widetilde{\tau}_j = \varepsilon_j^{\#}$ to obtain (6.24). The inequality $|x-y|^2 \le 2|x|^2 + 2|y|^2$, followed by

$$\sum_{r=1}^{m_j} \theta_{j,r}^2 = \frac{m_j(m_j^2-1)}{12}, \tag{6.29}$$

proves (6.25). □

Definition 6.7. Define the atomic-midpoint and repeated-index square sums by

$$\mathscr{E}_2^{\#}(T) = \sum_{0<\tau_j \le T} m_j |\varepsilon_j^{\#}|^2, \tag{6.30}$$

$$\mathscr{E}_2^{\mathrm{idx}}(T) = \sum_{\gamma_k \le T} |\varepsilon_k|^2, \tag{6.31}$$

and define their windowed analogues by restricting the ordinates to $(T, T+L]$.

Status. Bounds deduced from classical estimates – unconditional unless RH is explicitly assumed.

Proposition 6.8 (Crude global and short-window bounds)**.** *For $1 \le L \le T$,*

$$\mathscr{E}_2^{\#}(T),\ \mathscr{E}_2^{\mathrm{idx}}(T) = O(T \log T), \tag{6.32}$$

$$\mathscr{E}_2^{\#}(T;L),\ \mathscr{E}_2^{\mathrm{idx}}(T;L) = O(L \log T + \log T). \tag{6.33}$$

Under RH,

$$\mathscr{E}_2^{\#}(T),\ \mathscr{E}_2^{\mathrm{idx}}(T) = O\Big(\frac{T \log T}{(\log\log T)^2}\Big), \tag{6.34}$$

$$\mathscr{E}_2^{\#}(T;L),\ \mathscr{E}_2^{\mathrm{idx}}(T;L) = O\Big(\frac{L \log T + \log T}{(\log\log T)^2}\Big). \tag{6.35}$$

The implied constants may depend on a fixed lower height. None of these bounds makes the unnormalized quadratic phase remainder small.

Corollary 6.9 (What the available square-sum bounds actually imply)**.** *With $N(T) \sim (T/2\pi)\log T$, the global bounds imply*

$$\frac{\mathscr{E}_2^{\#}(T)}{N(T)},\ \frac{\mathscr{E}_2^{\mathrm{idx}}(T)}{N(T)} = O(1), \tag{6.36}$$

$$\frac{\mathscr{E}_2^{\#}(T)}{N(T)},\ \frac{\mathscr{E}_2^{\mathrm{idx}}(T)}{N(T)} = O\Big(\frac{1}{(\log\log T)^2}\Big) \quad \textit{under RH.} \tag{6.37}$$

For a fixed-length window, the unconditional estimate remains only

$$\mathscr{E}_2^{\#}(T;L),\ \mathscr{E}_2^{\mathrm{idx}}(T;L) = O(\log T) \qquad (L \asymp 1), \tag{6.38}$$

and does not tend to zero. Hence the displacement is globally bounded in mean square, and under RH globally small on average, but no presently used bound supplies uniform local smallness on every short window.

Remark 6.10 (Why the quadratic phase remainder remains decisive). The exact estimate

$$|\mathcal{Q}_N(\omega)| \le \frac{\omega^2}{2} \sum_{k\le N} |\varepsilon_k|^2 \tag{6.39}$$

therefore cannot be turned into $o(1)$, or even into an error smaller than the first-order term, from the preceding bounds alone. Three distinct questions remain: global average control, control after localization by a smooth compactly supported weight, and uniform control in windows of length comparable with $1/\omega$. The last two are the ones required by the local routes to RH.

Proof. Proposition 6.5 and $N_0'(t) = (2\pi)^{-1}\log(t/2\pi)$ give, for large ordinates,

$$|\varepsilon_j^{\#}| \ll \frac{|S^{\#}(\tau_j)| + O(\tau_j^{-1})}{\log \tau_j}. \tag{6.40}$$

The classical unconditional estimate $S(T) = O(\log T)$ therefore gives $|\varepsilon_j^{\#}| = O(1)$. Under RH, Littlewood's bound gives $|\varepsilon_j^{\#}| = O(1/\log\log T)$. The local zero count

$$N(T+L) - N(T) = O(L\log T + \log T) \tag{6.41}$$

then proves the asserted atomic-midpoint bounds.

For the repeated-index sum, use Proposition 6.6. The multiplicity at height t obeys

$$m(t) \le N(t+1) - N(t-1) = O(\log t), \tag{6.42}$$

Under RH, the jump relation for S, together with Littlewood's bound, gives

$$m(t) = O\Big(\frac{\log t}{\log\log t}\Big). \tag{6.43}$$

Since every relevant derivative $d_j \asymp \log T$ on a fixed high window, one has

$$\sum_j \frac{m_j(m_j^2-1)}{d_j^2} \ll \frac{(\max_j m_j)^2}{(\log T)^2} \sum_j m_j. \tag{6.44}$$

Unconditionally $\max_j m_j = O(\log T)$, while under RH $\max_j m_j = O(\log T/\log\log T)$. Since $\sum_j m_j$ is the ordinary zero count in the relevant global interval or short window, (6.44) has precisely the orders asserted above. This proves the indexed bounds. The bounds for S, multiplicities, and zero counts are standard; see [14, Chapters 9 and 14]. □

Warning 6.11 (What these bounds do not establish). The ordinate-clock displacements ε_k and $\varepsilon_j^{\#}$ are not horizontal displacements $\beta - 1/2$. RH does not force them to vanish; even on RH the ordinates fluctuate around the RVM clock through $S(T)$. Consequently, a bound for these square sums is not an equivalent form of RH. Continuous mean-square information for $S(t)$ also does not automatically control its values at zero ordinates in windows of fixed length. A route based on (6.9) needs a genuinely discrete and local estimate.

Remark 6.12. Equations (6.8) and (6.18) explain why a comparison with the RVM clock is sensitive to the argument of zeta. The explicit formula then rewrites that argument contribution in terms of prime powers. This is the analytic route by which the prime-power branches entered the project.

Part 2. Finite Möbius interpolation

7. The first affine interpolation: numerical and symbolic forms

Definition 7.1 (Terminal affine interpolation)**.** Let $Q_N = \{q_1, \dots, q_{h_N}\}$ be the primes in $(N/2, N]$, and let A_N, P_N, B_N be defined in Section 8. For a prescribed terminal sum $m \in \mathbb{C}$, define

$$\boxed{\mathcal{I}_N(s; m) = A_N(s) + \frac{m - B_N}{h_N} P_N(s).} \tag{7.1}$$

This is the symmetric affine interpolation in which all terminal-prime coefficients are assigned their common mean

$$\lambda_N(m) = \frac{m - B_N}{h_N}. \tag{7.2}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 7.2 (Recovery of the true finite Möbius polynomial)**.** *For the true Mertens value $m = M(N)$,*

$$\mathcal{I}_N(s; M(N)) = M_s(N). \tag{7.3}$$

Proof. The set Q_N contains only primes, so every terminal Möbius coefficient is exactly -1. Therefore its total contribution at $s = 0$ is $-h_N$, which proves $M(N) = B_N - h_N$. Hence the affine mean coefficient for the true terminal value is

$$\frac{M(N) - B_N}{h_N} = \frac{-h_N}{h_N} = -1. \tag{7.4}$$

Substitution assigns coefficient -1 to every term in $P_N(s)$, so

$$\mathcal{I}_N(s; M(N)) = A_N(s) - P_N(s). \tag{7.5}$$

The right side is exactly the decomposition of $M_s(N)$. □

Example 7.3 (The numerical model $N = 10$)**.** The Möbius values up to ten are

$$(1, -1, -1, 0, -1, 1, -1, 0, 0, 1). \tag{7.6}$$

The only terminal prime is 7, so

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad P_{10}(s) = 7^{-s}, \tag{7.7}$$

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \qquad B_{10} = 0. \tag{7.8}$$

Thus

$$\mathcal{I}_{10}(s; m) = A_{10}(s) + m7^{-s}. \tag{7.9}$$

The actual terminal value is $M(10) = -1$, which restores the coefficient -1 of 7^{-s}.

Status. Exact result proved in this manuscript – unconditional.

Theorem 7.4 (Completion equation)**.** *At every point where $P_N(s) \neq 0$, the unique value of m satisfying*

$$\mathcal{I}_N(s; m) = \zeta(s)^{-1} \tag{7.10}$$

is

$$\boxed{m = \Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)}.} \tag{7.11}$$

Proof. Starting from the affine interpolation, the completion equation is

$$A_N(s) + \frac{m - B_N}{h_N} P_N(s) = \zeta(s)^{-1}. \tag{7.12}$$

Because $h_N \neq 0$ and $P_N(s) \neq 0$, this is a nondegenerate linear equation in m. Subtract $A_N(s)$, multiply by $h_N/P_N(s)$, and add B_N to obtain the displayed solution. If two values m_1, m_2 solved the equation, subtracting their equations would give

$$\frac{m_1 - m_2}{h_N} P_N(s) = 0, \tag{7.13}$$

so $m_1 = m_2$. This proves both existence and uniqueness. □

Remark 7.5. The function $\Phi_N(s)$ is therefore not introduced by an arbitrary quotient. It is the terminal value selected by requiring the affine finite interpolation to coincide with the completed reciprocal-zeta Dirichlet series at the spectral parameter s.

8. Terminal half-interval decomposition

Definition 8.1. For an integer $N \geq 2$, define

$$Q_N = \left\{ q \text{ prime} : \frac{N}{2} < q \leq N \right\}, \tag{8.1}$$

and put

$$h_N = |Q_N|, \qquad q_N = \min Q_N. \tag{8.2}$$

Bertrand's postulate guarantees $h_N \geq 1$.

Definition 8.2. Let

$$M_s(N) = \sum_{n \leq N} \frac{\mu(n)}{n^s}, \tag{8.3}$$

$$P_N(s) = \sum_{q \in Q_N} q^{-s}, \tag{8.4}$$

$$A_N(s) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \frac{\mu(n)}{n^s}, \tag{8.5}$$

$$B_N = A_N(0) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \mu(n). \tag{8.6}$$

Since every $q \in Q_N$ is prime and $\mu(q) = -1$,

$$M_s(N) = A_N(s) - P_N(s) \tag{8.7}$$

and

$$M(N) = B_N - h_N. \tag{8.8}$$

Definition 8.3. For $\Re s > 1$, define the exact tail

$$R_N(s) = \frac{1}{\zeta(s)} - M_s(N) = \sum_{n > N} \frac{\mu(n)}{n^s}. \tag{8.9}$$

The first equality supplies a meromorphic continuation of R_N.

Definition 8.4. Define

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} \tag{8.10}$$

at points at which the quotient is defined, and meromorphically elsewhere.

9. Canonical identity and endpoint values

Status. Exact result proved in this manuscript – unconditional.

Theorem 9.1 (Canonical orientation)**.** *For every $N \geq 2$,*

$$\boxed{\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = M(N) + h_N \frac{R_N(s)}{P_N(s)}.} \tag{9.1}$$

Proof. The functions $A_N(s)$ and $P_N(s)$ are finite Dirichlet polynomials and hence entire. In the half-plane $\Re s > 1$, absolute convergence of the reciprocal-zeta Dirichlet series gives

$$\zeta(s)^{-1} = M_s(N) + R_N(s). \tag{9.2}$$

The terminal decomposition $M_s(N) = A_N(s) - P_N(s)$ therefore yields

$$\zeta(s)^{-1} - A_N(s) = M_s(N) + R_N(s) - A_N(s) \tag{9.3}$$

$$= R_N(s) - P_N(s). \tag{9.4}$$

Divide by $P_N(s)$ at a point where it is nonzero, multiply by h_N, and add B_N:

$$\Phi_N(s) = B_N + h_N \left(\frac{R_N(s)}{P_N(s)} - 1 \right) \tag{9.5}$$

$$= B_N - h_N + h_N \frac{R_N(s)}{P_N(s)}. \tag{9.6}$$

Because the terminal primes all have Möbius value -1, $M(N) = B_N - h_N$. This proves the identity on $\Re s > 1$. Both sides are meromorphic continuations of the same function, so the identity extends to every connected component of their common domain and hence meromorphically to the plane. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 9.2 (Value at the origin)**.** *The meromorphic interpolation satisfies*

$$\boxed{\Phi_N(0) = -2.} \tag{9.7}$$

Proof. The point $s = 0$ is regular for every finite Dirichlet polynomial in the formula. The classical special value $\zeta(0) = -1/2$ gives

$$\zeta(0)^{-1} = -2. \tag{9.8}$$

Moreover, by definition,

$$A_N(0) = B_N, \qquad P_N(0) = \sum_{q \in Q_N} 1 = h_N. \tag{9.9}$$

Since $h_N \geq 1$, the denominator does not vanish. Direct substitution in the completion solution gives

$$\Phi_N(0) = B_N + h_N \frac{-2 - B_N}{h_N} \tag{9.10}$$

$$= B_N - 2 - B_N = -2. \tag{9.11}$$

No limiting argument is involved. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 9.3 (Large positive real limit)**.** *Along the positive real axis,*

$$\boxed{\lim_{\sigma \to +\infty} \Phi_N(\sigma) = M(N).} \tag{9.12}$$

Proof. For $\sigma > 1$, the tail satisfies

$$|R_N(\sigma)| \le \sum_{n>N} n^{-\sigma} \le \int_N^\infty x^{-\sigma}\, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{9.13}$$

The term $q_N^{-\sigma}$ occurs in $P_N(\sigma)$, and all terms are positive, hence

$$P_N(\sigma) \ge q_N^{-\sigma}. \tag{9.14}$$

Consequently,

$$\left| h_N \frac{R_N(\sigma)}{P_N(\sigma)} \right| \le \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^\sigma. \tag{9.15}$$

For $N \ge 3$, Bertrand's theorem gives a prime in $(N/2, N)$, so $q_N < N$, and the exponential factor tends to zero. For $N = 2$, $q_N/N = 1$, but the same bound is $2/(\sigma - 1)$, which also tends to zero. Hence the second expression in Theorem 9.1 tends to $M(N)$ for every $N \ge 2$. □

Example 9.4 (The $N = 10$ orientation check)**.** For $N = 10$,

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad M(10) = -1, \qquad B_{10} = 0. \tag{9.16}$$

The inverted quotient P_N/R_N would give $-1/2$ at $s = 0$. The quotient in (9.1) gives

$$0 + \frac{-2}{1} = -2. \tag{9.17}$$

This finite check confirms the orientation of the quotient in Theorem 9.1.

10. Exponential approach, rounding, and recovery of the first tail term

Definition 10.1. Let

$$a_N = \min\{n > N : \mu(n) \neq 0\}. \tag{10.1}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 10.2 (Leading exponential term)**.** *As $\sigma \to +\infty$,*

$$\boxed{\Phi_N(\sigma) - M(N) = h_N \mu(a_N) \left(\frac{q_N}{a_N} \right)^\sigma (1 + o(1)).} \tag{10.2}$$

Proof. Factor the least terminal prime from the denominator:

$$P_N(\sigma) = q_N^{-\sigma} \left[1 + \sum_{\substack{q \in Q_N \\ q > q_N}} \left(\frac{q_N}{q} \right)^\sigma \right]. \tag{10.3}$$

Every ratio in the bracket is strictly smaller than one, and there are only finitely many of them. Thus the bracket is $1 + o(1)$.

For the numerator, factor the first nonzero Möbius term after N:

$$R_N(\sigma) = \mu(a_N) a_N^{-\sigma} \left[1 + \sum_{n > a_N} \frac{\mu(n)}{\mu(a_N)} \left(\frac{a_N}{n} \right)^\sigma \right]. \tag{10.4}$$

For $\sigma \ge 2$, the absolute value of the summand is bounded by $(a_N/n)^2$, whose sum over $n > a_N$ is finite. Each fixed summand tends to zero as $\sigma \to \infty$. Dominated convergence therefore shows that the bracket is $1 + o(1)$. Division gives

$$\frac{R_N(\sigma)}{P_N(\sigma)} = \mu(a_N) \left(\frac{q_N}{a_N} \right)^\sigma (1 + o(1)). \tag{10.5}$$

Multiply by h_N and use the canonical identity. □

Remark 10.3 (Existence of the root limit and absence of leading cancellation). The limit in Corollary 10.8 is not an additional hypothesis. The integer a_N is the unique least index with nonzero tail coefficient. After factoring $\mu(a_N)a_N^{-\sigma}$, the remaining absolute tail is dominated by $\sum_{n>a_N}(a_N/n)^2$ and tends to zero by dominated convergence. Hence the leading term cannot be cancelled by later terms for all sufficiently large real σ.

Remark 10.4. The factor is $(q_N/a_N)^\sigma$, not $(a_N/q_N)^\sigma$. Since $q_N \le N < a_N$, the correct factor decays. The inverted factor would contradict Corollary 9.3.

Status. Exact result proved in this manuscript – unconditional.

Theorem 10.5 (Uniform real-axis error bound)**.** *For every real $\sigma > 1$,*

$$\boxed{|\Phi_N(\sigma) - M(N)| \le \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^\sigma.} \tag{10.6}$$

Proof. The reciprocal-zeta tail is absolutely convergent for $\sigma > 1$, so

$$|R_N(\sigma)| \le \sum_{n>N} |\mu(n)| n^{-\sigma} \tag{10.7}$$

$$\le \sum_{n>N} n^{-\sigma} \tag{10.8}$$

$$\le \int_N^\infty x^{-\sigma}\, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{10.9}$$

The denominator is a sum of positive real terms. Retaining only its largest term gives

$$P_N(\sigma) = \sum_{q \in Q_N} q^{-\sigma} \ge q_N^{-\sigma}. \tag{10.10}$$

The canonical identity now gives

$$|\Phi_N(\sigma) - M(N)| = h_N \frac{|R_N(\sigma)|}{P_N(\sigma)} \tag{10.11}$$

$$\le \frac{h_N N^{1-\sigma} q_N^\sigma}{\sigma - 1} \tag{10.12}$$

$$= \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^\sigma. \tag{10.13}$$

□

Status. Exact result proved in this manuscript – unconditional.

Corollary 10.6 (Finite rounding criterion)**.** *If*

$$\boxed{\frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^\sigma < \frac{1}{2},} \tag{10.14}$$

then $M(N)$ is the unique nearest integer to the real number $\Phi_N(\sigma)$.

Proof. Theorem 10.5 and (10.14) give

$$|\Phi_N(\sigma) - M(N)| < \frac{1}{2}. \tag{10.15}$$

For any integer $k \ne M(N)$, the reverse triangle inequality yields

$$|\Phi_N(\sigma) - k| \ge |k - M(N)| - |\Phi_N(\sigma) - M(N)| \tag{10.16}$$

$$> 1 - \frac{1}{2} = \frac{1}{2}. \tag{10.17}$$

Thus $M(N)$ is strictly closer to $\Phi_N(\sigma)$ than every other integer. □

Definition 10.7. Define

$$E_N(s) = \frac{\Phi_N(s) - M(N)}{h_N} = \frac{R_N(s)}{P_N(s)}. \tag{10.18}$$

Status. Exact result proved in this manuscript – unconditional.

Corollary 10.8 (Recovery of a_N)**.** *The first nonzero Möbius term after N is recovered by*

$$\boxed{a_N = q_N \lim_{\sigma\to+\infty} |E_N(\sigma)|^{-1/\sigma}.} \tag{10.19}$$

Proof. By Theorem 10.2 and Definition 10.7,

$$E_N(\sigma) = \mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}(1 + o(1)). \tag{10.20}$$

Since $|\mu(a_N)| = 1$, taking absolute values and then $-1/\sigma$-th powers gives

$$|E_N(\sigma)|^{-1/\sigma} = \frac{a_N}{q_N}\,|1 + o(1)|^{-1/\sigma} \longrightarrow \frac{a_N}{q_N}. \tag{10.21}$$

Multiplication by q_N proves (10.19). □

11. Feasible terminal values and non-distinction in $\Re s > 1$

Definition 11.1. For an integer parameter m, define

$$F_N(s; m) = M_s(N) + P_N(s)\left(1 + \frac{m - B_N}{h_N}\right). \tag{11.1}$$

A value m is called terminally feasible when

$$-1 \le \frac{m - B_N}{h_N} \le 1. \tag{11.2}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 11.2 (Exact substitution identity)**.** *The expression* (11.1) *is the exact result of substituting*

$$A_N(s) = M_s(N) + P_N(s) \tag{11.3}$$

into the terminal-value interpolation.

Proof. From $M_s(N) = A_N(s) - P_N(s)$ one has $A_N(s) = M_s(N) + P_N(s)$. Substitution in (7.1) gives

$$\mathcal{I}_N(s; m) = M_s(N) + P_N(s) + \frac{m - B_N}{h_N}P_N(s) \tag{11.4}$$

$$= F_N(s; m), \tag{11.5}$$

which is the claimed identity. □

Status. Exact result proved in this manuscript – unconditional.

Theorem 11.3 (The half-plane limit does not select $M(N)$)**.** *Fix s with $\Re s > 1$. For every sequence of feasible integers $m(N)$,*

$$\boxed{F_N(s; m(N)) \longrightarrow \frac{1}{\zeta(s)} \qquad (N \to \infty).} \tag{11.6}$$

Thus convergence in the absolutely convergent half-plane does not distinguish $m(N) = M(N)$ from other feasible terminal values.

Proof. Fix s with $\sigma = \Re s > 1$. Absolute convergence of $\sum \mu(n)n^{-s}$ gives

$$M_s(N) \longrightarrow \zeta(s)^{-1}. \tag{11.7}$$

For the terminal prime polynomial,

$$|P_N(s)| \le \sum_{N/2<q\le N} q^{-\sigma} \tag{11.8}$$

$$\le \sum_{N/2<n\le N} n^{-\sigma} \tag{11.9}$$

$$\le \int_{N/2-1}^{N} x^{-\sigma}\,\mathrm{d}x = O_\sigma(N^{1-\sigma}). \tag{11.10}$$

Feasibility means $|(m(N) - B_N)/h_N| \le 1$, so the multiplier $1 + (m(N) - B_N)/h_N$ has absolute value at most two. The entire terminal correction is therefore $O_\sigma(N^{1-\sigma})$, uniformly over all feasible choices. Adding it to $M_s(N)$ proves the common limit. □

Remark 11.4. The selection of the true value $M(N)$ is arithmetical and combinatorial. It is not produced by convergence to $1/\zeta(s)$ in $\Re s > 1$, because the entire feasible family has the same limit.

12. Poles and finite parts at zeta zeros

Definition 12.1. If a meromorphic function has a Laurent expansion

$$f(s) = \frac{a_{-1}}{s-\rho} + a_0 + O(s-\rho), \tag{12.1}$$

its finite part at ρ is

$$\mathrm{FP}_{s=\rho} f(s) = a_0. \tag{12.2}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 12.2 (Finite part of the reciprocal zeta function)**.** *Let ρ be a simple zero of ζ. Then*

$$\frac{1}{\zeta(s)} = \frac{1}{\zeta'(\rho)(s-\rho)} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(s-\rho), \tag{12.3}$$

and hence

$$\mathrm{FP}_{s=\rho} \frac{1}{\zeta(s)} = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2}. \tag{12.4}$$

Proof. Write $z = s - \rho$. Since the zero is simple,

$$\zeta(\rho+z) = \zeta'(\rho)z + \frac{1}{2}\zeta''(\rho)z^2 + O(z^3). \tag{12.5}$$

Factor the linear term:

$$\zeta(\rho+z) = \zeta'(\rho)z\left[1 + \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right]. \tag{12.6}$$

Using $(1+w)^{-1} = 1 - w + O(w^2)$,

$$\frac{1}{\zeta(\rho+z)} = \frac{1}{\zeta'(\rho)z}\left[1 - \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right] \tag{12.7}$$

$$= \frac{1}{\zeta'(\rho)z} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(z). \tag{12.8}$$

The constant Laurent coefficient is the finite part. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 12.3 (Finite part of Φ_N)**.** *Assume that ρ is a simple zero of ζ and $P_N(\rho) \neq 0$. Then*

$$\mathrm{FP}_{s=\rho}\,\Phi_N(s) = B_N + h_N \frac{-\dfrac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho)}{P_N(\rho)} \tag{12.9}$$

$$- h_N \frac{P_N'(\rho)}{\zeta'(\rho) P_N(\rho)^2}. \tag{12.10}$$

Proof. Set $z = s - \rho$. Proposition 12.2 and the Taylor expansion of the entire function A_N give

$$\zeta(s)^{-1} - A_N(s) = \frac{a_{-1}}{z} + a_0 + O(z), \tag{12.11}$$

where

$$a_{-1} = \frac{1}{\zeta'(\rho)}, \qquad a_0 = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho). \tag{12.12}$$

Likewise,

$$P_N(s) = p_0 + p_1 z + O(z^2), \qquad p_0 = P_N(\rho), \quad p_1 = P_N'(\rho). \tag{12.13}$$

Because $p_0 \neq 0$,

$$\frac{1}{P_N(s)} = \frac{1}{p_0} - \frac{p_1}{p_0^2} z + O(z^2). \tag{12.14}$$

Multiplication yields

$$\frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = \frac{a_{-1}}{p_0 z} + \frac{a_0}{p_0} - \frac{a_{-1} p_1}{p_0^2} + O(z). \tag{12.15}$$

Multiplying by h_N and adding B_N gives the asserted finite part. □

Definition 12.4. For a simple zero ρ, define the normalized finite-part observable

$$\mathcal{F}_N(\rho) = N^{-\rho}\,\mathrm{FP}_{s=\rho}\,\Phi_N(s). \tag{12.16}$$

13. Further analytic consequences of the finite Möbius interpolation

13.1. **Meromorphic structure and residues.** Status. Exact result proved in this manuscript – unconditional.

Proposition 13.1 (Sources of poles)**.** *The meromorphic function Φ_N can have poles only at zeros of ζ and at zeros of the exponential polynomial P_N. At a simple zero ρ of ζ satisfying $P_N(\rho) \neq 0$, the residue is*

$$\operatorname*{Res}_{s=\rho} \Phi_N(s) = \frac{h_N}{\zeta'(\rho) P_N(\rho)}. \tag{13.1}$$

At a simple zero s_0 of P_N that is not a zero of ζ and satisfies $\zeta(s_0)^{-1} \neq A_N(s_0)$, the residue is

$$\operatorname*{Res}_{s=s_0} \Phi_N(s) = h_N \frac{\zeta(s_0)^{-1} - A_N(s_0)}{P_N'(s_0)}. \tag{13.2}$$

Proof. Both statements follow directly from the quotient in (8.10). At a simple zeta zero, $\zeta(s)^{-1}$ has principal part $1/(\zeta'(\rho)(s-\rho))$. At a simple zero of P_N, the denominator has principal part $P_N'(s_0)(s - s_0)$. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 13.2 (Value at the pole of zeta)**.** *The reciprocal zeta function has a simple zero at $s = 1$, and therefore*

$$\Phi_N(1) = B_N - h_N \frac{A_N(1)}{P_N(1)}. \tag{13.3}$$

Proof. Use $\zeta(s)^{-1} \to 0$ as $s \to 1$ in (8.10). The denominator $P_N(1)$ is positive. □

13.2. **Eventually exact sign and quantitative rounding.** Status. Exact result proved in this manuscript – unconditional.

Corollary 13.3 (Eventually exact sign)**.** *For sufficiently large real σ,*

$$\operatorname{sgn}\big(\Phi_N(\sigma) - M(N)\big) = \mu(a_N). \tag{13.4}$$

Proof. The leading asymptotic has the form

$$\Phi_N(\sigma) - M(N) = A_N(\sigma)(1 + o(1)), \qquad A_N(\sigma) = h_N \mu(a_N) \left(\frac{q_N}{a_N}\right)^{\sigma}. \tag{13.5}$$

The magnitude multiplying $\mu(a_N)$ is strictly positive. Choose σ_0 so that the relative error has modulus less than one half for $\sigma \geq \sigma_0$. Then the full expression cannot change the sign of its leading term, and its sign is $\mu(a_N)$. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 13.4 (Existence of a finite rounding threshold)**.** *For every $N \geq 2$ there exists a finite $\sigma_N > 1$ such that*

$$\operatorname{nint} \Phi_N(\sigma) = M(N) \tag{13.6}$$

for all real $\sigma \geq \sigma_N$, where nint *denotes the nearest integer.*

Proof. Let

$$B_N(\sigma) = \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N}\right)^{\sigma}. \tag{13.7}$$

If $N \geq 3$, Bertrand's theorem gives $q_N < N$, so the exponential factor tends to zero and therefore $B_N(\sigma) \to 0$. The exceptional case $N = 2$ has $Q_2 = \{2\}$, $h_2 = 1$, and

$$B_2(\sigma) = \frac{2}{\sigma - 1} \longrightarrow 0. \tag{13.8}$$

Thus in every case one may choose σ_N so that $B_N(\sigma) < 1/2$ for all $\sigma \geq \sigma_N$. The rounding criterion then gives the conclusion. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 13.5 (A sufficient explicit threshold)**.** *Assume $N \geq 3$, and let*

$$\lambda_N = \log \frac{N}{q_N} > 0. \tag{13.9}$$

Every $\sigma > 1$ satisfying

$$\sigma \lambda_N > \log(2 h_N N) + \log \frac{1}{\sigma - 1} \tag{13.10}$$

is a valid rounding point. In particular, any $\sigma \geq 2$ satisfying

$$\sigma > \frac{\log(2 h_N N)}{\lambda_N} \tag{13.11}$$

is sufficient.

Proof. Rewrite the rounding condition as

$$\frac{h_N N}{\sigma-1}e^{-\sigma\lambda_N} < \frac{1}{2}. \tag{13.12}$$

All factors are positive. Taking logarithms gives

$$\log(h_N N) - \log(\sigma-1) - \sigma\lambda_N < -\log 2, \tag{13.13}$$

which rearranges to the implicit condition in the statement. When $\sigma \geq 2$, one has $\log(1/(\sigma-1)) \leq 0$. The stronger but simpler inequality $\sigma\lambda_N > \log(2h_N N)$ therefore implies the implicit condition and is sufficient. For $N = 2$, the direct estimate $2/(\sigma-1) < 1/2$ shows that every $\sigma > 5$ is sufficient. □

13.3. **Recursive recovery of the nonzero tail.**

Definition 13.6. Let

$$a_{N,1} < a_{N,2} < a_{N,3} < \cdots \tag{13.14}$$

be the increasing sequence of integers exceeding N for which $\mu(a_{N,j}) \neq 0$.

Status. Exact result proved in this manuscript – unconditional.

Theorem 13.7 (Recursive exponential extraction)**.** *The real-axis function $E_N(\sigma)$ determines the entire signed sequence*

$$\big(a_{N,j}, \mu(a_{N,j})\big)_{j\geq 1} \tag{13.15}$$

recursively. More precisely, once the first $J-1$ terms are known, define

$$E_N^{(J)}(\sigma) = E_N(\sigma) - \frac{\sum_{j<J}\mu(a_{N,j})a_{N,j}^{-\sigma}}{P_N(\sigma)}. \tag{13.16}$$

Then

$$a_{N,J} = q_N \lim_{\sigma\to\infty} |E_N^{(J)}(\sigma)|^{-1/\sigma}, \tag{13.17}$$

$$\mu(a_{N,J}) = \lim_{\sigma\to\infty} \operatorname{sgn} E_N^{(J)}(\sigma). \tag{13.18}$$

Proof. By definition,

$$E_N(\sigma) = \frac{1}{P_N(\sigma)}\sum_{j\geq 1}\mu(a_{N,j})a_{N,j}^{-\sigma}. \tag{13.19}$$

Subtracting the first $J-1$ known terms leaves

$$E_N^{(J)}(\sigma) = \frac{1}{P_N(\sigma)}\sum_{j\geq J}\mu(a_{N,j})a_{N,j}^{-\sigma}. \tag{13.20}$$

Factor the first remaining term and the least denominator term exactly as in Theorem 10.2. Dominated convergence gives

$$E_N^{(J)}(\sigma) = \mu(a_{N,J})\left(\frac{q_N}{a_{N,J}}\right)^{\sigma}(1+o(1)). \tag{13.21}$$

Taking the negative σ-th root recovers the ratio $a_{N,J}/q_N$, and eventual sign recovers $\mu(a_{N,J})$. Induction on J determines the entire signed support sequence uniquely. □

Remark 13.8. Theorem 13.7 is an exact uniqueness statement. It does not by itself yield a computational advantage, because evaluating E_N already requires the reciprocal zeta function with sufficient precision and control of exponentially small differences.

Part 3. Prime-power phase walks over zeta ordinates

14. Phase walks, endpoint conventions, and a universal half

Definition 14.1. Fix $q > 1$ and write

$$\omega = \log q. \tag{14.1}$$

For the distinct positive ordinates τ, with total multiplicity $m(\tau)$, define

$$S_q(T) = \sum_{0<\tau\le T} m(\tau) e^{i\omega\tau}. \tag{14.2}$$

Equivalently, if γ_k denotes the ordered list repeated with multiplicity, then

$$S_q(T) = \sum_{0<\gamma_k\le T} e^{i\omega\gamma_k}. \tag{14.3}$$

At an ordinate γ_k, the right, left, and midpoint values are

$$S_k^{\mathrm{right}} = \sum_{j\le k} b_j, \tag{14.4}$$

$$S_k^{\mathrm{left}} = \sum_{j<k} b_j, \tag{14.5}$$

$$S_k^{\mathrm{mid}} = \sum_{j<k} b_j + \frac{1}{2} b_k, \qquad b_k = e^{i\omega\gamma_k}. \tag{14.6}$$

Remark 14.2. The quantities $S_k^{\mathrm{right}}, S_k^{\mathrm{left}}$, and S_k^{mid} are indexwise: an ordinate of multiplicity m appears m times in the ordered list. The symmetric midpoint of the whole atom at a distinct ordinate τ instead contains $\frac{1}{2} m(\tau) e^{i\omega\tau}$.

Status. Exact result proved in this manuscript – unconditional.

Proposition 14.3 (Endpoint conversion)**.** *At every index k,*

$$\overline{b_k} S_k^{\mathrm{right}} = \overline{b_k} S_k^{\mathrm{mid}} + \frac{1}{2}, \tag{14.7}$$

$$\overline{b_k} S_k^{\mathrm{left}} = \overline{b_k} S_k^{\mathrm{mid}} - \frac{1}{2}. \tag{14.8}$$

Thus changing the endpoint convention translates the carrier coefficient by $\pm 1/2$ and does not change a centered variance.

Proof. The right sum differs from the indexwise midpoint sum by $\frac{1}{2} b_k$, while the left sum differs by $-\frac{1}{2} b_k$. Multiplication by $\overline{b}_k$ and $|b_k| = 1$ gives the two identities. Translating all sampled values by a fixed constant does not alter their centered variance. □

Status. Exact result proved in this manuscript – unconditional.

Theorem 14.4 (Exact triangular half identity)**.** *Let $|b_k| = 1$ and*

$$S_k = \sum_{j\le k} b_j. \tag{14.9}$$

Then

$$\boxed{\Re\left(\frac{1}{N}\sum_{k=1}^{N} \overline{b_k} S_k\right) = \frac{1}{2} + \frac{|S_N|^2}{2N}.} \tag{14.10}$$

Proof. Let $b = (b_1, \dots, b_N)^{\mathsf{T}}$, and let L be the lower triangular matrix with $L_{kj} = 1$ for $j \le k$. Then the vector of partial sums is Lb, and

$$\sum_{k=1}^{N} \overline{b_k} S_k = b^* L b. \tag{14.11}$$

Let J denote the all-ones matrix. Entrywise, $L + L^*$ equals $J + I$: off the diagonal exactly one of L_{jk}, L_{kj} equals one, while on the diagonal both do. Hence

$$2\Re(b^* L b) = b^*(L + L^*)b \tag{14.12}$$

$$= b^* J b + b^* b \tag{14.13}$$

$$= \left| \sum_{k=1}^{N} b_k \right|^2 + N, \tag{14.14}$$

because $|b_k| = 1$. Division by $2N$ proves the identity. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 14.5. *For midpoint evaluation,*

$$\Re\left(\frac{1}{N} \sum_{k=1}^{N} \overline{b_k} S_k^{\mathrm{mid}} \right) = \frac{|S_N|^2}{2N}. \tag{14.15}$$

Proof. By Proposition 14.3,

$$\overline{b}_k S_k^{\mathrm{mid}} = \overline{b}_k S_k^{\mathrm{right}} - \frac{1}{2}. \tag{14.16}$$

Average over k, take real parts, and apply Theorem 14.4. □

Warning 14.6. Theorem 14.4 is universal for unit phases. It does not identify prime-power structure and does not imply the Riemann Hypothesis. It applies exactly to the raw full-sample cumulative walk, not automatically to a residual after smooth and prime-power subtraction.

15. Complete-zero weights and the positive horizontal defect

Definition 15.1. For a positive ordinate τ, define

$$W_q(\tau) = \sum_{\rho : \Im\rho = \tau} q^{\beta - \frac{1}{2}}, \tag{15.1}$$

where zeros are counted with multiplicity.

Definition 15.2. Define

$$D_q(\tau) = \sum_{\substack{\rho = \beta + i\tau \\ \beta > 1/2}} \left[\cosh\left((\beta - \tfrac{1}{2}) \log q \right) - 1 \right]. \tag{15.2}$$

The use of functional-equation pairing and a nonnegative expression is closely related to the global positivity mechanisms in Weil's criterion and the generalized Li criterion of Bombieri–Lagarias [15, 8]. The formulation below is instead localized at each ordinate and then assembled as an atomic measure. The manuscript makes no exhaustive novelty claim for that localization.

Status. Exact local criterion proved here; unconditional. The positivity mechanism has classical global antecedents.

Theorem 15.3 (Positive defect at each ordinate)**.** *For every $q > 1$ and every positive ordinate τ,*

$$\boxed{W_q(\tau) = m(\tau) + 2D_q(\tau), \qquad D_q(\tau) \geq 0.} \tag{15.3}$$

Moreover,

$$D_q(\tau) = 0 \tag{15.4}$$

if and only if every zero at ordinate τ lies on the critical line.

Proof. The functional equation and conjugation symmetry imply that, at a fixed positive ordinate τ, every zero $\beta + i\tau$ off the critical line is accompanied with equal multiplicity by $1 - \beta + i\tau$; see [14, Chapter 2]. Put $\delta = \beta - 1/2 > 0$. The pair contributes to $W_q(\tau)$

$$q^{\beta-1/2} + q^{1/2-\beta} = q^{\delta} + q^{-\delta} \tag{15.5}$$

$$= 2\cosh(\delta \log q) \tag{15.6}$$

$$= 2 + 2\left[\cosh(\delta \log q) - 1\right]. \tag{15.7}$$

The first term 2 is exactly the contribution of two unit weights to the multiplicity $m(\tau)$; the second is the excess. A zero on the critical line has $\delta = 0$ and contributes one to both quantities. Summing all pairs proves $W_q = m + 2D_q$. Since $q > 1$, $\log q \neq 0$, and $\cosh x - 1 \geq 0$ with equality only at $x = 0$, the defect vanishes precisely when no positive δ occurs. □

Definition 15.4. Define the locally finite complex measures

$$\mu_q = \sum_{\tau>0} m(\tau) e^{i\tau \log q} \delta_\tau, \tag{15.8}$$

$$\nu_q = \sum_{\tau>0} W_q(\tau) e^{i\tau \log q} \delta_\tau. \tag{15.9}$$

Status. Exact result proved in this manuscript – unconditional.

Corollary 15.5 (Atomic positive criterion)**.** *The exact measure identity*

$$\boxed{\frac{1}{2} e^{-iT \log q}(\nu_q - \mu_q) = \sum_{\tau>0} D_q(\tau)\delta_\tau} \tag{15.10}$$

holds. For any one fixed $q > 1$,

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{15.11}$$

Proof. At the atom $T = \tau$, the coefficient of $\nu_q - \mu_q$ is

$$\left[W_q(\tau) - m(\tau)\right] e^{i\tau \log q} = 2D_q(\tau) e^{i\tau \log q}. \tag{15.12}$$

Multiplication by $\frac{1}{2} e^{-iT \log q}$ cancels the phase at each atom and gives the positive measure in (15.10). If $\nu_q = \mu_q$, this measure is zero, so every nonnegative mass $D_q(\tau)$ is zero. Theorem 15.3 then places every zero on the critical line. Conversely, under RH, every weight $q^{\beta-1/2}$ equals one, so $W_q(\tau) = m(\tau)$ and the two measures coincide. □

Remark 15.6. No value of $q > 1$ is logically privileged. For a small displacement $\delta = \beta - 1/2$,

$$\cosh(\delta \log q) - 1 = \frac{1}{2}\delta^2 (\log q)^2 + O\left(\delta^4 (\log q)^4\right). \tag{15.13}$$

Larger q amplifies small horizontal displacements but also enlarges the analytic weights.

16. The explicit-formula origin of the two prime-power branches

The classical Fourier-explicit-formula line begins with Guinand's prime-number transforms and Fourier reciprocities and reaches Weil's general formulation [5, 6, 7, 15]. For admissible test functions, the arithmetic side is supported by the von Mangoldt function $\Lambda(n)$, hence by prime powers. Landau's formula and Gonek's uniform refinements isolate the same prime-power resonances in exponential sums over zeros [12, 11, 10].

Definition 16.1. Let $n = p^r$ be a prime power, put

$$\ell_n = \log n, \tag{16.1}$$

and suppose $\ell_n \neq \omega$. Define

$$c_n^- = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega - \ell_n)}, \tag{16.2}$$

$$c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega + \ell_n)}. \tag{16.3}$$

The associated oscillators are

$$P_n^-(T) = c_n^- e^{i(\omega-\ell_n)T}, \tag{16.4}$$

$$P_n^+(T) = c_n^+ e^{i(\omega+\ell_n)T}. \tag{16.5}$$

Warning 16.2 (Resonant carrier). The coefficient model (16.2)–(16.3) is nonresonant. All infinite sums built from these coefficients, including D_∞ and $\Delta_\omega(Q)$, are understood under the standing assumption

$$e^\omega \neq p^r \qquad \text{for every prime power } p^r. \tag{16.6}$$

If $e^\omega = p^r$, the quotient $(1 - e^{-i(\ell_n-\omega)T})/(\ell_n - \omega)$ must be replaced by its limit iT; the corresponding branch is polynomial in T, and the nonresonant formulas below do not apply without a separate subtraction. The numerical case $\omega = 1$ is nonresonant.

Heuristic 16.3 (Sharp-window prime-spectral ansatz)**.** For a sharp counting window, formal integration of the explicit formula suggests a decomposition

$$S_q(T) = C_q + A_{\mathrm{smooth},q}(T) + \frac{1}{2}e^{i\omega T} + \sum_{n=p^r} \left(P_n^-(T) + P_n^+(T)\right), \tag{16.7}$$

where the infinite prime-power sum requires a specified truncation or regularization. This is a heuristic argument when used with the discontinuous indicator $\mathbf{1}_{\gamma\leq T}$.

Definition 16.4 (Smooth term used in the computation)**.** The MATLAB program uses

$$A_{\mathrm{smooth}}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega}, \tag{16.8}$$

up to a branch-dependent additive constant. The constant is treated by a joint constant–carrier fit.

Warning 16.5. Equation (16.8) is the antiderivative of the smooth Riemann–von Mangoldt density multiplied by $e^{i\omega T}$. It is not, by itself, a complete sharp-window explicit formula. Gamma-factor, pole, endpoint, branch, and regularization conventions must be retained when an exact global identity is claimed.

17. Exact algebra of each prime-power mode

Status. Exact result proved in this manuscript – unconditional.

Proposition 17.1 (Sum, difference, and ratio identities)**.** *For every nonresonant prime power n,*

$$c_n^- + c_n^+ = \frac{i\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}, \tag{17.1}$$

$$c_n^- - c_n^+ = \frac{i\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}, \tag{17.2}$$

$$\frac{c_n^-}{c_n^+} = \frac{\omega + \ell_n}{\omega - \ell_n}. \tag{17.3}$$

Proof. Starting from the definitions, combine denominators:

$$c_n^- + c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega - \ell_n} + \frac{1}{\omega + \ell_n}\right] \tag{17.4}$$

$$= \frac{i\Lambda(n)}{2\pi\sqrt{n}}\frac{2\omega}{\omega^2 - \ell_n^2}, \tag{17.5}$$

which is (17.1). Replacing the plus by a minus in the bracket gives $2\ell_n/(\omega^2 - \ell_n^2)$, proving (17.2). Finally the common factor cancels in the quotient:

$$\frac{c_n^-}{c_n^+} = \frac{\omega + \ell_n}{\omega - \ell_n}. \tag{17.6}$$

All identities are exact whenever $\ell_n \neq \omega$. □

Definition 17.2. Write

$$c_n^- = ia_n^-, \qquad c_n^+ = ia_n^+, \qquad a_n^\pm \in \mathbb{R}, \tag{17.7}$$

and define

$$u_n = a_n^- + a_n^+, \tag{17.8}$$

$$v_n = a_n^- - a_n^+, \tag{17.9}$$

$$d_n = |c_n^-|^2 - |c_n^+|^2. \tag{17.10}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 17.3 (Signed mode invariant)**.** *The quantities in Definition 17.2 satisfy*

$$u_n = -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2 - \omega^2)}, \tag{17.11}$$

$$v_n = -\frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\ell_n^2 - \omega^2)}, \tag{17.12}$$

$$\boxed{d_n = u_n v_n = \frac{\omega\Lambda(n)^2\ell_n}{\pi^2 n(\ell_n^2 - \omega^2)^2}.} \tag{17.13}$$

Proof. Because the denominators in $c_n^\pm$ are real, the coefficients are purely imaginary. Taking their real scalar factors and using the sum and difference identities gives

$$u_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega - \ell_n} + \frac{1}{\omega + \ell_n}\right] \tag{17.14}$$

$$= -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2 - \omega^2)}, \tag{17.15}$$

$$v_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}-\frac{1}{\omega+\ell_n}\right] \tag{17.16}$$

$$= -\frac{\ell_n\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}. \tag{17.17}$$

The algebraic factorization

$$(a_n^-)^2-(a_n^+)^2=(a_n^-+a_n^+)(a_n^--a_n^+)=u_nv_n \tag{17.18}$$

then gives the displayed formula for d_n. Its positivity follows from $\omega>0$, $\Lambda(n)^2\geq 0$, $\ell_n>0$, and the squared denominator. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 17.4 (Elliptic form and anisotropy)**.** *After demodulation by $e^{-i\omega T}$, the mode is*

$$g_n(T)=c_n^-e^{-i\ell_nT}+c_n^+e^{i\ell_nT}=\frac{\Lambda(n)}{\pi\sqrt{n}(\omega^2-\ell_n^2)}\left[\ell_n\sin(\ell_nT)+i\omega\cos(\ell_nT)\right]. \tag{17.19}$$

Its unsigned elliptic area is

$$\pi d_n. \tag{17.20}$$

For uniform phase, its anisotropy is

$$\mathcal{A}_n=\frac{|\ell_n^2-\omega^2|}{\ell_n^2+\omega^2}\longrightarrow 1 \qquad (n\to\infty). \tag{17.21}$$

Proof. Using $e^{ix}=\cos x+i\sin x$, expand $c_n^-e^{-i\ell_nT}+c_n^+e^{i\ell_nT}$ and collect its real and imaginary parts. This gives

$$g_n(T)=A_n\sin(\ell_nT)+iB_n\cos(\ell_nT), \tag{17.22}$$

with

$$A_n=\frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \qquad B_n=\frac{\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2-\ell_n^2)}. \tag{17.23}$$

This is an ellipse with semiaxes $|A_n|$ and $|B_n|$. Their product is $|d_n|=d_n$, so its unsigned area is πd_n. Under uniform phase, the coordinate variances are $A_n^2/2$ and $B_n^2/2$. The normalized difference of principal variances is therefore

$$\frac{|A_n^2-B_n^2|}{A_n^2+B_n^2}=\frac{|\ell_n^2-\omega^2|}{\ell_n^2+\omega^2}, \tag{17.24}$$

which tends to one as $\ell_n\to\infty$. □

Status. Exact coefficient-level result proved here; unconditional for $\omega=1$.

Theorem 17.5 (Unique analytic dominance of the $n=3$ mode)**.** *Assume $\omega=1$. Among all prime powers $n\geq 2$:*

(i) *$n=3$ uniquely minimizes the detuning $|1-\log n|$;*
(ii) *$n=3$ uniquely maximizes $|c_n^-|$;*
(iii) *$n=3$ uniquely maximizes the two-branch envelope*

$$\mathscr{E}_n^{\mathrm{env}}=|c_n^-|+|c_n^+|; \tag{17.25}$$

(iv) *$n=3$ uniquely maximizes the individual-mode Bohr energy*

$$\mathcal{B}_n=|c_n^-|^2+|c_n^+|^2; \tag{17.26}$$

(v) *its complete two-branch contribution satisfies, for every real T,*

$$|c_3^-| - |c_3^+| \le |P_3^-(T) + P_3^+(T)| \le |c_3^-| + |c_3^+|, \tag{17.27}$$

and

$$\lim_{X\to\infty} \frac{1}{X} \int_0^X |P_3^-(T) + P_3^+(T)|^2 \, \mathrm{d}T = \mathcal{B}_3. \tag{17.28}$$

Numerically,

$$0.97559755 \le |P_3^-(T) + P_3^+(T)| \le 1.07180346, \tag{17.29}$$

$$\frac{\mathscr{E}_3^{\mathrm{env}}}{\mathscr{E}_2^{\mathrm{env}}} \approx 3.569, \qquad \frac{\mathcal{B}_3}{\mathcal{B}_2} \approx 15.735. \tag{17.30}$$

Proof. Because prime powers are integers at least two and $2 < e < 3$, the nearest prime power to e in logarithmic distance is 3:

$$\log 3 - 1 < 1 - \log 2, \qquad \log n - 1 \ge \log 4 - 1 > \log 3 - 1 \quad (n \ge 4). \tag{17.31}$$

This proves (i).

For $n = p^r \ge 3$, $\Lambda(n) = \log p \le \log n$. Hence

$$|c_n^-| \le f(n) := \frac{\log n}{2\pi\sqrt{n}(\log n - 1)}. \tag{17.32}$$

Writing $L = \log x > 1$ and differentiating logarithmically gives

$$x\frac{f'(x)}{f(x)} = \frac{1}{L} - \frac{1}{2} - \frac{1}{L-1} = -\frac{1}{L(L-1)} - \frac{1}{2} < 0. \tag{17.33}$$

Thus f decreases strictly for $x > e$. Equality in (17.32) holds at the prime 3, so 3 dominates every prime power $n \ge 4$. Direct evaluation gives $|c_3^-| = 1.0237005\ldots > |c_2^-| = 0.2542146\ldots$, proving (ii).

For the total envelope and $n \ge 3$,

$$\mathscr{E}_n^{\mathrm{env}} \le \frac{\log n}{2\pi\sqrt{n}} \left(\frac{1}{\log n - 1} + \frac{1}{\log n + 1} \right) \tag{17.34}$$

$$= \frac{(\log n)^2}{\pi\sqrt{n}((\log n)^2 - 1)} =: g(n). \tag{17.35}$$

For $L > 1$,

$$\frac{\mathrm{d}}{\mathrm{d}L} \log\left(\frac{L^2 e^{-L/2}}{L^2 - 1} \right) = -\frac{2}{L(L^2-1)} - \frac{1}{2} < 0. \tag{17.36}$$

Hence g decreases, equality holds at $n = 3$, and $\mathscr{E}_3^{\mathrm{env}} = 1.0718034\ldots > \mathscr{E}_2^{\mathrm{env}} = 0.3002865\ldots$. This proves (iii).

For the Bohr energy and $n \ge 3$,

$$\mathcal{B}_n = \frac{\Lambda(n)^2}{4\pi^2 n} \left[\frac{1}{(\log n - 1)^2} + \frac{1}{(\log n + 1)^2} \right] \tag{17.37}$$

$$\le h(n) := \frac{(\log n)^2((\log n)^2 + 1)}{2\pi^2 n((\log n)^2 - 1)^2}. \tag{17.38}$$

With $L = \log x > 1$,

$$\frac{\mathrm{d}}{\mathrm{d}L} \log h(e^L) = -\frac{L^5 + 6L^2 - L + 2}{L(L-1)(L+1)(L^2+1)} < 0. \tag{17.39}$$

Equality in (17.38) holds at the prime 3, and direct evaluation gives $\mathcal{B}_3 = 1.0502766\ldots > \mathcal{B}_2 = 0.0667477\ldots$. This proves (iv).

Finally, the carrier factor has modulus one, so the reverse and ordinary triangle inequalities give (17.27). The two frequencies $1-\log 3$ and $1+\log 3$ are distinct; their cross term has zero Bohr mean, proving (17.28). The displayed numerical bounds and ratios are direct evaluations of the exact coefficients. □

Corollary 17.6 (What is analytically established about $\log 3$)**.** *At $\omega = 1$, the explicit coefficient system itself singles out* $\log 3$*: it gives the unique nearest prime-power resonance, the largest minus-branch amplitude, the largest two-branch pointwise envelope, and the largest individual-mode continuous mean-square. Thus a coefficient-level influence of* $\log 3$ *is proved independently of the radial histogram.*

Status. Exact geometric model theorem – unconditional; application to the full zeta background is conditional.

Theorem 17.7 (Two-edge radial law for a circular mode against a fixed background)**.** *Let $a, b > 0$, $a \neq b$, let Θ be uniform on $[0, 2\pi]$, and put*

$$Z(\Theta) = b + ae^{i\Theta}, \qquad R = |Z(\Theta)|. \tag{17.40}$$

Then R is supported on $(|a-b|, a+b)$ and has density

$$\boxed{f_{a,b}(r) = \frac{2r}{\pi\sqrt{((a+b)^2 - r^2)(r^2 - (a-b)^2)}}}. \tag{17.41}$$

The density has integrable square-root singularities at both endpoints. Thus a single large circular oscillator added to a nonzero fixed background produces an exact two-edge radial law.

Proof. The squared radius satisfies

$$R^2 = a^2 + b^2 + 2ab\cos\Theta. \tag{17.42}$$

The random variable $X = \cos\Theta$ has the arcsine density

$$f_X(x) = \frac{1}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1. \tag{17.43}$$

Solving (17.42) for X gives

$$x(r) = \frac{r^2 - a^2 - b^2}{2ab}, \qquad \frac{\mathrm{d}x}{\mathrm{d}r} = \frac{r}{ab}. \tag{17.44}$$

The change-of-variables formula therefore yields

$$f_R(r) = \frac{r}{\pi ab\sqrt{1 - x(r)^2}} \tag{17.45}$$

$$= \frac{2r}{\pi\sqrt{4a^2b^2 - (r^2 - a^2 - b^2)^2}} \tag{17.46}$$

$$= \frac{2r}{\pi\sqrt{((a+b)^2 - r^2)(r^2 - (a-b)^2)}}. \tag{17.47}$$

The factorization also gives the support and the endpoint asymptotics. □

Remark 17.8 (What this explains and what it does not). The demodulated $n = 3$ mode is an ellipse whose semiaxis ratio is $\log 3$, so it is close to, but not exactly, the circular model in Theorem 17.7. The theorem supplies a rigorous mechanism by which a large near-circular slow mode can create two radial edges. Applying it to the actual zeta walk would require quantitative control showing that the remaining background is sufficiently concentrated and sufficiently decorrelated over the $n = 3$ phase. That premise is not proved.

Warning 17.9 (What the theorem does not prove). Theorem 17.5 does not force the radial histogram of the full sum to be bimodal. Modality depends on the joint distribution and correlation of the remaining background with the $n = 3$ term. The observed bimodal-to-unimodal transition after subtracting $n = 3$ is therefore a finite numerical statement. A deterministic matched-amplitude non-arithmetic frequency scan is reported in the numerical chapter and implemented in Annex A. It supplies strong finite evidence for frequency specificity, but it remains a numerical control rather than an analytic theorem about the complete background.

18. The convergent signed tail

Definition 18.1. Define

$$D_\infty = \sum_{n=p^r} d_n, \tag{18.1}$$

$$\Delta_\omega(Q) = \sum_{\substack{n=p^r \\ n>Q}} d_n = D_\infty - \sum_{\substack{n=p^r \\ n\le Q}} d_n. \tag{18.2}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 18.2 (Asymptotic signed tail)**.** *For fixed $\omega > 0$ satisfying* (16.6)*,*

$$\boxed{\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q} \qquad (Q \to \infty).} \tag{18.3}$$

Proof. Separate primes from proper prime powers. For a prime p, $\Lambda(p) = \ell_p = \log p$, and the exact formula becomes

$$d_p = \frac{\omega}{\pi^2 p \log p}\left(1 - \frac{\omega^2}{(\log p)^2}\right)^{-2}. \tag{18.4}$$

For $p > e^{2\omega}$, expand $(1 - z)^{-2} = 1 + O(z)$ with $z = \omega^2/(\log p)^2$. Uniformly in that range,

$$d_p = \frac{\omega}{\pi^2 p \log p} + O_\omega\left(\frac{1}{p(\log p)^3}\right). \tag{18.5}$$

The error tail is $O_\omega((\log Q)^{-3})$ by the prime number theorem and partial summation.

For the main term, write the prime sum as a Stieltjes integral:

$$\sum_{p>Q} \frac{1}{p \log p} = \int_Q^\infty \frac{1}{x \log x}\, \mathrm{d}\pi(x). \tag{18.6}$$

The prime number theorem $\pi(x) = x/\log x + o(x/\log x)$, followed by partial summation, gives

$$\int_Q^\infty \frac{1}{x \log x}\, \mathrm{d}\pi(x) = \int_Q^\infty \frac{\mathrm{d}x}{x(\log x)^2} + o\left(\frac{1}{\log Q}\right) = \frac{1}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{18.7}$$

It remains to bound $n = p^r$ with $r \ge 2$. Outside a fixed finite set depending on ω, the denominator satisfies $(r^2(\log p)^2 - \omega^2)^2 \gg_\omega r^4(\log p)^4$. Hence

$$d_{p^r} \ll_\omega \frac{1}{r^3 p^r \log p}. \tag{18.8}$$

Summing over $p^r > Q$, $r \ge 2$, gives $O_\omega(Q^{-1/2})$, after separating the finitely many small primes. This is $o(1/\log Q)$. Combining the prime and proper-power parts proves the theorem. □

Remark 18.3. For $\omega = 1$,

$$\pi\Delta_1(Q) \sim \frac{1}{\pi \log Q}, \tag{18.9}$$

$$\frac{3}{4}\Delta_1(Q) \sim \frac{3}{4\pi^2 \log Q}. \tag{18.10}$$

These constants match two finite numerical quantities, but the matching relations themselves remain conjectural.

19. Finite zero-sampled carrier identities

Definition 19.1. Fix a finite multiset of positive ordinates

$$\Gamma_H = \{\gamma_k : 0 < \gamma_k \le H\}, \qquad N(H) = |\Gamma_H|, \tag{19.1}$$

and let Y be any complex-valued function on Γ_H. For a cutoff Q, define

$$R_Q(\gamma) = Y(\gamma) - \sum_{\substack{n=p^r \\ n\le Q}} \left(P_n^-(\gamma) + P_n^+(\gamma)\right). \tag{19.2}$$

The raw carrier coefficient and zero kernel are

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} R_Q(\gamma), \tag{19.3}$$

$$M_H(x) = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} x^{i\gamma}, \qquad x > 0. \tag{19.4}$$

Also put

$$\widehat{c}_{Y,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} Y(\gamma). \tag{19.5}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 19.2 (Exact finite-sample carrier formula)**.** *For the objects in Definition 19.1,*

$$\boxed{\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right).} \tag{19.6}$$

Proof. For each sampled ordinate, demodulation gives

$$e^{-i\omega\gamma_k} P_n^-(\gamma_k) = c_n^- e^{-i\ell_n\gamma_k} = c_n^- n^{-i\gamma_k}, \tag{19.7}$$

$$e^{-i\omega\gamma_k} P_n^+(\gamma_k) = c_n^+ e^{i\ell_n\gamma_k} = c_n^+ n^{i\gamma_k}. \tag{19.8}$$

Insert the definition of the residual into the sample average defining $\widehat{c}_{Q,H}$. Finite sums may be interchanged without any convergence issue. The average of $n^{i\gamma_k}$ is $M_H(n)$, and the average of $n^{-i\gamma_k}$ is its complex conjugate because $n > 0$. Collecting the terms yields the formula exactly. □

Status. Exact result proved in this manuscript – unconditional.

Theorem 19.3 (Exact imaginary-part formula)**.** *Write*

$$M_H(n) = x_n + iy_n. \tag{19.9}$$

Then

$$\Im\left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right) = u_n x_n, \tag{19.10}$$

and consequently

$$\boxed{\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n).} \tag{19.11}$$

Proof. Write $M_H(n) = x_n + iy_n$ and $c_n^{\pm} = ia_n^{\pm}$ with real $a_n^{\pm}$. Then

$$c_n^- \overline{M_H(n)} + c_n^+ M_H(n) = ia_n^-(x_n - iy_n) + ia_n^+(x_n + iy_n) \tag{19.12}$$

$$= (a_n^- - a_n^+)y_n + i(a_n^- + a_n^+)x_n \tag{19.13}$$

$$= v_n y_n + iu_n x_n. \tag{19.14}$$

Its imaginary part is therefore $u_n x_n = u_n \Re M_H(n)$. Taking imaginary parts in the exact carrier formula and summing proves the result. □

Definition 19.4. Define the Q-independent quantity

$$C_H = \Im\widehat{c}_{Y,H} - \pi D_\infty. \tag{19.15}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 19.5 (Exact global defect identity)**.** *For the finite-sample raw coefficient,*

$$\boxed{\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = C_H - \sum_{\substack{n=p^r \\ n\le Q}} u_n \left[\Re M_H(n) - \pi v_n\right].} \tag{19.16}$$

Proof. The exact carrier identity gives

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{n\le Q} u_n \Re M_H(n). \tag{19.17}$$

On the other hand, $d_n = u_n v_n$ and $\Delta_\omega(Q) = D_\infty - \sum_{n\le Q} d_n$, so

$$\pi\Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n\le Q} u_n v_n. \tag{19.18}$$

Subtract the second equation from the first and group the cutoff-independent terms:

$$\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = \left(\Im\widehat{c}_{Y,H} - \pi D_\infty\right) \tag{19.19}$$

$$- \sum_{n\le Q} u_n \left[\Re M_H(n) - \pi v_n\right]. \tag{19.20}$$

The parenthesis is exactly C_H. Every manipulation is finite except for D_∞, whose convergence follows from Theorem 18.2. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 19.6 (Equivalent raw transport target)**.** *Along any specified joint sequence $H = H(Q)$ with $Q \to \infty$, the conjectural raw relation*

$$\Im\widehat{c}_{Q,H} = \pi\Delta_\omega(Q) + o\left(\frac{1}{\log Q}\right) \tag{19.21}$$

is exactly equivalent to

$$\boxed{C_H - \sum_{n\le Q} u_n \left[\Re M_H(n) - \pi v_n\right] = o\left(\frac{1}{\log Q}\right).} \tag{19.22}$$

Proof. Subtract $\pi\Delta_\omega(Q)$ from $\Im\widehat{c}_{Q,H}$ and apply Theorem 19.5. The dependence $H = H(Q)$ must be specified because C_H and every kernel value $M_H(n)$ depend on the sample height. No limit interchange is involved: the equivalence holds pointwise along the chosen sequence. □

Status. Exact result proved in this manuscript – unconditional.

Theorem 19.7 (Exact increment identity)**.** *When Q crosses a prime power n,*

$$\Delta\Im\widehat{c}_{Q,H} = -u_n\Re M_H(n), \tag{19.23}$$

$$\Delta\left(\pi\Delta_\omega(Q)\right) = -\pi u_n v_n. \tag{19.24}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta\Im\widehat{c}_{Q,H}}{\Delta(\pi\Delta_\omega(Q))} = \frac{\Re M_H(n)}{\pi v_n}.} \tag{19.25}$$

Proof. Let Q_- and Q_+ be cutoffs immediately before and immediately after the prime power n, so that no other prime power lies between them. The exact raw coefficient formula (19.6) gives

$$\widehat{c}_{Q_+,H} - \widehat{c}_{Q_-,H} = -\left(c_n^-\overline{M_H(n)} + c_n^+ M_H(n)\right). \tag{19.26}$$

Taking imaginary parts and using (19.10) yields (19.23). By definition, $\Delta_\omega(Q)$ loses exactly the single summand $d_n = u_n v_n$ when the cutoff crosses n, which proves (19.24). Division of the two nonzero increments gives (19.25). □

20. Exact joint constant–carrier identities

The coefficient reported by the corrected numerical program is not the raw coefficient $\widehat{c}_{Q,H}$. It is the coefficient obtained by jointly projecting onto the constant vector and the carrier. The projection correction can be written exactly in the same zero kernels.

Definition 20.1 (Joint carrier data)**.** Put

$$q = e^\omega, \qquad \mu_H = M_H(q), \qquad \mathfrak{d}_H = 1 - |\mu_H|^2. \tag{20.1}$$

Assume $\mathfrak{d}_H > 0$. For a sampled function Z, define its joint carrier coefficient by

$$\mathfrak{c}_H[Z] = \frac{\langle e^{i\omega\gamma}, Z\rangle_H - \overline{\mu_H}\langle 1, Z\rangle_H}{\mathfrak{d}_H}. \tag{20.2}$$

Thus the coefficient reported at cutoff Q is

$$c_{Q,H} = \mathfrak{c}_H[R_Q], \qquad c_{Y,H} = \mathfrak{c}_H[Y]. \tag{20.3}$$

Definition 20.2 (Joint contribution of one prime power)**.** For a prime power n, define

$$A_H(n) = c_n^-\overline{M_H(n)} + c_n^+ M_H(n), \tag{20.4}$$

$$B_H(n) = c_n^- M_H(q/n) + c_n^+ M_H(qn), \tag{20.5}$$

$$K_H(n) = \frac{A_H(n) - \overline{\mu_H}B_H(n)}{\mathfrak{d}_H}. \tag{20.6}$$

Also define the real quantity

$$J_H(n) = \Im K_H(n). \tag{20.7}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 20.3 (Exact reported-carrier formula)**.** *For every finite cutoff Q,*

$$\boxed{c_{Q,H} = c_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} K_H(n).} \tag{20.8}$$

Moreover,

$$\boxed{J_H(n) = \frac{u_n \Re M_H(n) - \Re\left[\overline{\mu_H}\left(a_n^- M_H(q/n) + a_n^+ M_H(qn)\right)\right]}{\mathfrak{d}_H}.} \tag{20.9}$$

Consequently,

$$\boxed{\Im c_{Q,H} = \Im c_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} J_H(n).} \tag{20.10}$$

Proof. The joint carrier functional $\mathfrak{c}_H$ is complex linear in its argument. The residual decomposition

$$R_Q = Y - \sum_{\substack{n=p^r \\ n\le Q}} \left(P_n^- + P_n^+\right) \tag{20.11}$$

therefore gives

$$c_{Q,H} = c_{Y,H} - \sum_{n\le Q} \mathfrak{c}_H[P_n^- + P_n^+]. \tag{20.12}$$

For a single prime power,

$$\begin{aligned} \left\langle e^{i\omega\gamma}, P_n^- + P_n^+ \right\rangle_H &= c_n^- \overline{M_H(n)} + c_n^+ M_H(n) \\ &= A_H(n), \end{aligned} \tag{20.13}$$

whereas

$$\begin{aligned} \left\langle 1, P_n^- + P_n^+ \right\rangle_H &= c_n^- M_H(q/n) + c_n^+ M_H(qn) \\ &= B_H(n). \end{aligned} \tag{20.14}$$

Substitution in (20.2) proves $\mathfrak{c}_H[P_n^- + P_n^+] = K_H(n)$, and hence (20.8).

To compute the imaginary part, write $c_n^\pm = i a_n^\pm$. The first projection satisfies

$$\Im A_H(n) = u_n \Re M_H(n) \tag{20.15}$$

by Theorem 19.3. Furthermore,

$$B_H(n) = i\left[a_n^- M_H(q/n) + a_n^+ M_H(qn)\right], \tag{20.16}$$

and for any complex number Z, $\Im(iZ) = \Re Z$. Therefore

$$\Im(\overline{\mu_H} B_H(n)) = \Re\left[\overline{\mu_H}\left(a_n^- M_H(q/n) + a_n^+ M_H(qn)\right)\right]. \tag{20.17}$$

Since $\mathfrak{d}_H$ is real and positive, taking imaginary parts in (20.6) gives (20.9). Finally, taking imaginary parts in (20.8) proves (20.10). □

Definition 20.4 (Joint global constant)**.** Define

$$C_H^{\text{joint}} = \Im c_{Y,H} - \pi D_\infty. \tag{20.18}$$

Status. Exact result proved in this manuscript – unconditional.

Theorem 20.5 (Exact joint defect and increment identities)**.** *For every finite cutoff Q,*

$$\boxed{\Im c_{Q,H} - \pi\Delta_\omega(Q) = C_H^{\mathrm{joint}} - \sum_{\substack{n=p^r\\ n\le Q}} [J_H(n) - \pi u_n v_n]\,.} \tag{20.19}$$

When the cutoff crosses a single prime power n,

$$\Delta\Im c_{Q,H} = -J_H(n), \tag{20.20}$$

$$\Delta(\pi\Delta_\omega(Q)) = -\pi u_n v_n. \tag{20.21}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta\Im c_{Q,H}}{\Delta(\pi\Delta_\omega(Q))} = \frac{J_H(n)}{\pi u_n v_n}.} \tag{20.22}$$

Proof. Equation (20.10) gives

$$\Im c_{Q,H} = \Im c_{Y,H} - \sum_{n\le Q} J_H(n). \tag{20.23}$$

The signed tail identity remains

$$\pi\Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n\le Q} u_n v_n. \tag{20.24}$$

Subtracting these two equations proves (20.19). Crossing a single prime power adds exactly one term to each finite cutoff sum, yielding (20.20) and (20.21). Their quotient is (20.22). □

Status. Exact result proved in this manuscript – unconditional.

Corollary 20.6 (Correct transport target for the reported coefficient)**.** *Along any specified joint sequence $H = H(Q)$ with $Q \to \infty$, the assertion*

$$\Im c_{Q,H} = \pi\Delta_\omega(Q) + o\Big(\frac{1}{\log Q}\Big) \tag{20.25}$$

is equivalent to

$$\boxed{C_H^{\mathrm{joint}} - \sum_{n\le Q} [J_H(n) - \pi u_n v_n] = o\Big(\frac{1}{\log Q}\Big)\,.} \tag{20.26}$$

Proof. This is an immediate reformulation of (20.19). The specification of a joint sequence is necessary because C_H^{joint}, $J_H(n)$, and the sample kernel all depend on H. □

Warning 20.7 (Raw and reported increments are different). The raw ratio (19.25) is not the increment statistic of the coefficient printed by the corrected program. The reported statistic is (20.22). They coincide only when the exact constant-projection correction in (20.9) is negligible and $\mathfrak{d}_H = 1 + o(1)$.

21. Landau scaling and the conditional obstruction to termwise area transport

Landau's formula concerns sums of x^ρ, not sums of $x^{i\gamma}$ without horizontal weights. Passing from the former to the latter requires RH, or a separate control of the real parts. Uniformity when x grows with H requires a uniform Landau–Gonek formula [12, 11].

Heuristic 21.1 (Conditional Landau estimate for the ordinate kernel)**.** Assume RH and a uniform Landau–Gonek estimate in the range under consideration. With

$$L_H = \log(H/2\pi), \tag{21.1}$$

one expects at a prime power n

$$\Re M_H(n) \approx -\frac{\Lambda(n)}{\sqrt{n}\,L_H}. \tag{21.2}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 21.2 (Required termwise target)**.** *Individual transport of the signed area of mode n would require*

$$\Re M_H(n) \sim \pi v_n. \tag{21.3}$$

For a prime p,

$$\pi v_p = -\frac{(\log p)^2}{\sqrt{p}((\log p)^2 - \omega^2)} \sim -\frac{1}{\sqrt{p}}. \tag{21.4}$$

Proof. The increment identity (19.25) shows that exact mode-by-mode transport is equivalent to $\Re M_H(n)/(\pi v_n) \to 1$, which is (21.3). For a prime p, substitute $\Lambda(p) = \log p$ and $\ell_p = \log p$ into (17.12); division of numerator and denominator by $(\log p)^2$ gives (21.4). □

Heuristic 21.3 (Termwise ratio)**.** Combining (21.2) and (21.4) gives

$$\frac{\Re M_H(p)}{\pi v_p} \approx \frac{\log p}{L_H}. \tag{21.5}$$

If $p \asymp H^\theta$, this predicts

$$\frac{\Re M_H(p)}{\pi v_p} \approx \theta. \tag{21.6}$$

Thus termwise transport cannot be asymptotically exact at a fixed $\theta < 1$; it becomes plausible only for $p = H^{1-o(1)}$.

Warning 21.4. The fixed-p Landau formula alone does not prove (21.6). The latter is a falsifiable scaling prediction that requires uniformity in the joint range $p = p(H)$.

Conjecture 21.5 (Local transport near the height scale)**.** *Let*

$$Q_1, Q_2 = H^{1-o(1)}. \tag{21.7}$$

In a suitable uniform regime,

$$\Im\widehat{c}_{Q_2,H} - \Im\widehat{c}_{Q_1,H} \sim \pi\left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]. \tag{21.8}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 21.6 (Exact local defect)**.** *The defect in Conjecture 21.5 is exactly*

$$\begin{aligned}&[\Im\widehat{c}_{Q_2,H} - \Im\widehat{c}_{Q_1,H}] - \pi\left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]\\ &\quad = -\sum_{Q_1 < n \le Q_2} u_n\left[\Re M_H(n) - \pi v_n\right].\end{aligned} \tag{21.9}$$

Proof. Subtract (19.16) at Q_1 and Q_2. The constant C_H cancels. □

22. Mixed carrier area versus prime self-area

Status. Exact result proved in this manuscript – unconditional.

Proposition 22.1 (Mixed form measured by the centroid coefficient)**.** *The imaginary part of the raw carrier coefficient is*

$$\Im \widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} \Im \left[e^{-i\omega\gamma_k} R_Q(\gamma_k) \right] \tag{22.1}$$

$$= -\frac{1}{N(H)} \sum_{\gamma_k \le H} \Im \left[\overline{R_Q(\gamma_k)} e^{i\omega\gamma_k} \right]. \tag{22.2}$$

It is a mixed bilinear pairing between the residual position and the actual phase increment.

Proof. Take the imaginary part of the defining average (19.3). For the second equality, use

$$\Im(e^{-i\omega\gamma} R) = -\Im(\overline{R}\, e^{i\omega\gamma}), \tag{22.3}$$

which follows by complex conjugation. □

Definition 22.2. For a finite prime-power model G, define the formal self-area

$$\mathfrak{A}_H(G) = -\frac{\pi}{H} \int \Im \left(\overline{G(t)}\, \mathcal{J}G(t) \right) \mathrm{d}t, \tag{22.4}$$

where $\mathcal{J}$ is a specified frequency-sign or Hilbert-type operator.

Warning 22.3. The mixed pairing (22.2) is not identical to the self-pairing (22.4). Such an identification would require a justified replacement

$$e^{i\omega\gamma_k} \longleftrightarrow \mathcal{J}G_Q(\gamma_k). \tag{22.5}$$

The exact termwise defect of this replacement is encoded by $\Re M_H(n) - \pi v_n$.

23. Exact quadratic form for the sampled residual variance

The numerical variance is the squared norm of the *full residual* after jointly removing the constant and carrier. It is therefore not the projected norm of the prime-power model alone. The background vector, the mixed background–prime term, and the projected prime Gram form all enter exactly.

Definition 23.1 (Sampled inner product)**.** For sampled vectors on Γ_H, define

$$\langle \mathbf{f}, \mathbf{g} \rangle_H = \frac{1}{N(H)} \mathbf{f}^* \mathbf{g} = \frac{1}{N(H)} \sum_{\gamma_k \le H} \overline{f(\gamma_k)} g(\gamma_k). \tag{23.1}$$

Definition 23.2 (Sampled vectors and prime-mode matrix)**.** Let

$$\mathbf{1}_H = (1)_{\gamma_k \in \Gamma_H}, \tag{23.2}$$

$$\mathbf{a}_H = (e^{i\omega\gamma_k})_{\gamma_k \in \Gamma_H}, \tag{23.3}$$

$$\mathbf{y}_H = (Y(\gamma_k))_{\gamma_k \in \Gamma_H}. \tag{23.4}$$

For a prime power n and $\sigma \in \{-1, +1\}$, define the original, non-demodulated branch vector

$$\mathbf{w}_{n,\sigma} = \left(e^{i(\omega + \sigma \log n)\gamma_k} \right)_{\gamma_k \in \Gamma_H}. \tag{23.5}$$

For a cutoff Q, let

$$\mathcal{I}_Q = \{(n, \sigma) : n = p^r \le Q,\ \sigma \in \{-1, +1\}\}, \tag{23.6}$$

let V_Q be the matrix whose columns are the vectors $\mathbf{w}_{n,\sigma}$ in any fixed order, and let $\mathbf{c}_Q$ be the coefficient vector with entries c_n^σ. Then

$$\mathbf{p}_Q = V_Q \mathbf{c}_Q \tag{23.7}$$

is the sampled prime-power model, and

$$\mathbf{r}_Q = \mathbf{y}_H - V_Q \mathbf{c}_Q \tag{23.8}$$

is the sampled residual used by the joint fit.

Definition 23.3 (Joint projection and numerical variance)**.** Put

$$B_H = \begin{pmatrix} \mathbf{1}_H & \mathbf{a}_H \end{pmatrix}. \tag{23.9}$$

Assume $|M_H(e^\omega)| < 1$, equivalently that the two columns of B_H are linearly independent, and define

$$\Pi_H = I - B_H (B_H^* B_H)^{-1} B_H^*. \tag{23.10}$$

The variance reported after the joint constant–carrier fit is

$$\boxed{V(Q,H) = \frac{1}{N(H)} \mathbf{r}_Q^* \Pi_H \mathbf{r}_Q.} \tag{23.11}$$

Remark 23.4. If $|M_H(e^\omega)| = 1$, the same definition is made with the orthogonal projection onto the complement of $\operatorname{span}\{\mathbf{1}_H, \mathbf{a}_H\}$, or equivalently with the Moore–Penrose inverse. The data used in the numerical chapter are in the nondegenerate case.

Status. Exact result proved in this manuscript – unconditional.

Proposition 23.5 (Exact unprojected branch Gram kernel)**.** *For $j = (n, \sigma)$ and $k = (m, \tau)$ in $\mathcal{I}_Q$,*

$$K_{Q,H}^{(0)}(j,k) := \frac{1}{N(H)} \mathbf{w}_{n,\sigma}^* \mathbf{w}_{m,\tau} = M_H(m^\tau n^{-\sigma}). \tag{23.12}$$

Thus the four branch interactions between p^r and q^s involve

$$M_H(q^s/p^r), \quad M_H(q^s p^r), \quad M_H((q^s p^r)^{-1}), \quad M_H(p^r/q^s). \tag{23.13}$$

Proof. The common carrier cancels in the inner product. More explicitly,

$$\frac{1}{N(H)} \mathbf{w}_{n,\sigma}^* \mathbf{w}_{m,\tau} = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{-i(\omega + \sigma \log n)\gamma_k} e^{i(\omega + \tau \log m)\gamma_k} \tag{23.14}$$

$$= \frac{1}{N(H)} \sum_{\gamma_k \le H} (m^\tau n^{-\sigma})^{i\gamma_k} \tag{23.15}$$

$$= M_H(m^\tau n^{-\sigma}). \tag{23.16}$$

The four sign choices give (23.13). □

Status. Exact result proved in this manuscript – unconditional.

Proposition 23.6 (Exact projected branch Gram kernel)**.** *Put $q = e^\omega$ and*

$$G_H = \frac{1}{N(H)} B_H^* B_H = \begin{pmatrix} 1 & M_H(q) \\ M_H(q) & 1 \end{pmatrix}. \tag{23.17}$$

For $j = (n, \sigma)$ define

$$\mathbf{b}_j = \frac{1}{N(H)} B_H^* \mathbf{w}_{n,\sigma} = \begin{pmatrix} M_H(qn^\sigma) \\ M_H(n^\sigma) \end{pmatrix}. \tag{23.18}$$

Then

$$\boxed{\widetilde{K}_{Q,H}(j,k) := \frac{1}{N(H)} \mathbf{w}_j^* \Pi_H \mathbf{w}_k = K_{Q,H}^{(0)}(j,k) - \mathbf{b}_j^* G_H^{-1} \mathbf{b}_k.} \tag{23.19}$$

Proof. From (23.10),

$$\frac{1}{N(H)}V_Q^*\Pi_H V_Q = \frac{1}{N(H)}V_Q^*V_Q - \frac{1}{N(H)}V_Q^*B_H(B_H^*B_H)^{-1}B_H^*V_Q. \tag{23.20}$$

Since $B_H^*B_H = N(H)G_H$, the (j,k) entry of the second term is $\mathbf{b}_j^*G_H^{-1}\mathbf{b}_k$. The two components of $\mathbf{b}_j$ follow from

$$\frac{1}{N(H)}\mathbf{1}_H^*\mathbf{w}_{n,\sigma} = M_H(qn^\sigma), \tag{23.21}$$

$$\frac{1}{N(H)}\mathbf{a}_H^*\mathbf{w}_{n,\sigma} = M_H(n^\sigma). \tag{23.22}$$

□

Definition 23.7 (Projected background and mixed vector)**.** Define

$$V_Y(H) = \frac{1}{N(H)}\mathbf{y}_H^*\Pi_H\mathbf{y}_H, \tag{23.23}$$

$$\mathbf{d}_{Q,H} = \frac{1}{N(H)}V_Q^*\Pi_H\mathbf{y}_H, \tag{23.24}$$

$$\widetilde{K}_{Q,H} = \frac{1}{N(H)}V_Q^*\Pi_H V_Q. \tag{23.25}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 23.8 (Entrywise formula for the mixed vector)**.** *Let*

$$\mathbf{s}_{Y,H} = \frac{1}{N(H)}B_H^*\mathbf{y}_H. \tag{23.26}$$

For $j = (n,\sigma)$,

$$\boxed{d_{Q,H}(j) = \frac{1}{N(H)}\mathbf{w}_j^*\mathbf{y}_H - \mathbf{b}_j^*G_H^{-1}\mathbf{s}_{Y,H}.} \tag{23.27}$$

Proof. Insert (23.10) in (23.24). The normalization $B_H^*B_H = N(H)G_H$ gives the second term in (23.27). □

Status. Exact result proved in this manuscript – unconditional.

Theorem 23.9 (Exact decomposition of the numerical residual variance)**.** *For every finite cutoff Q,*

$$\boxed{V(Q,H) = V_Y(H) - 2\Re\left(\mathbf{c}_Q^*\mathbf{d}_{Q,H}\right) + \mathbf{c}_Q^*\widetilde{K}_{Q,H}\mathbf{c}_Q.} \tag{23.28}$$

Grouping the coefficient vector, mixed vector, and Gram matrix by prime base gives the exact identity

$$V(Q,H) = V_Y(H) - 2\Re\sum_p \mathbf{c}_p^*\mathbf{d}_{p,H} \tag{23.29}$$

$$+ \sum_p \mathbf{c}_p^*\widetilde{K}_{p,p;H}\mathbf{c}_p + \sum_{p\neq r}\mathbf{c}_p^*\widetilde{K}_{p,r;H}\mathbf{c}_r. \tag{23.30}$$

Proof. By (23.8) and (23.11),

$$\begin{aligned} V(Q,H) &= \frac{1}{N(H)}(\mathbf{y}_H - V_Q\mathbf{c}_Q)^*\Pi_H(\mathbf{y}_H - V_Q\mathbf{c}_Q) \\ &= \frac{1}{N(H)}\mathbf{y}_H^*\Pi_H\mathbf{y}_H - \frac{1}{N(H)}\mathbf{y}_H^*\Pi_H V_Q\mathbf{c}_Q \end{aligned} \tag{23.31}$$

$$-\frac{1}{N(H)}\mathbf{c}_Q^*V_Q^*\Pi_H\mathbf{y}_H+\frac{1}{N(H)}\mathbf{c}_Q^*V_Q^*\Pi_HV_Q\mathbf{c}_Q. \tag{23.32}$$

The projection is Hermitian, so the two middle terms are complex conjugates. Definitions 23.7 therefore turn (23.32) into (23.28). Partitioning the finite index set by prime base yields (23.29)–(23.30). □

Status. Exact result proved in this manuscript – unconditional.

Corollary 23.10 (Prime model alone)**.** *The identity*

$$\frac{1}{N(H)}(V_Q\mathbf{c}_Q)^*\Pi_H(V_Q\mathbf{c}_Q)=\mathbf{c}_Q^*\widetilde{K}_{Q,H}\mathbf{c}_Q \tag{23.33}$$

holds for the projected prime model alone. It equals the numerical residual variance only when

$$V_Y(H)-2\Re(\mathbf{c}_Q^*\mathbf{d}_{Q,H})=0. \tag{23.34}$$

Proof. Equation (23.33) follows directly from (23.25). Comparison with (23.28) gives the necessary and sufficient condition (23.34). □

Warning 23.11 (What the diagonal question actually contains)*.* The scalar Parseval sum

$$\sum_{p,r}\left(|c_{p^r}^-|^2+|c_{p^r}^+|^2\right) \tag{23.35}$$

is neither the full numerical variance nor the diagonal of the complete residual problem. Before any asymptotic assertion about a "diagonal" is made, one must control all four exact contributions in (23.29)– (23.30): the projected background energy, the background–prime mixed term, the within-tower prime blocks, and the cross-tower prime blocks. The fact that the arguments inside a fixed prime tower remain powers of that prime is exact; a Landau–Gonek interpretation of the ordinate-only kernel is an additional analytic step.

Question 23.12 (Exact variance components)**.** Determine, uniformly for a growing cutoff, the separate sizes of

$$V_Y(H),\qquad -2\Re\sum_p\mathbf{c}_p^*\mathbf{d}_{p,H},\qquad \sum_p\mathbf{c}_p^*\widetilde{K}_{p,p;H}\mathbf{c}_p,\qquad \sum_{p\neq r}\mathbf{c}_p^*\widetilde{K}_{p,r;H}\mathbf{c}_r. \tag{23.36}$$

Only their complete sum is the variance reported by the program.

24. Finite logarithmic shells in Bohr mean

Definition 24.1. For fixed $a>0$, define the finite multiplicative shell

$$\mathcal{S}_a(Q)=\{n=p^r : Q<n\le e^aQ\} \tag{24.1}$$

and its demodulated sum

$$G_{Q,a}(T)=\sum_{n\in\mathcal{S}_a(Q)}g_n(T). \tag{24.2}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 24.2 (Bohr energy of a fixed logarithmic shell)**.** *For fixed $a>0$,*

$$\lim_{X\to\infty}\frac{1}{X}\int_0^X|G_{Q,a}(T)|^2\,\mathrm{d}T=\frac{a}{2\pi^2\log Q}+o\left(\frac{1}{\log Q}\right) \tag{24.3}$$

as $Q\to\infty$.

Proof. Distinct frequencies $\pm \log n$ are orthogonal in Bohr mean. For primes,

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{24.4}$$

The prime number theorem and partial summation give [2, 3]

$$\sum_{Q<p\le e^a Q} \frac{1}{p} = \frac{a}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{24.5}$$

Proper prime powers contribute a lower-order term. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 24.3. *The choice $a = 3/2$ gives*

$$\frac{3}{4\pi^2 \log Q} + o\left(\frac{1}{\log Q}\right). \tag{24.6}$$

Proof. Substitute $a = 3/2$ in Proposition 24.2. □

Warning 24.4. Corollary 24.3 proves only the Bohr energy of a deliberately selected hard shell. It does not prove that the actual residual is supported in $(Q, e^{3/2}Q]$. Many transfer profiles have the same quadratic mass, and the projected zero-sampled variance is not the Bohr norm in Proposition 24.2.

25. Regularization and the remaining explicit-formula bridge

Definition 25.1. A regularized counting window is a family $H_{T,\eta}(u)$ tending to the midpoint indicator of $[0, T]$ as $\eta \to \infty$. A formal Gaussian example is

$$H_{T,\eta}(u) = \int_0^T \frac{\eta}{\sqrt{\pi}} e^{-\eta^2(u-v)^2}\, \mathrm{d}v. \tag{25.1}$$

Heuristic 25.2 (Fourier displacement)**.** For

$$h_{T,\eta,\omega}(u) = e^{i\omega u} H_{T,\eta}(u), \tag{25.2}$$

formal Fourier transformation gives

$$\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)}. \tag{25.3}$$

Insertion into the explicit formula generates the two branches in Definition 16.1, multiplied by explicit regularizing weights.

Warning 25.3. To turn Section 25 into a theorem one must specify a test class for which the explicit formula is valid, retain every archimedean and boundary term, and control jointly:

$$H, \qquad Q, \qquad \eta. \tag{25.4}$$

A formal Gaussian manipulation is not a proof of a sharp-window formula.

Conjecture 25.4 (Uniform explicit formula for the ordinate walk)**.** *There exist choices $Q = Q(H) \to \infty$ and $\eta = \eta(H) \to \infty$ for which the regularized prime-power formula, including its complete archimedean and endpoint terms, converges locally to the midpoint ordinate walk, with an error tending to zero in a topology strong enough to identify the atomic measure* (15.10).

Remark 25.5. The ordinary explicit formula reconstructs complete-zero weights, which retain $q^{\beta-1/2}$. Replacing that object by the unweighted ordinate walk is precisely where the positive defect $D_q(\tau)$ enters. Truncation control alone does not remove the horizontal zero information.

26. Energy exhaustion and distributional convergence

Conjecture 26.1 (Energy exhaustion)**.** *For a suitable joint regime $Q = Q(H) \to \infty$,*

$$\mathrm{Var}_{\gamma \le H}(F_{Q,H}) \le \varepsilon(Q,H), \qquad \varepsilon(Q(H),H) \to 0. \tag{26.1}$$

The variance is the full projected residual form (23.28)*, not the prime Gram term alone and not an unprojected Lebesgue L^2-norm.*

Status. Exact result proved in this manuscript – unconditional.

Theorem 26.2 (Abstract local distributional implication)**.** *Suppose a sequence F_j satisfies*

$$F_j - \frac{1}{2} \longrightarrow 0 \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}), \tag{26.2}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right)\right] \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{26.3}$$

Then RH *holds.*

Proof. Define

$$G_j(T) = e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right). \tag{26.4}$$

Multiplication by the bounded smooth phase preserves local L^2 convergence, so $G_j \to 0$ in L^2_{loc}. On every compact set, Cauchy–Schwarz then gives $G_j \to 0$ in L^1. For $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle G_j', \varphi\rangle = -\int_{\mathbb{R}} G_j(T)\varphi'(T)\,\mathrm{d}T \longrightarrow 0. \tag{26.5}$$

The assumed identification of the derivatives therefore implies $\nu_q - \mu_q = 0$ in $\mathcal{D}'$. A locally finite measure that is zero as a distribution is the zero measure. Multiplying by $\frac{1}{2}e^{-iT\log q}$ and applying Corollary 15.5 gives $D_q(\tau) = 0$ at every ordinate, hence RH. □

Warning 26.3. A normalized mean-square limit on intervals $[0,H]$ does not imply Theorem 26.2. A fixed compact defect, or even a sparse family of atomic defects, is diluted by division by H. Local or distributional convergence is indispensable.

27. Additional structure of the positive horizontal defect

27.1. Even-moment expansion.

Status. Exact result proved in this manuscript – unconditional.

Proposition 27.1 (Power-series expansion in $\log q$)**.** *At a fixed ordinate τ, put*

$$\alpha_\rho = \beta - \frac{1}{2} \tag{27.1}$$

for zeros with $\beta > 1/2$. Then

$$D_q(\tau) = \sum_{m=1}^{\infty} \frac{(\log q)^{2m}}{(2m)!} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \alpha_\rho^{2m}. \tag{27.2}$$

Proof. At a fixed ordinate there are finitely many zeros, counted with finite multiplicity. Put $y = \log q$ and $\alpha_\rho = \beta - 1/2 > 0$. The entire expansion

$$\cosh(\alpha_\rho y) - 1 = \sum_{m=1}^{\infty} \frac{\alpha_\rho^{2m} y^{2m}}{(2m)!} \tag{27.3}$$

converges absolutely for every complex y. Because the outer zero sum is finite at the fixed ordinate, it may be interchanged with the power series. Summing the coefficients gives the asserted even-moment expansion, with no uniform-convergence subtlety. □

STATUS. Exact result proved in this manuscript – unconditional.

Corollary 27.2 (Quadratic lower bound)**.** *For every $q > 1$,*

$$D_q(\tau) \geq \frac{(\log q)^2}{2} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{27.4}$$

Proof. The Taylor series

$$\cosh x - 1 = \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \tag{27.5}$$

has nonnegative terms for real x. Dropping every term except the first gives $\cosh x - 1 \geq x^2/2$. Apply this with $x = (\beta - 1/2)\log q$ to every off-line pair at the fixed ordinate. Since all summands remain nonnegative, summing preserves the inequality and produces the stated quadratic moment. □

STATUS. Exact result proved in this manuscript – unconditional.

Proposition 27.3 (Monotonicity and convexity)**.** *For fixed τ, the function*

$$t \longmapsto D_{e^t}(\tau), \qquad t \geq 0, \tag{27.6}$$

is nonnegative, increasing, and convex. It is identically zero if and only if every zero at ordinate τ lies on the critical line.

Proof. Set $t = \log q \geq 0$. For a horizontal displacement $\delta > 0$, define $f_\delta(t) = \cosh(\delta t) - 1$. Then

$$f_\delta'(t) = \delta \sinh(\delta t) \geq 0, \qquad f_\delta''(t) = \delta^2 \cosh(\delta t) > 0. \tag{27.7}$$

Thus every summand is increasing and strictly convex. Their finite sum has the same properties whenever at least one summand is present. If no off-line pair is present, the sum is identically zero. Conversely, an identically zero sum of nonnegative summands cannot contain a positive summand, so no off-line pair exists. □

STATUS. Exact result proved in this manuscript – unconditional.

Corollary 27.4 (One parameter and many parameters)**.** *One fixed $q > 1$ suffices for the logical criterion* (15.11)*. A continuum of values of q additionally encodes all even power sums of the horizontal displacements at each ordinate through* (27.2)*.*

Proof. The fixed-parameter assertion is Corollary 15.5. For the family assertion, Proposition 27.1 writes the analytic function of $y = \log q$ as

$$D_{e^y}(\tau) = \sum_{m \geq 1} \frac{y^{2m}}{(2m)!} P_{2m}(\tau), \tag{27.8}$$

where $P_{2m}(\tau)$ is the $2m$-th power sum of the positive horizontal displacements. Differentiating $2m$ times at zero gives $P_{2m}(\tau)$. Thus a neighborhood of one in the q-parameter encodes all even moments, whereas positivity makes a single nontrivial parameter enough merely to test whether they all vanish. □

27.2. **Jump formulation.**

Definition 27.5. Define the right-continuous cumulative functions

$$U_q(T) = \sum_{0<\tau\le T} m(\tau)e^{i\tau\log q}, \tag{27.9}$$

$$V_q(T) = \sum_{0<\tau\le T} W_q(\tau)e^{i\tau\log q}. \tag{27.10}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 27.6 (Exact jump defect)**.** *At every ordinate τ,*

$$\boxed{\frac{1}{2}e^{-i\tau\log q}\Delta(V_q-U_q)(\tau) = D_q(\tau).} \tag{27.11}$$

Proof. The jump of the complete-zero walk at τ is

$$\Delta V_q(\tau) = W_q(\tau)e^{i\tau\log q}, \tag{27.12}$$

whereas the jump of the ordinate-only walk is

$$\Delta U_q(\tau) = m(\tau)e^{i\tau\log q}. \tag{27.13}$$

Subtracting and using $W_q - m = 2D_q$ gives

$$\Delta(V_q-U_q)(\tau) = 2D_q(\tau)e^{i\tau\log q}. \tag{27.14}$$

Demodulation by $\frac{1}{2}e^{-i\tau\log q}$ leaves the nonnegative scalar $D_q(\tau)$. □

28. Regularized explicit formula with explicit prime-power weights

28.1. **The explicit formula in the Fourier normalization used here.** Status. Classical result – unconditional.

Theorem 28.1 (Guinand–Weil explicit formula)**.** *Let h be holomorphic in a strip $|\Im z| \le 1/2+\epsilon$, real-valued on $\mathbb{R}$, and suppose that, uniformly in the strip, $h(z) \ll (1+|\Re z|)^{-1-\delta}$ for some $\delta>0$. Define*

$$\widehat{h}(v) = \int_{\mathbb{R}} h(t)e^{-ivt}\,\mathrm{d}t. \tag{28.1}$$

Then, with zeros counted with multiplicity,

$$\sum_{\rho} h\Big(\frac{\rho-1/2}{i}\Big) = h\Big(\frac{1}{2i}\Big) + h\Big(-\frac{1}{2i}\Big) - \frac{\widehat{h}(0)}{2\pi}\log\pi \tag{28.2}$$

$$+ \frac{1}{2\pi}\int_{-\infty}^{\infty} h(u)\Re\frac{\Gamma'}{\Gamma}\Big(\frac{1}{4}+\frac{iu}{2}\Big)\,\mathrm{d}u \tag{28.3}$$

$$- \frac{1}{2\pi}\sum_{n=2}^{\infty}\frac{\Lambda(n)}{\sqrt{n}}\Big[\widehat{h}(\log n)+\widehat{h}(-\log n)\Big]. \tag{28.4}$$

The formula is understood by the usual limiting procedure whenever the chosen test lies at the boundary of the stated class [7, 15].

Proof. Consider the completed function $\xi(s)$ from (4.7) and the contour integral

$$\frac{1}{2\pi i}\int_{\partial\mathcal{R}} \frac{\xi'}{\xi}(s)h\Big(\frac{s-1/2}{i}\Big)\,\mathrm{d}s \tag{28.5}$$

over a rectangle whose vertical sides are $\Re s = 1/2 \pm a$, with $a > 1/2$, and whose horizontal sides tend to infinite height through values avoiding zeros. The decay assumptions on h make the horizontal

integrals vanish. By the residue theorem, the enclosed zeros of ξ contribute the spectral sum on the left, while the factors $s(s-1)$ produce the two evaluations at $\pm 1/(2i)$.

On the right vertical side, decompose the logarithmic derivative as

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}\left(\frac{s}{2}\right) + \frac{\zeta'}{\zeta}(s). \tag{28.6}$$

The functional equation $\xi(s) = \xi(1-s)$ transforms the left vertical side into the conjugate contribution of the right side. After the two sides are combined and parametrized by $s = 1/2 + iu$, the constant $-\frac{1}{2}\log\pi$ gives $-\widehat{h}(0)\log\pi/(2\pi)$, and the gamma logarithmic derivative gives the integral in (28.3).

For $\Re s > 1$, the absolutely convergent Dirichlet series

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s} \tag{28.7}$$

permits termwise integration. The contribution of n^{-s} on the two paired vertical sides is

$$\frac{\Lambda(n)}{\sqrt{n}}\left[\widehat{h}(\log n) + \widehat{h}(-\log n)\right], \tag{28.8}$$

with the prefactor $-1/(2\pi)$ arising from the contour normalization and the parametrization $\mathrm{d}s = i\,\mathrm{d}u$. Summing the contributions gives (28.4). Absolute convergence in the initial test class justifies all interchanges; the standard limiting argument extends the identity to the stated boundary cases. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 28.2 (Complex-linear extension)**.** *The formula in Theorem 28.1 remains valid for complex-valued h satisfying the same analytic and decay conditions.*

Proof. Define

$$h_1(z) = \frac{1}{2}\left(h(z) + \overline{h(\overline{z})}\right), \tag{28.9}$$

$$h_2(z) = \frac{1}{2i}\left(h(z) - \overline{h(\overline{z})}\right). \tag{28.10}$$

Both functions are holomorphic in the strip, real-valued on the real axis, satisfy the same decay, and obey $h = h_1 + ih_2$. Apply the real formula separately to h_1 and h_2, then combine by complex linearity. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 28.3 (Why only prime powers occur)**.** *The arithmetic side of the formula is supported exactly on prime powers, because*

$$\Lambda(n) \neq 0 \quad \Longleftrightarrow \quad n = p^r \tag{28.11}$$

for a prime p and integer $r \geq 1$.

Proof. This is the defining support property of the von Mangoldt function: $\Lambda(p^r) = \log p$ and $\Lambda(n) = 0$ otherwise. Applying it to the prime sum (28.4) removes every non-prime-power integer. □

28.2. **Gaussian convolution of a finite interval.**

Definition 28.4. For $\eta > 0$, define

$$g_\eta(t) = \frac{\eta}{\sqrt{\pi}} e^{-\eta^2 t^2}, \tag{28.12}$$

and

$$H_{T,\eta}(z) = \int_0^T g_\eta(z-u)\,\mathrm{d}u. \tag{28.13}$$

The oscillatory test is

$$h_{T,\eta,\omega}(z) = e^{i\omega z} H_{T,\eta}(z). \tag{28.14}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 28.5 (Fourier transform)**.** *With the convention*

$$\widehat{f}(v) = \int_{\mathbb{R}} f(t) e^{-ivt}\,\mathrm{d}t, \tag{28.15}$$

one has

$$\boxed{\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1-e^{-i(v-\omega)T}}{i(v-\omega)},} \tag{28.16}$$

where the quotient is interpreted continuously as T at $v=\omega$.

Proof. With the Fourier convention $\widehat{f}(v) = \int_{\mathbb{R}} f(t)e^{-ivt}\,\mathrm{d}t$, the normalized Gaussian has transform

$$\widehat{g}_\eta(v) = e^{-v^2/(4\eta^2)}. \tag{28.17}$$

The regularized interval is the convolution $H_{T,\eta} = \mathbf{1}_{[0,T]} * g_\eta$, so the convolution theorem gives

$$\widehat{H}_{T,\eta}(v) = e^{-v^2/(4\eta^2)} \frac{1-e^{-ivT}}{iv}, \tag{28.18}$$

with the quotient interpreted continuously at $v=0$. Multiplication by $e^{i\omega t}$ translates the Fourier transform: $\widehat{e^{i\omega t}H}(v) = \widehat{H}(v-\omega)$. Substitution proves the formula. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 28.6 (Regularized two-branch prime contribution)**.** *In the prime-power side of the Guinand–Weil explicit formula, the test* (28.14) *produces, apart from terms independent of T, the oscillatory sum*

$$\mathcal{P}^{\mathrm{osc}}_{\eta,\omega}(T) = \sum_{n=p^r} c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} \tag{28.19}$$

$$+ \sum_{n=p^r} c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T}. \tag{28.20}$$

Proof. By Corollary 28.2, the explicit formula applies to the complex oscillatory test. Insert $v=\ell_n$ and $v=-\ell_n$ into the exact Fourier transform. For the first argument,

$$\widehat{h}(\ell_n) = w_{n,-} \frac{1-e^{-i(\ell_n-\omega)T}}{i(\ell_n-\omega)}, \tag{28.21}$$

where $w_{n,-} = e^{-(\ell_n-\omega)^2/(4\eta^2)}$. Multiplication by the prime-side factor $-\Lambda(n)/(2\pi\sqrt{n})$ separates a constant and the oscillatory term

$$w_{n,-} c_n^- e^{i(\omega-\ell_n)T}. \tag{28.22}$$

The argument $-\ell_n$ similarly gives

$$w_{n,+}c_n^+ e^{i(\omega+\ell_n)T}, \qquad w_{n,+} = e^{-(\ell_n+\omega)^2/(4\eta^2)}. \tag{28.23}$$

The remaining terms are independent of T and belong to the additive constant that must be retained or fitted before demodulation. □

Warning 28.7 (Exact regularized branches versus a complete sharp-residual model). For each admissible regularized test, Proposition 28.6 is an exact specialization of Guinand–Weil. Three later operations are not automatic consequences of that theorem: removing the Gaussian weights, passing to a discontinuous counting window, and declaring the resulting unweighted modes to be a complete spectral basis for the residual. The finite identities obtained after choosing those modes are exact identities of the chosen finite model; completeness of the model is a heuristic or a conjecture.

28.3. **A uniform bound for the regularized prime tail.** STATUS. Exact result proved in this manuscript – unconditional.

Theorem 28.8 (Uniform Gaussian tail). *Assume $Q > e^{2\omega}$, put $L = \log Q$, and suppose $L > \omega + \eta^2$. Then*

$$\sup_{T\in\mathbb{R}} \left| \sum_{\substack{n=p^r \\ n>Q}} \left[c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} + c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T} \right] \right|$$

$$\ll_\omega Q^{-1/2} \exp\left[-\frac{(L-\omega)^2}{4\eta^2} \right] \tag{28.24}$$

$$+ \frac{\eta^2}{L-\omega-\eta^2} \exp\left(\frac{\eta^2}{4} - \frac{(L-\omega-\eta^2)^2}{4\eta^2} \right). \tag{28.25}$$

Proof. For $n > Q > e^{2\omega}$,

$$|\ell_n - \omega| \geq \frac{1}{2}\ell_n, \qquad \ell_n + \omega \geq \ell_n. \tag{28.26}$$

Since $\Lambda(n) \leq \log n$, the absolute value of the two oscillatory branches is bounded, up to a constant depending on ω, by

$$\sum_{n>Q} f(n), \qquad f(x) = x^{-1/2} \exp\left[-\frac{(\log x - \omega)^2}{4\eta^2} \right]. \tag{28.27}$$

The function f is decreasing for $\log x > \omega - \eta^2$, hence in particular on $[Q, \infty)$. The integral test therefore gives the endpoint-sensitive estimate

$$\sum_{n>Q} f(n) \leq f(Q) + \int_Q^\infty f(x)\ \mathrm{d}x. \tag{28.28}$$

The first term is exactly the expression in (28.24).

For the integral $I = \int_Q^\infty f(x)\ \mathrm{d}x$, put $u = \log x$. Completing the square gives

$$I = \int_L^\infty \exp\left[\frac{u}{2} - \frac{(u-\omega)^2}{4\eta^2} \right] \mathrm{d}u \tag{28.29}$$

$$= \exp\left(\frac{\omega}{2} + \frac{\eta^2}{4} \right) \int_L^\infty \exp\left[-\frac{(u-\omega-\eta^2)^2}{4\eta^2} \right] \mathrm{d}u. \tag{28.30}$$

For $r = L - \omega - \eta^2 > 0$,

$$\int_r^\infty e^{-v^2/(4\eta^2)}\,\mathrm{d}v \le \frac{2\eta^2}{r} e^{-r^2/(4\eta^2)}. \tag{28.31}$$

The factor $e^{\omega/2}$ is absorbed in the ω-dependent implied constant. This gives (28.25). No estimate depends on T, because all oscillatory exponentials have modulus one. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 28.9 (One explicit joint choice)**.** *Assume* $\eta^2 > \omega/3$ *and*

$$\log Q = \omega + 3\eta^2. \tag{28.32}$$

Then the complete right-hand side of (28.24)– (28.25) *is*

$$O_\omega(e^{-3\eta^2/4}). \tag{28.33}$$

Proof. The condition $\eta^2 > \omega/3$ ensures $Q > e^{2\omega}$. Moreover,

$$L - \omega - \eta^2 = 2\eta^2. \tag{28.34}$$

The integral term has exponent $\eta^2/4 - \eta^2 = -3\eta^2/4$. The endpoint term has exponent

$$-\frac{L}{2} - \frac{(L-\omega)^2}{4\eta^2} = -\frac{\omega}{2} - \frac{15}{4}\eta^2, \tag{28.35}$$

and is therefore smaller. This proves the stated bound. □

Warning 28.10. Theorem 28.8 controls the prime side of a regularized explicit formula. It does not identify the complete-zero spectral side with the unweighted ordinate walk. That identification is exactly where the positive measure (15.10) appears.

29. Mean-square statements and their exact logical strength

29.1. **Finite Dirichlet-polynomial tails in Lebesgue mean.** Status. Classical result – unconditional.

Proposition 29.1 (Finite-tail mean-value estimate)**.** *Let*

$$E_{Q,R}(T) = \sum_{\substack{Q<n\le R\\ n=p^r}} \left(c_n^- n^{-iT} + c_n^+ n^{iT}\right). \tag{29.1}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q,R}(T)|^2\,\mathrm{d}T \ll_\omega \left(1 + \frac{R}{X}\right)\left[\log\frac{\log R}{\log Q} + \frac{1}{\log Q} + Q^{-1/2}\right] \tag{29.2}$$

for $e^{2\omega} < Q < R$.

Proof. Apply the Montgomery–Vaughan mean-value theorem for Dirichlet polynomials [4, Chapter 7] separately to

$$\sum_{Q<n\le R} c_n^- n^{-iT} \quad \text{and} \quad \sum_{Q<n\le R} c_n^+ n^{iT}. \tag{29.3}$$

It gives an upper bound of the form

$$\int_0^X \left|\sum_{n\le R} a_n n^{-iT}\right|^2 \mathrm{d}T \ll (X+R)\sum_{n\le R}|a_n|^2. \tag{29.4}$$

The inequality $|u+v|^2 \le 2|u|^2 + 2|v|^2$ combines the branches. For $n = p^r > e^{2\omega}$,

$$|c_n^-|^2 + |c_n^+|^2 \ll_\omega \frac{\Lambda(n)^2}{n(\log n)^2} = \frac{1}{r^2 p^r}. \tag{29.5}$$

The prime terms sum to $\log(\log R/\log Q) + O(1/\log Q)$ by Mertens' theorem for primes, while proper powers contribute $O(Q^{-1/2})$. Division by X yields the stated estimate. □

STATUS. Exact result proved in this manuscript – unconditional.

Corollary 29.2 (A finite logarithmic tail can be exhausted)**.** *Let*

$$R(X) = X^{1/2}, \qquad Q(X) = R(X)^{1-1/\sqrt{\log\log X}}. \tag{29.6}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q(X),R(X)}(T)|^2 \,\mathrm{d}T \longrightarrow 0. \tag{29.7}$$

Proof. For the choice (29.6),

$$\frac{R(X)}{X} = X^{-1/2} \longrightarrow 0. \tag{29.8}$$

If $\delta_X = 1/\sqrt{\log\log X}$, then $\log Q = (1-\delta_X)\log R$, so

$$\log \frac{\log R}{\log Q} = -\log(1-\delta_X) = O(\delta_X) \longrightarrow 0. \tag{29.9}$$

Also $1/\log Q \to 0$ and $Q^{-1/2} \to 0$. Apply Proposition 29.1. □

Warning 29.3. Corollary 29.2 concerns a finite tail in Lebesgue measure. The numerical variance is sampled on zeta ordinates and contains the projected prime kernel (23.25); the two norms are not interchangeable.

29.2. **Normalized global convergence does not detect fixed atoms.** STATUS. Exact result proved in this manuscript – unconditional.

Proposition 29.4 (Counterexample to global normalized sufficiency)**.** *There exist functions F_X such that*

$$\frac{1}{X}\int_0^X \left|F_X(T) - \frac{1}{2}\right|^2 \mathrm{d}T \longrightarrow 0, \tag{29.10}$$

but whose distributional derivatives retain a nonzero fixed atomic part.

Proof. Take $F_X(T) = 1/2 + \mathbf{1}_{[0,1]}(T)$. Then for every $X \ge 1$,

$$\frac{1}{X}\int_0^X \left|F_X(T) - \frac{1}{2}\right|^2 \mathrm{d}T = \frac{1}{X} \longrightarrow 0. \tag{29.11}$$

However, on the fixed compact interval $[0,1]$, the defect remains identically one. In the sense of distributions,

$$\frac{\mathrm{d}}{\mathrm{d}T}\mathbf{1}_{[0,1]} = \delta_0 - \delta_1, \tag{29.12}$$

which does not tend to zero. Thus normalization by the length of an expanding interval can erase a fixed local or atomic obstruction. □

STATUS. Exact result proved in this manuscript – unconditional.

Corollary 29.5. *Any route from energy exhaustion to the vanishing of the positive atomic measure* (15.10) *must use a local topology, atomwise convergence, or distributional convergence against compactly supported test functions.*

Proof. Proposition 29.4 shows that global normalized convergence alone permits a persistent compact atomic defect. Therefore any sufficient argument must contain additional local information of one of the stated forms. □

30. Weighted prime-power profiles

Definition 30.1. Let $W \in C_c(\mathbb{R})$ satisfy $\operatorname{supp} W \subset [0,\infty)$. Define

$$G_{Q,W}(T) = \sum_{n=p^r} W\left(\log \frac{n}{Q}\right) g_n(T). \tag{30.1}$$

Because $W(u) = 0$ for $u < 0$ and has compact support, the sum is finite and contains only prime powers in a fixed multiplicative interval above Q.

Status. Exact result proved in this manuscript – unconditional.

Theorem 30.2 (Weighted Bohr energy)**.** *As $Q \to \infty$,*

$$\lim_{X\to\infty} \frac{1}{X} \int_0^X |G_{Q,W}(T)|^2 \, \mathrm{d}T = \frac{1}{2\pi^2 \log Q} \int_0^\infty |W(u)|^2 \, \mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{30.2}$$

Proof. Expand the squared modulus of the finite weighted sum. Every diagonal term contributes

$$|W(\log(n/Q))|^2 \left(|c_n^-|^2 + |c_n^+|^2\right). \tag{30.3}$$

Every unequal pair has a nonzero frequency difference and therefore zero Bohr mean. For primes, use

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{30.4}$$

Write $p = Qe^u$ and apply the prime number theorem by partial summation. For compactly supported continuous W,

$$\sum_p \frac{1}{p} \left| W\left(\log \frac{p}{Q}\right)\right|^2 = \frac{1}{\log Q} \int_0^\infty |W(u)|^2 \, \mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{30.5}$$

The coefficient error and all proper prime powers are lower order. This proves the formula. □

Status. Exact result proved in this manuscript – unconditional.

Corollary 30.3 (Interpretation of the constant 3/4)**.** *A weighted Bohr model has leading energy $3/(4\pi^2 \log Q)$ precisely when*

$$\int_0^\infty |W(u)|^2 \, \mathrm{d}u = \frac{3}{2}. \tag{30.6}$$

The constant determines a quadratic profile mass, not a unique hard support.

Proof. Compare the leading term in Theorem 30.2 with $3/(4\pi^2 \log Q)$. Equality of the coefficients is equivalent to $\int_0^\infty |W(u)|^2 \, \mathrm{d}u = 3/2$. □

31. Detailed projected Gram algebra

31.1. **Four branch expansion.** Status. Exact result proved in this manuscript – unconditional.

Proposition 31.1 (Full unprojected quadratic expansion)**.** *For a finite set $\mathcal{N}$ of prime powers, put*

$$G_{\mathcal{N}}(\gamma) = \sum_{n\in\mathcal{N}} \left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma} \right). \tag{31.1}$$

Then

$$\|G_{\mathcal{N}}\|_H^2 = \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^- M_H(m/n) \tag{31.2}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^+ M_H(mn) \tag{31.3}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^- \overline{M_H(mn)} \tag{31.4}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^+ M_H(n/m). \tag{31.5}$$

Proof. Write the finite model as

$$G(\gamma) = \sum_{n} \left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma} \right). \tag{31.6}$$

Multiplying by its conjugate produces four sign choices. For example,

$$\overline{c_m^- m^{-i\gamma}}\, c_n^- n^{-i\gamma} = \overline{c_m^-} c_n^- \left(\frac{m}{n} \right)^{i\gamma}, \tag{31.7}$$

whose sample average is $\overline{c_m^-} c_n^- M_H(m/n)$. The other three products give the arguments mn, $(mn)^{-1}$, and n/m. Summing all pairs yields the displayed expansion exactly. □

31.2. **Prime-tower blocks.** Status. Exact result proved in this manuscript – unconditional.

Proposition 31.2 (Same-tower multiplicative closure)**.** *For $m = p^r$ and $n = p^s$, every argument of M_H in (31.2)–(31.5) is one of*

$$p^{s-r}, \qquad p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}. \tag{31.8}$$

Thus the exact Gram arguments of a two-branch prime tower remain inside the multiplicative set generated by p; the block is not reduced to the scalar diagonal $r = s$.

Proof. Substitute $m = p^r$ and $n = p^s$ into the four multiplicative arguments. They become

$$p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}, \qquad p^{s-r}. \tag{31.9}$$

Each is one, a prime power, or the reciprocal of a prime power. This is an exact algebraic closure statement. Interpreting these entries by the main term of a Landau–Gonek formula for the ordinate-only kernel M_H additionally requires the Riemann Hypothesis, or an independent replacement of $x^{i\gamma}$ by $x^{\rho-1/2}$. □

Status. Exact result proved in this manuscript – unconditional.

Proposition 31.3 (Projection correction)**.** *Let V be the matrix whose columns are the sampled prime-power modes, and let B be the two-column matrix with columns 1 and $e^{i\omega\gamma}$. Then the residual Gram matrix is*

$$\widetilde{K}_H = \frac{1}{N(H)} V^*V - \frac{1}{N(H)^2} V^*B \left(\frac{1}{N(H)} B^*B \right)^{-1} B^*V. \tag{31.10}$$

In particular, even the norm of a single frequency is altered by the constant–carrier projection.

Proof. Let B be the matrix whose columns are the sampled constant and carrier vectors. Since these columns are linearly independent when $|M_H(e^\omega)| < 1$, the orthogonal projection onto their span is

$$P_B = B(B^*B)^{-1}B^*. \tag{31.11}$$

The complementary projection is $I - P_B$. If V contains the sampled prime frequency vectors, the projected Gram matrix is

$$\frac{1}{N(H)}V^*(I-P_B)V = \frac{1}{N(H)}V^*V \tag{31.12}$$

$$- \frac{1}{N(H)}V^*B(B^*B)^{-1}B^*V. \tag{31.13}$$

Writing $B^*B = N(H)G_H$ gives the normalized Schur-complement form in the proposition. In particular, projection changes both diagonal and off-diagonal entries. □

32. Analytic targets isolated by the exact identities

Conjecture 32.1 (Collective law for the reported carrier)**.** *In a specified joint range $Q = Q(H) \to \infty$,*

$$\Im c_{Q,H} = \pi\Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{32.1}$$

By Corollary 20.6, this is equivalent to a collective cancellation involving the three exact kernel families

$$M_H(n), \qquad M_H(q/n), \qquad M_H(qn). \tag{32.2}$$

together with the nonorthogonality factor $1-|M_H(q)|^2$. The raw law for $\widehat{c}_{Q,H}$ is a different conjecture.

Conjecture 32.2 (Projected variance law)**.** *For a suitable joint range,*

$$V(Q,H) = \frac{3}{4}\Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{32.3}$$

The left-hand side is the full projected residual expression (23.28)*. Its prime Gram term is only one of three cutoff-dependent contributions in addition to the projected background energy.*

Question 32.3 (Background, mixed, tower, and cross-tower separation)**.** Can one prove, uniformly in a growing cutoff, asymptotic formulas for

$$-2\Re\sum_p \mathbf{c}_p^*\mathbf{d}_{p,H}, \tag{32.4}$$

$$\sum_p \mathbf{c}_p^*\widetilde{K}_{p,p;H}\mathbf{c}_p, \tag{32.5}$$

$$\sum_{p\neq r} \mathbf{c}_p^*\widetilde{K}_{p,r;H}\mathbf{c}_r \tag{32.6}$$

relative to the projected background energy $V_Y(H)$? These are the exact components that must be separated before any assertion about a "diagonal" can be made.

33. Conditional routes from the exponential zero walk to the Riemann Hypothesis

Proposition 33.1 (Precisely established components of the architecture)**.** *The following statements are proved in the preceding sections.*

(i) *For every fixed $q > 1$, the complete-zero and ordinate-only measures differ by a nonnegative atomic defect, and equality of those measures is equivalent to RH.*
(ii) *For admissible regularized test functions, the two prime-power branches are exact terms of the Guinand–Weil explicit formula.*
(iii) *Equality on a nontrivial interval of two finite exponential polynomials forces equality of their grouped frequencies and coefficients. Thus the spectral-alignment implication is unconditional, although its interval-equality premise is open in the present application.*
(iv) *Each demodulated prime-power mode has an exact elliptic parametrization, a positive signed invariant d_n, and the tail law $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$.*
(v) *The seven statements below are proved as conditional implications: whenever the explicitly stated local reconstruction or convergence hypothesis holds, RH follows.*

None of these assertions proves completeness of the unweighted sharp prime-power ansatz or verifies any of the open hypotheses below.

Each route below has two layers. The logical implication is proved. The analytic estimate needed to activate it is stated separately and is not claimed. The ordinate-clock displacement ε_k and the horizontal displacement $\alpha_\rho = \beta - 1/2$ are never identified.

33.1. Route I: equality of complete and ordinate-only atomic measures.

Status. Conditional route. The fixed-q equivalence is exact; prime-side identification of the two measures is open.

Theorem 33.2 (Fixed-q atomic criterion)**.** *For any fixed $q > 1$,*

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{33.1}$$

Equivalently, equality of the jumps at every positive ordinate is enough.

Proof. By Corollary 15.5,

$$\frac{1}{2} e^{-iT \log q} (\nu_q - \mu_q) = \sum_{\tau > 0} D_q(\tau) \delta_\tau, \tag{33.2}$$

with nonnegative masses. Equality of the measures forces every mass to vanish and therefore every horizontal displacement to be zero. Conversely, RH makes every complete-zero weight equal to its multiplicity. A locally finite atomic measure is determined by its jumps. □

Missing analytic step. One must reconstruct the complete-zero measure from the prime side and then identify it with the ordinate-only measure. A direct implementation through the RVM clock would require control on windows fine enough to resolve individual jumps of $S^{\#}(T) + R_\vartheta(T)$. Global bounds for $S(T)$ do not provide that atomic control.

33.2. Route II: local distributional convergence.

Status. Conditional route. The distributional implication is exact; the local convergence theorem is open.

Theorem 33.3 (Test-function criterion)**.** *Let η_j be locally finite complex measures such that*

$$\eta_j \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{33.3}$$

If $\langle \eta_j, \varphi \rangle \to 0$ for every $\varphi \in C_c^\infty(\mathbb{R})$, then RH holds.

Proof. Both hypotheses determine the limit of $\langle \eta_j, \varphi \rangle$. Uniqueness of the complex limit gives $\langle \nu_q - \mu_q, \varphi \rangle = 0$ for every test function. Thus the measure difference vanishes, and Route I applies. □

Missing analytic step. For the phase-displacement expansion, local convergence requires control of weighted sums such as $\sum \varepsilon_k e^{i\omega\widetilde{\gamma_k}}$ in compact windows, whose natural oscillation scale is $O(1/\omega)$. Such estimates interact with Guinand–Weil and zero correlations, but no equivalence with Montgomery's pair-correlation conjecture [13] is asserted or proved here.

33.3. **Route III: small-q curvature at each ordinate.** Status. Conditional route. The local curvature criterion is exact; differentiating a prime-side reconstruction twice at $q = 1$ is open.

Theorem 33.4 (Quadratic curvature criterion)**.** *Put $q = e^y$ and $\alpha_\rho = \beta - 1/2$. For every fixed ordinate τ,*

$$D_{e^y}(\tau) = \frac{y^2}{2} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \alpha_\rho^2 + O_\tau(y^4) \qquad (y \to 0). \tag{33.4}$$

Consequently,

$$\text{RH} \iff D_{e^y}(\tau) = o(y^2) \text{ for every } \tau. \tag{33.5}$$

Proof. At a fixed ordinate only finitely many zeros occur. Taylor expansion of $\cosh(\alpha_\rho y) - 1$ may therefore be summed term by term. Division by y^2 and positivity show that $o(y^2)$ is equivalent to the vanishing of every α_ρ^2. □

Missing analytic step. A prime-side proof would need a regularized explicit formula uniform for y near zero, two justified differentiations in y, and interchange of the y-limit with the prime-power sum, the regularization limit, and local atom extraction. The endpoint half atom from Warning 5.3 must remain present.

33.4. **Route IV: large-q amplification.** Status. Conditional route. The growth exponent is exact at each ordinate; a prime-side bound uniform in q is open.

Definition 33.5. Define

$$\alpha_{\max}(\tau) = \max_{\rho=\beta+i\tau} |\beta - 1/2|. \tag{33.6}$$

Status. Conditional route: the implication is proved; the analytic premise is open.

Theorem 33.6 (Recovery of the maximal horizontal displacement)**.** *For every positive ordinate τ,*

$$\lim_{q\to\infty} \frac{\log(1 + D_q(\tau))}{\log q} = \alpha_{\max}(\tau). \tag{33.7}$$

Proof. If $\alpha_{\max} = 0$, the defect is identically zero. Otherwise, terms with maximal displacement contribute a positive constant multiple of $q^{\alpha_{\max}}$, while every other term is bounded by the same power. Taking logarithms and applying the squeeze theorem proves the limit. □

Missing analytic step. One needs a prime-side estimate for $D_q(\tau)$ that is uniform as $q \to \infty$ and local in τ. A bound for $\sum |\varepsilon_k|^2$ does not supply it and is not equivalent to RH, because ε_k measures ordinate fluctuations around RVM rather than horizontal displacement from the critical line. Even a sparse global bound for $\sum |\alpha_\rho|^2$ would not eliminate isolated off-line zeros without local positivity.

33.5. **Route V: reconstruction from a family of frequencies.** STATUS. Conditional route. Finite exponential-polynomial uniqueness is proved; reconstruction of the local data from the prime side is open.

Theorem 33.7 (Finite exponential-polynomial reconstruction)**.** *At a fixed ordinate τ,*

$$\mathcal{W}_\tau(y) = \sum_{\rho:\Im\rho=\tau} e^{(\beta-1/2)y} \tag{33.8}$$

is an entire exponential polynomial. Its values on any real set with an accumulation point determine the multiset of horizontal displacements. Moreover, $\mathcal{W}_\tau(y) \equiv m(\tau)$ if and only if all zeros at τ lie on the critical line.

Proof. The sum is finite. The identity theorem gives uniqueness from an accumulation set. Derivatives at zero are the power sums of the finite multiset of exponents, which determine that multiset through Newton identities. Constancy forces the nonnegative second moment to vanish. □

STATUS. Classical linear-independence principle, proved here in the normalization used by the manuscript.

Theorem 33.8 (Uniqueness and alignment of finite exponential sums)**.** *Let $I \subset \mathbb{R}$ be a nontrivial interval and suppose*

$$\sum_{j=1}^{J} a_j e^{i\lambda_j t} = \sum_{k=1}^{K} b_k e^{i\mu_k t} \qquad (t \in I), \tag{33.9}$$

where the frequencies are real and the sums are finite. After grouping repeated frequencies and deleting zero grouped coefficients, the two frequency sets coincide and the corresponding grouped coefficients are equal.

Proof. Move the right side of (33.9) to the left and group equal frequencies. The result is

$$F(t) = \sum_{r=1}^{R} c_r e^{i\nu_r t} = 0 \qquad (t \in I), \tag{33.10}$$

with distinct real numbers ν_r. The function F is entire, so the identity theorem gives $F \equiv 0$ on $\mathbb{C}$. Fix $t_0 \in I$. For $m = 0, \dots, R-1$,

$$0 = F^{(m)}(t_0) = \sum_{r=1}^{R} c_r (i\nu_r)^m e^{i\nu_r t_0}. \tag{33.11}$$

This is a homogeneous linear system whose determinant is a nonzero product of $e^{i\nu_r t_0}$ and the Vandermonde determinant

$$\prod_{1\le r<s\le R} i(\nu_s - \nu_r). \tag{33.12}$$

Hence every $c_r = 0$, proving equality of the grouped supports and masses. □

Corollary 33.9 (Finite prime–zero spectral alignment)**.** *Let*

$$P_{q,Q}(t) = \sum_{n\in\mathcal{P}_Q} A_{q,Q}(n) e^{i(\log n)t}, \tag{33.13}$$

$$Z_{q,H}(t) = \sum_{\tau\in\Gamma_H^{\mathrm{dist}}} B_{q,H}(\tau) e^{i\tau t}, \tag{33.14}$$

where both index sets are finite. If $P_{q,Q} = Z_{q,H}$ on a nontrivial interval, then the nonzero frequencies $\log n$ and τ coincide as multisets after grouping, and their grouped coefficients coincide. The same statement applies to the shifted frequencies $\omega \pm \log n$ after a common carrier modulation or demodulation.

Proof. Apply Theorem 33.8. Multiplication by a nonvanishing common exponential merely translates every frequency by the same amount and does not affect uniqueness. □

Warning 33.10 (The interval premise is the unresolved step). Theorem 33.8 proves the alignment implication, not the existence of an interval on which a prime-side and a zero-side polynomial coincide. In the actual explicit formula, the sums are regularized and the complete-zero coefficients retain the horizontal weights $q^{\beta-1/2}$. To infer RH one must additionally identify those grouped coefficients with the ordinate-only multiplicities, for one fixed $q > 1$ or for a family of q-values with an accumulation point.

Missing analytic step. The missing issue is not spectral uniqueness; that is already proved. It is the local reconstruction of $\mathcal{W}_\tau(y)$ from prime-side quantities for enough values of y, uniformly through the regularization and sharp-window limits.

33.6. **Route VI: local energy exhaustion plus derivative identification.** Status. Conditional route. The sufficient implication is exact; the required local estimates are open and stronger than a global variance decay.

Theorem 33.11 (A sufficient two-part energy route). *Let F_j be regularized residuals. Suppose that for every compact interval I,*

$$\int_I |F_j(T) - 1/2|^2 \, \mathrm{d}T \longrightarrow 0, \tag{33.15}$$

and that

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}(F_j(T) - 1/2)\right] \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'. \tag{33.16}$$

Then RH holds.

Proof. Local L^2 convergence implies local L^1 convergence. Therefore the distributional derivative of the displayed product tends to zero. The second hypothesis identifies the same limit with $\nu_q - \mu_q$, which must therefore vanish. Route I applies. □

Missing analytic step. The measured global normalized variance can dilute any fixed or sparse defect. The needed statement is local in T and must be accompanied by derivative identification. This two-part premise is sufficient for RH; it is not asserted to be equivalent to RH for an arbitrary regularization.

33.7. **Route VII: one-sided positive test families.** Status. Conditional route. The abstract positive-measure implication is exact; a prime-compatible local test family with vanishing pairings is open.

Theorem 33.12 (Exhausting positive tests). *Let*

$$\mathcal{D}_q = \sum_{\tau>0} D_q(\tau)\delta_\tau. \tag{33.17}$$

Suppose $\varphi_j \in C_c^\infty(0,\infty)$ are nonnegative, eventually bounded below on every compact subset of $(0,\infty)$, and $\langle \mathcal{D}_q, \varphi_j\rangle \to 0$. Then RH holds.

Proof. Fix an ordinate τ and a compact interval containing it. Eventual positive lower bounds on that compact give $0 \le cD_q(\tau) \le \langle \mathcal{D}_q, \varphi_j\rangle$. The right side tends to zero, so every atomic mass vanishes. □

Missing analytic step. Abstract exhausting positive tests are easy to construct. The open problem is to choose tests that are simultaneously local enough to see individual positive atoms and compatible with a controlled prime-power explicit formula. Weil and Bombieri–Lagarias provide global positivity frameworks, but they do not by themselves supply this ordinate-local prime-side estimate [15, 8].

34. Open problems and falsifiable next computations

Question 34.1 (Raw and reported increment audit)**.** For every prime power n in a growing range, compute both exact ratios

$$\mathcal{I}_H^{\mathrm{raw}}(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{34.1}$$

$$\mathcal{I}_H^{\mathrm{joint}}(n) = \frac{J_H(n)}{\pi u_n v_n}. \tag{34.2}$$

Compare the raw ratio with the conditional leading prediction

$$\frac{\log n}{\log(H/2\pi)} \tag{34.3}$$

and measure separately the exact projection correction that converts the raw ratio into the reported joint ratio. This tests the local Landau mechanism without fitting a global law and prevents the two carrier statistics from being conflated.

Question 34.2 (Height–cutoff matrix)**.** Repeat the complete computation for a two-dimensional grid of (H, Q), including regimes

$$Q = H^{\theta} \tag{34.4}$$

with several fixed θ and a near-height regime $Q = H^{1-o(1)}$. Record raw and joint coefficients separately.

Question 34.3 (Numerical audit of the exact variance components)**.** For every cutoff and height, compute separately

$$V_Y(H), \quad -2\Re\sum_p \mathbf{c}_p^* \mathbf{d}_{p,H}, \quad \sum_p \mathbf{c}_p^* \widetilde{K}_{p,p;H} \mathbf{c}_p, \quad \sum_{p\neq r} \mathbf{c}_p^* \widetilde{K}_{p,r;H} \mathbf{c}_r, \tag{34.5}$$

and verify numerically that their sum equals the directly fitted residual variance in (23.11).

Question 34.4 (Local test-function convergence)**.** For compactly supported smooth φ, determine whether the regularized residual satisfies

$$\int \left(F_{Q,H}(T) - \frac{1}{2}\right) \varphi(T)\, \mathrm{d}T \longrightarrow 0 \tag{34.6}$$

under a joint limiting regime, while the prime-side derivative is identified with the complete spectral measure.

Part 4. Numerical support

35. Computational object and exact program identities

35.1. Input data and sampled height.

Experiment 35.1 (Zero data)**.** The full-data run used 2,001,052 positive ordinates from `zeross.bin`, with

$$\gamma_1 = 14.134725142, \qquad \gamma_{2{,}001{,}052} = 1{,}132{,}490.6587144111. \tag{35.1}$$

The phase parameter was

$$\omega = 1, \qquad q = e. \tag{35.2}$$

All ordinates were used; the stride was one.

35.2. **Exact definition of the program.**

Definition 35.2 (Program variables)**.** At each sampled ordinate, the code forms

$$Y(\gamma_k) = S_q(\gamma_k) - A_{\mathrm{smooth}}(\gamma_k), \tag{35.3}$$

and

$$R_Q(\gamma_k) = Y(\gamma_k) - P_Q^-(\gamma_k) - P_Q^+(\gamma_k), \tag{35.4}$$

where

$$P_Q^-(\gamma) = \sum_{\substack{n=p^r\\ n\le Q}} c_n^- e^{i(\omega-\ell_n)\gamma}, \tag{35.5}$$

$$P_Q^+(\gamma) = \sum_{\substack{n=p^r\\ n\le Q}} c_n^+ e^{i(\omega+\ell_n)\gamma}. \tag{35.6}$$

The raw carrier statistic is

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k\le H} e^{-i\omega\gamma_k} R_Q(\gamma_k). \tag{35.7}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 35.3 (Identity evaluated by the raw program)**.** *The program variables satisfy exactly*

$$\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r\\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n) \right). \tag{35.8}$$

Its imaginary part is

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r\\ n\le Q}} u_n \Re M_H(n). \tag{35.9}$$

Proof. The arrays used by the program are finite, so the abstract identity in Theorem 19.2 applies without any limiting operation. The code generates exactly the prime powers $n \le Q$, uses the coefficients in Definition 16.1, and forms the sample kernel by averaging $n^{i\gamma_k}$. Substitution of these arrays in the theorem gives the program identity entry by entry. □

35.3. **Joint constant–carrier fit used for the reported values.**

Definition 35.4. For every cutoff, the code fits

$$R_Q(\gamma_k) = \alpha_{Q,H} + c_{Q,H} e^{i\omega\gamma_k} + E_{Q,H}(\gamma_k) \tag{35.10}$$

by least squares in the sampled inner product (23.1). The reported variance is

$$V(Q,H) = \frac{1}{N(H)} \sum_{\gamma_k\le H} |E_{Q,H}(\gamma_k)|^2. \tag{35.11}$$

Status. Exact result proved in this manuscript – unconditional.

Proposition 35.5 (Closed form of the reported carrier)**.** *Let*

$$\mu_H = M_H(e^{\omega}). \tag{35.12}$$

Then

$$\boxed{c_{Q,H} = \frac{\langle e^{i\omega\gamma}, R_Q\rangle_H - \overline{\mu_H}\langle 1, R_Q\rangle_H}{1-|\mu_H|^2}.} \tag{35.13}$$

The fitted constant is

$$\alpha_{Q,H} = \frac{\langle 1, R_Q\rangle_H - \mu_H \langle e^{i\omega\gamma}, R_Q\rangle_H}{1 - |\mu_H|^2}. \tag{35.14}$$

Proof. The fit minimizes

$$\sum_{k=1}^{N(H)} |R_k - a - cb_k|^2 \tag{35.15}$$

over $a, c \in \mathbb{C}$. Differentiating with respect to the conjugate parameters, or equivalently imposing orthogonality of the residual to the basis vectors 1 and b, gives the two normal equations. Their Gram matrix is $\left(\begin{smallmatrix} 1 & \mu_H \\ \overline{\mu}_H & 1 \end{smallmatrix}\right)$. Its inverse is explicit, and multiplication by the right-hand side gives the displayed formulas for the constant and carrier coefficients. □

35.4. **Prime-power generation and exact coefficients.**

Experiment 35.6 (Prime powers)**.** The cutoffs and counts were

Q	number of prime powers $n = p^r \le Q$
0	0
3	2
10	7
100	35
1000	193
10000	1280

Every coefficient was generated from (16.2)–(16.3); no frequency or amplitude was fitted individually.

36. Radial geometry and low prime-power resonances

36.1. **Direct radial audit.**

Experiment 36.1 (Original walk and theoretical subtractions)**.** After subtraction of the empirical complex center, the radial statistics were

case	mean radius	radial standard deviation	smoothed radial modes
original	1.9203104476	0.7265477726	1.35731059, 2.49789956
after $n = 3$	1.8079247921	0.3083120307	1.68017926
after $n = 2, 3$	1.7884459085	0.2214048337	1.81300495

The theoretical $n = 3$ coefficients were

$$c_3^- = -1.02370050729i, \qquad c_3^+ = 0.0481029537851i. \tag{36.1}$$

Subtracting $n = 3$ reduced the radial variance by approximately 81.99%; subtracting $n = 2, 3$ reduced it by approximately 90.71%. Within this deterministic subtraction audit, removing the theoretical $n = 3$ term accounts for 81.99% of the measured radial variance and changes the selected smoothed density from two detected modes to one. This is a finite algebraic observation about the actual data; frequency-specific causality is reserved for the matched null control below.

Remark 36.2 (Global-center statistic)*.* For each compared full-sample series, the radial audit subtracts that series' single empirical center; no moving center is used. It is therefore not the local moving-center statistic whose arcsine-type geometry can occur in generic planar walks. The analytic role of $n = 3$ is established by Theorem 17.5; the number of modes in the finite radial density remains a numerical property of the complete dataset.

36.2. **Geometric interpretation.**

Experiment 36.3 (Anisotropy)**.** At $Q = 10000$, the centered residual anisotropy was

$$0.981212. \tag{36.2}$$

This is consistent with the exact single-mode limit (17.21): after low frequencies are removed, large-n modes are strongly elongated in one direction.

36.3. **Matched-amplitude non-arithmetic null control.**

Experiment 36.4 (Required causal control)**.** Annex A contains a deterministic-seed null experiment. It replaces $\log 3$ by non-prime-power frequencies in a neighbourhood of $\log 3$, retains the exact amplitudes c_3^- and c_3^+, and applies the same full-sample recentering and radial peak detector. The test records the radial variance reduction and the number of detected peaks for each null frequency. No numerical result from this added test is asserted here because the zero file is not embedded in the manuscript build. The finite claim that the histogram collapse is frequency-specific should be made only after comparing the observed $n = 3$ statistic with this null distribution.

37. Cutoff sweep and carrier stabilization

37.1. **Global cutoff table.**

Experiment 37.1 (Full-data cutoff sweep)**.** The jointly fitted carrier and residual energy were

Q	terms	$\Re c_{Q,H}$	$\Im c_{Q,H}$	$V(Q,H)$	$\sqrt{V(Q,H)}$
0	0	0.4999989123	0.1381673175	1.2420277147	1.1144630
3	2	0.4999998679	0.0956865241	0.1339324657	0.3659678
10	7	0.4999999862	0.0755224108	0.0596099186	0.2441514
100	35	0.5000000416	0.0545919879	0.0220396104	0.1484574
1000	193	0.5000000755	0.0426449212	0.0112822119	0.1062178
10000	1280	0.4999998710	0.0343125459	0.0082559398	0.0908622

At the largest cutoff the finite variance reduction relative to $Q = 0$ was

$$1 - \frac{0.0082559398}{1.2420277147} = 0.99335\ldots. \tag{37.1}$$

37.2. **Normalized $1/\log Q$ comparisons.**

Experiment 37.2 (Two normalized quantities)**.** For $Q \geq 3$, the finite normalized values were

Q	$\pi \log Q\, \Im c_{Q,H}$	$\frac{4\pi^2}{3} \log Q\, V(Q,H)$
3	0.33025	1.93628
10	0.54631	1.80623
100	0.78981	1.33564
1000	0.92545	1.02558
10000	0.99284	1.00065

At $Q = 10000$,

$$\frac{1}{\pi \log Q} = 0.0345601\ldots, \tag{37.2}$$

$$\frac{3}{4\pi^2 \log Q} = 0.00825061\ldots. \tag{37.3}$$

The corresponding measured values were 0.0343125459 and 0.0082559398. These are finite coincidences to be interpreted through the exact defect identities, not proofs of asymptotic laws.

38. Height dependence

38.1. **Twenty-four block audit.**

Experiment 38.1 (Block ranges)**.** For $Q = 10000$ and 24 consecutive height blocks,

$$\Re\,\mathrm{mean}_{\mathrm{block}}\,F_Q \in [0.499994991361, 0.500003656384], \tag{38.1}$$

$$\Im\,\mathrm{mean}_{\mathrm{block}}\,F_Q \in [0.033825249952, 0.035105891286], \tag{38.2}$$

$$V_{\mathrm{block}} \in [0.0054744557, 0.0089851833]. \tag{38.3}$$

The standard deviation of the block means was

$$0.0003812589. \tag{38.4}$$

The carrier is stable across height at this scale, while the block variance has a visible drift. Consequently the global factor 3/4 cannot yet be regarded as height-uniform.

38.2. **Termwise increment scale.**

Experiment 38.2 (Present range of the raw Landau scale)**.** For

$$H = 1.13249 \times 10^6, \qquad Q = 10^4, \tag{38.5}$$

one has approximately

$$\frac{\log Q}{\log(H/2\pi)} \approx 0.76. \tag{38.6}$$

This number is the leading prediction for the *raw* increment ratio (19.25), under the additional uniform Landau hypothesis stated in Section 21. The printed coefficient is instead the joint coefficient $c_{Q,H}$, for which the exact prime-power increment is (20.22). Its global finite ratio is

$$\frac{\Im c_{Q,H}}{\pi\Delta_1(Q)} \approx 0.986. \tag{38.7}$$

No comparison of 0.986 with 0.76 is valid until the projection terms

$$M_H(q/n), \qquad M_H(qn), \qquad M_H(q). \tag{38.8}$$

in (20.9) are evaluated. The direct falsifiable test for the reported program is therefore (20.22), prime power by prime power.

39. Branch decomposition and omitted frequencies

39.1. **Branch energies.**

Experiment 39.1 (Minus and plus branches)**.** At $Q = 10000$,

$$\|P_Q^-\|^2 = 1.2433344613, \tag{39.1}$$

$$\|P_Q^+\|^2 = 0.0309962015, \tag{39.2}$$

$$2\Re\langle P_Q^-, P_Q^+\rangle = -0.0005963448. \tag{39.3}$$

The minus branch dominates, and direct plus–minus energy cancellation is small.

39.2. **Omitted prime-power shell.**

Experiment 39.2 (Exact-frequency omitted fit)**.** For the 56 omitted prime powers immediately above $Q = 10000$, the median amplitude-to-null ratios were

$$3.35349 \quad \text{and} \quad 2.67559 \tag{39.4}$$

for the minus and plus branches. The absolute correlation between the jointly fitted coefficient vector and the theoretical coefficient vector was

$$0.916474. \tag{39.5}$$

The narrow shell explained

$$0.0144446 \tag{39.6}$$

of the remaining residual energy. Thus the next prime-power frequencies are present, but this narrow shell does not account for the full residual.

40. Signed invariant and finite-area comparisons

40.1. **Direct evaluation of $\Delta_1(Q)$.**

Experiment 40.1 (Common invariant)**.** At $Q = 10000$,

$$\Delta_1(10000) \approx 0.01108. \tag{40.1}$$

Hence

$$\pi\Delta_1(10000) \approx 0.03481, \qquad \frac{3}{4}\Delta_1(10000) \approx 0.00831. \tag{40.2}$$

The same exact signed invariant therefore reproduces both observed scales to finite accuracy. The analytic claims that the centroid and projected variance equal these multiples remain Conjectures 32.1 and 32.2.

40.2. **Coefficient prefix sums.**

Experiment 40.2 (Prefix sums)**.** The coefficient combinations

$$A_Q = \sum_{n \le Q} (c_n^- + c_n^+), \qquad B_Q = \sum_{n \le Q} (c_n^- - c_n^+) \tag{40.3}$$

were, at $Q = 10000$,

$$A_Q = -2.5552137617i, \qquad B_Q = -10.6932626383i. \tag{40.4}$$

They do not account for the real carrier value $1/2$.

41. Finite-part computations for Φ_N

41.1. **Direct algebraic verification.**

Experiment 41.1 (Finite-part identity)**.** A numerical comparison between the direct tail expression and the value reconstructed from Proposition 12.3 gave

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{41.1}$$

This verifies the implemented identity at the working precision.

41.2. **Scale scan.**

Experiment 41.2 (Normalized finite-part differences)**.** For

$$N = 2000, 4000, 8000, 16000, 32000, 64000, \tag{41.2}$$

the successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \tag{41.3}$$

were approximately

$$1.12, \quad 0.75, \quad 0.80, \quad 0.75, \quad 0.44. \tag{41.4}$$

No monotone cancellation law is established by these six scales.

42. Numerical tests dictated by the analytic identities

42.1. **Prime-power increment audit.**

Experiment 42.1 (Required output)**.** For every prime power n in the selected range, compute and store

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{42.1}$$

together with

$$\theta_H(n) = \frac{\log n}{\log(H/2\pi)}. \tag{42.2}$$

The comparison is pointwise and requires no global fitting.

42.2. **Two-parameter height–cutoff scan.**

Experiment 42.2 (Required grid)**.** For several heights H_j and exponents θ, set

$$Q_{j,\theta} = H_j^{\theta} \tag{42.3}$$

and record separately:

$$\widehat{c}_{Q,H}, \quad c_{Q,H}, \quad V(Q,H), \quad \Delta_\omega(Q), \quad \sum_p a_p^* \widetilde{K}_{p,p;H} a_p, \quad \sum_{p\neq q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{42.4}$$

This directly separates the raw carrier, the joint fitted carrier, the projected background energy, the mixed background–prime term, the tower blocks, and the cross-tower interactions.

43. Conclusion

The manuscript begins with the discrepancy between the exponential zero walk and the inverse Riemann–von Mangoldt clock. The exact perturbation formula separates a first-order oscillatory sum from a quadratic remainder. The bound

$$|\mathcal{Q}_N(\omega)| \le \frac{\omega^2}{2} \sum_{k\le N} |\varepsilon_k|^2 \tag{43.1}$$

is rigorous, but the available global and short-window bounds for the square sum are too weak to make it a small error. In particular, ε_k is an ordinate-clock displacement and does not vanish under RH. Any route that uses the first-order phase expansion must therefore supply new discrete local control rather than rely on a formal Taylor expansion.

For regularized test functions, the two prime-power branches are exact Fourier displacements of Guinand–Weil. The manuscript does not identify this fact with completeness of the unweighted

sharp-residual modes. What is proved unconditionally is their coefficient algebra, the positive invariant d_n, and

$$\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q}. \tag{43.2}$$

At $\omega = 1$, $n = 3$ is analytically singled out: it is the unique closest prime-power resonance, it has the largest minus-branch coefficient and the largest two-branch envelope, and its contribution has pointwise size close to one. This proves a genuine coefficient-level influence of $\log 3$. It does not, by itself, prove that the radial histogram must be bimodal; that finite causal claim requires the matched-frequency null comparison provided in the MATLAB annex.

The finite Möbius interpolation is exact and independent of the sharp-mode conjectures. It is determined by

$$\Phi_N(s) = M(N) + h_N \frac{R_N(s)}{P_N(s)}, \tag{43.3}$$

with $\Phi_N(0) = -2$ and $\Phi_N(\sigma) \to M(N)$. The least nonzero tail coefficient cannot be cancelled by later terms on the positive real axis; this proves both the exponential rate and the recovery formula for a_N. The fact that every feasible affine interpolation converges to $1/\zeta(s)$ in $\Re s > 1$ also shows exactly what the construction does not do: absolute convergence alone does not select the true terminal Möbius value.

The seven RH routes have the following final status.

Route I. Equality of complete and ordinate-only atomic measures is exactly equivalent to RH for one fixed $q > 1$. The missing theorem is local prime-side identification of the jumps; a direct RVM implementation needs control of $S^{\#} + R_\vartheta$ on atom-resolving windows.

Route II. Local distributional convergence is sufficient. The missing estimate is local control of the phase-displacement sums and simultaneous identification of the prime-side derivative; global normalized convergence cannot detect fixed atoms.

Route III. Small-q curvature recovers the horizontal second moment at each ordinate. The missing step is a twice-differentiable prime-side formula uniform at $q = 1$, with all limit interchanges and the endpoint atom justified.

Route IV. Large-q growth recovers the maximal horizontal displacement. The missing step is a local prime-side bound uniform in q. Bounds for RVM ordinate displacements are irrelevant to this horizontal criterion and are not equivalent to RH.

Route V. A family of frequencies determines the finite horizontal multiset by ordinary exponential-polynomial uniqueness. The missing theorem is not uniqueness, but local reconstruction of that function from prime-side data.

Route VI. Local energy exhaustion plus derivative identification is sufficient. The observed global variance is not the required local energy, and the sufficient hypothesis is not claimed to be equivalent to RH.

Route VII. Positive tests eliminate the nonnegative atomic defect. The missing object is a test family that is both atom-sensitive and controlled by a prime-power explicit formula; abstract positive tests already exist.

These routes must not be described as seven completed approaches to RH. What is complete is the logical reduction in each case. Route I isolates an atom-resolving measure identity; Route II isolates local distributional control on the natural phase scale; Route III isolates a twice-differentiable small-q prime-side formula; Route IV isolates a uniform large-q growth theorem; Route V separates the already-proved uniqueness theorem from the open reconstruction premise; Route VI shows why global mean-square decay is insufficient without local derivative identification; and Route VII reduces the problem to constructing positive tests that remain compatible with a controlled explicit formula. Claims that these missing estimates are already equivalent to Montgomery's conjecture, or that

bounds such as $\sum \alpha_\rho^2 \ll T^\varepsilon$ are independently equivalent to RH, are not made: such equivalences require additional hypotheses and are not established here.

The exact finite-sample formulas also settle the diagonal issue. The reported variance is the sum of projected background energy, a mixed background–prime term, within-tower blocks, and cross-tower blocks. An ordinary Parseval diagonal is not the diagonal of this system. The remaining analytic problem is to estimate these four contributions in a joint height–cutoff–regularization regime while retaining enough locality to identify the positive atomic defect.

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Appendix A. Complete MATLAB implementation

Remark A.1. The program below is the self-contained implementation used for the prime-power audit, augmented by the matched-amplitude non-arithmetic null test of Section 36.3. It reads the original binary zero file, generates prime powers internally, retains all figures, and performs the joint constant–carrier fit. The listings use no frames or line-number columns.

Listing 1. Complete MATLAB implementation, including the matched-frequency $n = 3$ null control.

```
%% ZETA_PRIME_POWER_ANALYTIC_AUDIT
%
% Script autocontenido para los seis tests de la descomposicion
%
%   S_q(T) = sum_{gamma_j <= T} exp(i*gamma_j*log(q))
%
% No usa ningun .mat, .csv ni resultado anterior.
```

```
% El unico dato externo necesario es el fichero original de ordenadas:
%
%   zeross.bin
%
% Si no esta en la carpeta actual, MATLAB abre un selector para localizarlo.
%
% No resta 0.5*exp(i*T). Ajusta conjuntamente una constante aditiva y
% el coeficiente de exp(i*log(q)*T). La constante se elimina antes de
% demodular.
% Todos los resumenes principales se muestran con fprintf/disp.
% No se cierra ninguna figura.
%
% TESTS
% -----
% 1. Separacion P_Q^- y P_Q^+ y termino cruzado.
% 2. Identidades de c_n^-, c_n^+ y sumas A_Q, B_Q.
% 3. Busqueda de telescopado dentro de p,p^2,...,p^R.
% 4. Funcionales positivos y matrices de Gram por bloques.
% 5. Ecuacion funcional del residuo demodulado F_Q.
% 6. Unicidad: bloques, anisotropia, gaps y frecuencias omitidas.
%
% Requiere una version de MATLAB que permita funciones locales al final
% de un script (R2016b o posterior).

clear;
clc;
format long g;

%% ------------------------------------------------------------------------
% CONFIGURACION
% ------------------------------------------------------------------------

zeroBinaryFile = 'zeross.bin';

% q=e reproduce exp(i*gamma).
qPhase = exp(1);
omega = log(qPhase);

% stride=10: usa todos los ceros para construir S_q, pero analiza 1 de cada 10.
% stride=1: analiza los 2,001,052 puntos completos, mucho mas lento para Q=10000.
stride = 1;

Qlist = [0, 3, 10, 100, 1000, 10000];
Qlist = unique(sort(Qlist(:).'));
Qmax = max(Qlist);

numHeightBlocks = 24;
numGramBlocks = 20;
maxGramPoints = 60000;

omittedTestWidth = 500;
maxSpectralPoints = 50000;
numFrequencyGridPoints = 401;
```

```
gapMedianWindow = 1001;
resonanceTolerance = 1e-12;
minimumVariance = 1e-30;

% Si se pone true, guarda tablas y MAT al final. No son necesarios para correr.
saveResults = false;
outputDirectory = 'ZETA_PRIME_POWER_STRUCTURE_RESULTS';

%% ------------------------------------------------------------------------
% LOCALIZAR Y LEER zeross.bin
% ------------------------------------------------------------------------

if ~isfile(zeroBinaryFile)
    [selectedFile,selectedPath] = uigetfile( ...
        {'*.bin','Binary doubles (*.bin)';'*.*','Todos los archivos'}, ...
        'Selecciona zeross.bin');

    if isequal(selectedFile,0)
        error(['No se encontro zeross.bin y no se selecciono ningun archivo. ' ...
            'Este es el unico fichero de datos necesario.']);
    end

    zeroBinaryFile = fullfile(selectedPath,selectedFile);
end

fileID = fopen(zeroBinaryFile,'rb');
if fileID == -1
    error('No se pudo abrir %s.',zeroBinaryFile);
end

cleanupObject = onCleanup(@() fclose(fileID));
gammaAll = fread(fileID,Inf,'double=>double');
clear cleanupObject;

gammaAll = gammaAll(:);
gammaAll = gammaAll(isfinite(gammaAll) & gammaAll>0);

if numel(gammaAll)<2
    error('El fichero no contiene suficientes ordenadas validas.');
end

if any(diff(gammaAll)<=0)
    error('Las ordenadas no estan estrictamente ordenadas.');
end

Nfull = numel(gammaAll);

fprintf('\n------------------------------------------------------------\n');
fprintf('ZETA PRIME-POWER STRUCTURE: ANALYTIC AUDIT\n');
fprintf('------------------------------------------------------------\n');
fprintf('Archivo de ceros       : %s\n',zeroBinaryFile);
fprintf('Ceros disponibles      : %d\n',Nfull);
fprintf('q                      : %.17g\n',qPhase);
fprintf('omega=log(q)           : %.17g\n',omega);
```

```
fprintf('stride                 : %d\n',stride);
fprintf('Qlist                  :');
fprintf(' %g',Qlist);
fprintf('\n');

%% ------------------------------------------------------------------------
% CONSTRUIR S_q CON TODOS LOS CEROS; MUESTREAR DESPUES
% ------------------------------------------------------------------------

carrierAll = exp(1i*omega*gammaAll);
SphaseAll = cumsum(carrierAll);

% Parte suave analitica, salvo constante aditiva:
%
% int exp(i*omega*T)/(2*pi)*log(T/(2*pi)) dT
%
% = [exp(i omega T)log(T/2pi)-Ei(i omega T)]/(2*pi*i*omega).
%
% MATLAB: Ei(z) = -expint(-z), salvo la constante de rama, que solo
% modifica una constante aditiva y se absorbe en los ajustes.
Ei_iomegaT_all = -expint(-1i*omega*gammaAll);
AsmoothAll = (carrierAll.*log(gammaAll/(2*pi))-Ei_iomegaT_all) ...
    /(2*pi*1i*omega);

indices = (1:stride:Nfull).';
gamma = gammaAll(indices);
carrier = carrierAll(indices);
Sphase = SphaseAll(indices);
Asmooth = AsmoothAll(indices);

gapBeforeAll = [NaN;diff(gammaAll)];
gapLocalMedianAll = movmedian(gapBeforeAll,gapMedianWindow, ...
    'omitmissing','Endpoints','shrink');
gapResidualAll = gapBeforeAll-gapLocalMedianAll;

gapBefore = gapBeforeAll(indices);
gapResidual = gapResidualAll(indices);

clear carrierAll SphaseAll AsmoothAll Ei_iomegaT_all;
clear gapBeforeAll gapLocalMedianAll gapResidualAll;

Y = Sphase-Asmooth;
N = numel(gamma);

fprintf('Puntos analizados       : %d\n',N);
fprintf('gamma inicial           : %.15g\n',gamma(1));
fprintf('gamma final             : %.15g\n',gamma(end));

%% ------------------------------------------------------------------------
% GENERAR TODAS LAS POTENCIAS PRIMAS <= Qmax
% ------------------------------------------------------------------------

[qVec,basePrimeVec,powerExponentVec,lambdaVec] = makePrimePowers(Qmax);
numTerms = numel(qVec);
```

```
ellVec = log(qVec);
deltaMinusVec = omega-ellVec;
deltaPlusVec = omega+ellVec;

resonantIndex = find(abs(deltaMinusVec)<=resonanceTolerance,1);
if ~isempty(resonantIndex)
    error(['qPhase es resonante con n=%d. Esta version no oculta la ' ...
        'singularidad: debe introducirse el termino resonante regularizado.'], ...
        qVec(resonantIndex));
end

cMinusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaMinusVec);

cPlusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaPlusVec);

%% ------------------------------------------------------------------------
% TEST 2: IDENTIDADES ALGEBRAICAS DE LOS COEFICIENTES
% ------------------------------------------------------------------------

sumIdentityRHS = 1i*lambdaVec.*omega ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

differenceIdentityRHS = 1i*lambdaVec.*ellVec ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

ratioIdentityRHS = (omega+ellVec)./(omega-ellVec);

sumIdentityError = abs((cMinusVec+cPlusVec)-sumIdentityRHS);
differenceIdentityError = abs((cMinusVec-cPlusVec)-differenceIdentityRHS);
ratioIdentityError = abs((cMinusVec./cPlusVec)-ratioIdentityRHS);

AQ = cumsum(cMinusVec+cPlusVec);
BQ = cumsum(cMinusVec-cPlusVec);

fprintf('\n-------------------------------------------------------------\n');
fprintf('TEST 2: IDENTIDADES DE COEFICIENTES\n');
fprintf('-------------------------------------------------------------\n');
fprintf('Potencias primas <=Qmax                  : %d\n',numTerms);
fprintf('max |(c^-+c^+)-formula|                : %.6e\n', ...
    max(sumIdentityError));
fprintf('max |(c^--c^+)-formula|                : %.6e\n', ...
    max(differenceIdentityError));
fprintf('max |c^-/c^+-formula|                  : %.6e\n', ...
    max(ratioIdentityError));

coefficientCutoffRows = cell(numel(Qlist),9);

for jj = 1:numel(Qlist)
    currentQ = Qlist(jj);
    lastIndex = find(qVec<=currentQ,1,'last');
```

```
    if isempty(lastIndex)
        currentA = 0;
        currentB = 0;
        numberTerms = 0;
    else
        currentA = AQ(lastIndex);
        currentB = BQ(lastIndex);
        numberTerms = lastIndex;
    end

    coefficientCutoffRows(jj,:) = { ...
        currentQ,numberTerms, ...
        real(currentA),imag(currentA),abs(currentA-0.5), ...
        real(currentB),imag(currentB),abs(currentB-0.5), ...
        abs(currentB+0.5)};
end

coefficientCutoffDisplay = cell2table(coefficientCutoffRows, ...
    'VariableNames',{ ...
    'Q','num_terms','Re_AQ','Im_AQ','abs_AQ_minus_half', ...
    'Re_BQ','Im_BQ','abs_BQ_minus_half','abs_BQ_plus_half'});

disp(coefficientCutoffDisplay);


%% ------------------------------------------------------------------------
% AUDITORIA RADIAL DIRECTA: ORIGINAL, n=3, n=2+3
% ------------------------------------------------------------------------

fprintf('\n------------------------------------------------------------\n');
fprintf('AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD\n');
fprintf('------------------------------------------------------------\n');

numRadialBins = 220;
radialSmoothWidth = 9;

% Control nulo de frecuencia para el modo n=3.
numMatchedNulls = 199;
matchedNullSeed = 314159265;
matchedNullMaxPoints = 100000;
matchedNullHalfWidth = 0.25;
matchedNullPrimeLogExclusion = 5e-4;

index3 = find(qVec==3,1);
index23 = find(qVec<=3);

if isempty(index3)
    error('No se encontro n=3 entre las potencias primas.');
end

ell3 = ellVec(index3);
P3minus = cMinusVec(index3).*carrier.*exp(-1i*ell3*gamma);
P3plus = cPlusVec(index3).*carrier.*exp(1i*ell3*gamma);
P3 = P3minus+P3plus;
```

```
P23 = complex(zeros(N,1));
for auditIndex = index23(:).'
    ellAudit = ellVec(auditIndex);
    P23 = P23 ...
        +cMinusVec(auditIndex).*carrier.*exp(-1i*ellAudit*gamma) ...
        +cPlusVec(auditIndex).*carrier.*exp(1i*ellAudit*gamma);
end

[radialOriginal,radialCentersOriginal,radialDensityOriginal, ...
    radialPeaksOriginal] = radialDistributionSummary( ...
    Sphase,numRadialBins,radialSmoothWidth);

[radialAfter3,radialCentersAfter3,radialDensityAfter3, ...
    radialPeaksAfter3] = radialDistributionSummary( ...
    Sphase-P3,numRadialBins,radialSmoothWidth);

[radialAfter23,radialCentersAfter23,radialDensityAfter23, ...
    radialPeaksAfter23] = radialDistributionSummary( ...
    Sphase-P23,numRadialBins,radialSmoothWidth);

radialAuditTable = table( ...
    ["original";"after_n3";"after_n2_n3"], ...
    [mean(radialOriginal);mean(radialAfter3);mean(radialAfter23)], ...
    [std(radialOriginal);std(radialAfter3);std(radialAfter23)], ...
    [numel(radialPeaksOriginal);numel(radialPeaksAfter3); ...
        numel(radialPeaksAfter23)], ...
    [joinNumericVector(radialPeaksOriginal); ...
        joinNumericVector(radialPeaksAfter3); ...
        joinNumericVector(radialPeaksAfter23)], ...
    'VariableNames',{'case_name','mean_radius','std_radius', ...
    'num_detected_peaks','detected_peak_radii'});

disp(radialAuditTable);

fprintf('Coeficiente teorico total n=3:\n');
fprintf('  c3^- = %.12g %+.12gi\n',real(cMinusVec(index3)), ...
    imag(cMinusVec(index3)));
fprintf('  c3^+ = %.12g %+.12gi\n',real(cPlusVec(index3)), ...
    imag(cPlusVec(index3)));

figure('Name','Auditoria radial: original y sustracciones pequenas', ...
    'Color','w');
plot(radialCentersOriginal,radialDensityOriginal,'LineWidth',1.3);
hold on;
plot(radialCentersAfter3,radialDensityAfter3,'LineWidth',1.3);
plot(radialCentersAfter23,radialDensityAfter23,'LineWidth',1.3);
grid on;
xlabel('Radio respecto del centro medio');
ylabel('Densidad suavizada');
legend('Original','Despues de n=3','Despues de n=2,3', ...
    'Location','best');
title('Prueba directa de la bimodalidad radial');
```

```
%% ----------------------------------------------------------------------
% CONTROL NULO DE AMPLITUD EMPAREJADA PARA n=3
% ----------------------------------------------------------------------
%
% Mantiene exactamente c3^- y c3^+, pero sustituye log(3) por frecuencias
% no aritmeticas cercanas. Para cada serie se aplica el mismo recentering
% global de muestra completa y el mismo detector de picos. El objetivo es comprobar si el colapso
    bimodal es especifico de
% log(3) o si cualquier oscilador de amplitud comparable produce lo mismo.

fprintf('\n-------------------------------------------------------------\n');
fprintf('CONTROL NULO EMPAREJADO PARA EL MODO n=3\n');
fprintf('-------------------------------------------------------------\n');

rng(matchedNullSeed,'twister');
nullAuditIndices = unique(round(linspace(1,N,min(N,matchedNullMaxPoints)))).';
gammaNull = gamma(nullAuditIndices);
carrierNull = carrier(nullAuditIndices);
SphaseNull = Sphase(nullAuditIndices);
P3NullSample = P3(nullAuditIndices);

[radialOriginalNull,~,~,radialPeaksOriginalNull] = ...
    radialDistributionSummary(SphaseNull,numRadialBins,radialSmoothWidth);
[radialAfter3Null,~,~,radialPeaksAfter3Null] = ...
    radialDistributionSummary(SphaseNull-P3NullSample, ...
    numRadialBins,radialSmoothWidth);

observedNullAuditVarianceReduction = 1- ...
    var(radialAfter3Null,1)/max(var(radialOriginalNull,1),eps);
observedNullAuditPeakCount = numel(radialPeaksAfter3Null);

nullEll = zeros(numMatchedNulls,1);
nullVarianceReduction = zeros(numMatchedNulls,1);
nullPeakCount = zeros(numMatchedNulls,1);

ellLow = max(log(2),ell3-matchedNullHalfWidth);
ellHigh = min(log(4),ell3+matchedNullHalfWidth);
accepted = 0;
maxAttempts = 100000;
attempts = 0;

while accepted<numMatchedNulls && attempts<maxAttempts
    attempts = attempts+1;
    ellCandidate = ellLow+(ellHigh-ellLow)*rand;

    if min(abs(ellCandidate-ellVec))<=matchedNullPrimeLogExclusion
        continue;
    end

    accepted = accepted+1;
    nullEll(accepted) = ellCandidate;

    Pnull = cMinusVec(index3).*carrierNull.*exp(-1i*ellCandidate*gammaNull) ...
```

```
        +cPlusVec(index3).*carrierNull.*exp(1i*ellCandidate*gammaNull);

    [radialAfterNull,~,~,peaksAfterNull] = ...
        radialDistributionSummary(SphaseNull-Pnull, ...
        numRadialBins,radialSmoothWidth);

    nullVarianceReduction(accepted) = 1- ...
        var(radialAfterNull,1)/max(var(radialOriginalNull,1),eps);
    nullPeakCount(accepted) = numel(peaksAfterNull);
end

if accepted<numMatchedNulls
    error('No se pudieron generar suficientes frecuencias nulas admisibles.');
end

empiricalPVariance = (1+sum(nullVarianceReduction>= ...
    observedNullAuditVarianceReduction))/(numMatchedNulls+1);
empiricalPUnimodal = (1+sum(nullPeakCount<= ...
    observedNullAuditPeakCount))/(numMatchedNulls+1);

matchedNullTable = table( ...
    nullEll,nullVarianceReduction,nullPeakCount, ...
    'VariableNames',{'ell_null','radial_variance_reduction', ...
    'detected_peak_count'});

fprintf('Puntos usados en el control nulo       : %d\n',numel(nullAuditIndices));
fprintf('Numero de frecuencias nulas            : %d\n',numMatchedNulls);
fprintf('Reduccion observada al quitar n=3      : %.12g\n', ...
    observedNullAuditVarianceReduction);
fprintf('Picos observados despues de quitar n=3 : %d\n', ...
    observedNullAuditPeakCount);
fprintf('Mediana reduccion nula                 : %.12g\n', ...
    median(nullVarianceReduction));
fprintf('Maxima reduccion nula                  : %.12g\n', ...
    max(nullVarianceReduction));
fprintf('p empirica reduccion >= observada      : %.12g\n', ...
    empiricalPVariance);
fprintf('p empirica picos <= observado          : %.12g\n', ...
    empiricalPUnimodal);

%% ------------------------------------------------------------------------
% PREPARAR MATRIZ DE GRAM POR BLOQUES LOGARITMICOS DE n
% ------------------------------------------------------------------------

gramIndices = unique(round(linspace(1,N,min(N,maxGramPoints)))).';
numGramPoints = numel(gramIndices);

logEdges = linspace(log(2),log(Qmax)+100*eps(log(Qmax)), ...
    numGramBlocks+1);

gramBlockOfTerm = discretize(ellVec,logEdges);
gramBlockOfTerm(isnan(gramBlockOfTerm)) = numGramBlocks;

gramMinus = complex(zeros(numGramPoints,numGramBlocks,'single'));
```

```
gramPlus = complex(zeros(numGramPoints,numGramBlocks,'single'));

%% ------------------------------------------------------------------------
% BUCLE PRINCIPAL: TESTS 1, 4, 5 Y PARTE DE 6
% ------------------------------------------------------------------------

Pminus = complex(zeros(N,1));
Pplus = complex(zeros(N,1));

nextTerm = 1;
maxIncrementIdentityError = 0;
maxConstantCancellationIdentityError = 0;

summaryRows = cell(numel(Qlist),43);
positivityRows = cell(numel(Qlist),17);

blockRows = cell(numel(Qlist)*numHeightBlocks,16);
blockRow = 0;

[constantY,cY] = fitConstantAndCarrier(Y,carrier);

RbothAtQmax = [];
FAtQmax = [];
EAtQmax = [];
PminusAtQmax = [];
PplusAtQmax = [];
cBothAtQmax = NaN;

for qi = 1:numel(Qlist)

    Q = Qlist(qi);

    while nextTerm<=numTerms && qVec(nextTerm)<=Q

        ell = ellVec(nextTerm);
        z = exp(1i*ell*gamma);

        termMinus = cMinusVec(nextTerm).*carrier.*conj(z);
        termPlus = cPlusVec(nextTerm).*carrier.*z;

        oldCombinedResidual = Y-Pminus-Pplus;

        Pminus = Pminus+termMinus;
        Pplus = Pplus+termPlus;

        newCombinedResidual = Y-Pminus-Pplus;

        incrementError = max(abs( ...
            (newCombinedResidual-oldCombinedResidual) ...
            +termMinus+termPlus));

        maxIncrementIdentityError = max( ...
            maxIncrementIdentityError,incrementError);
```

```
        gramBlock = gramBlockOfTerm(nextTerm);

        gramMinus(:,gramBlock) = gramMinus(:,gramBlock) ...
            +single(termMinus(gramIndices));

        gramPlus(:,gramBlock) = gramPlus(:,gramBlock) ...
            +single(termPlus(gramIndices));

        nextTerm = nextTerm+1;
    end

    numTermsUsed = nextTerm-1;

    % Cuatro residuos pedidos.
    RminusOnlyRaw = Y-Pminus;
    RplusOnlyRaw = Y-Pplus;
    RbothRaw = Y-Pminus-Pplus;

    % CORRECCION: la parte suave esta definida salvo constante aditiva.
    % Se ajustan [constante + portadora] conjuntamente en cada residuo.
    [constantMinus,cMinusResidual] = ...
        fitConstantAndCarrier(RminusOnlyRaw,carrier);

    [constantPlus,cPlusResidual] = ...
        fitConstantAndCarrier(RplusOnlyRaw,carrier);

    [jointConstant,cBothResidual] = ...
        fitConstantAndCarrier(RbothRaw,carrier);

    RminusOnly = RminusOnlyRaw-constantMinus;
    RplusOnly = RplusOnlyRaw-constantPlus;
    Rboth = RbothRaw-jointConstant;

    E = Rboth-cBothResidual*carrier;
    F = conj(carrier).*Rboth;
    Fc = F-cBothResidual;

    [constantPminus,projectionMinus] = ...
        fitConstantAndCarrier(Pminus,carrier);

    [constantPplus,projectionPlus] = ...
        fitConstantAndCarrier(Pplus,carrier);

    carrierCancellationIdentityError = abs( ...
        cBothResidual-(cY-projectionMinus-projectionPlus));

    constantCancellationIdentityError = abs( ...
        jointConstant-(constantY-constantPminus-constantPplus));

    maxConstantCancellationIdentityError = max( ...
        maxConstantCancellationIdentityError, ...
        constantCancellationIdentityError);

    % TEST 1: energias y termino cruzado.
```

```
energyMinus = mean(abs(Pminus).^2);
energyPlus = mean(abs(Pplus).^2);
crossMinusPlus = 2*real(mean(conj(Pminus).*Pplus));
energyCombined = mean(abs(Pminus+Pplus).^2);

crossEnergyIdentityError = abs( ...
    energyCombined-(energyMinus+energyPlus+crossMinusPlus));

% Forma centrada.
PminusCentered = Pminus-mean(Pminus);
PplusCentered = Pplus-mean(Pplus);

centeredEnergyMinus = mean(abs(PminusCentered).^2);
centeredEnergyPlus = mean(abs(PplusCentered).^2);

centeredCross = 2*real(mean( ...
    conj(PminusCentered).*PplusCentered));

centeredEnergyCombined = mean(abs( ...
    PminusCentered+PplusCentered).^2);

% TEST 5: ecuacion funcional discreta y diferencial.
dgamma = diff(gamma);

derivativeDefect = diff(Rboth)./dgamma ...
    -1i*omega*0.5*(Rboth(2:end)+Rboth(1:end-1));

discreteDefect = Rboth(2:end) ...
    -exp(1i*omega*dgamma).*Rboth(1:end-1);

Fvariance = mean(abs(Fc).^2);
FmaxDeviation = max(abs(Fc));
FmeanAbsIncrement = mean(abs(diff(F)));
FtotalVariation = sum(abs(diff(F)));
Fanisotropy = complexAnisotropy(Fc,minimumVariance);

corrGapReal = safeCorrelation(real(Fc),gapBefore);
corrGapImag = safeCorrelation(imag(Fc),gapBefore);
corrGapAbs = safeCorrelation(abs(Fc),gapBefore);

corrGapResidualReal = safeCorrelation(real(Fc),gapResidual);
corrGapResidualImag = safeCorrelation(imag(Fc),gapResidual);
corrGapResidualAbs = safeCorrelation(abs(Fc),gapResidual);

% Bloques de altura.
heightEdges = round(linspace(1,N+1,numHeightBlocks+1));
blockMeans = complex(NaN(numHeightBlocks,1));

for hb = 1:numHeightBlocks

    idx = heightEdges(hb):(heightEdges(hb+1)-1);

    Fb = F(idx);
    Rb = Rboth(idx);
```

```
        blockMean = mean(Fb);
        blockMeans(hb) = blockMean;
        Fbc = Fb-blockMean;

        if numel(idx)>=2
            dgb = diff(gamma(idx));

            discreteBlock = Rb(2:end) ...
                -exp(1i*omega*dgb).*Rb(1:end-1);

            derivativeBlock = diff(Rb)./dgb ...
                -1i*omega*0.5*(Rb(2:end)+Rb(1:end-1));

            meanAbsDeltaBlock = mean(abs(diff(Fb)));

            variationPerHeight = sum(abs(diff(Fb))) ...
                /max(gamma(idx(end))-gamma(idx(1)),eps);

            discreteBlockRMS = sqrt(mean(abs(discreteBlock).^2));
            derivativeBlockRMS = sqrt(mean(abs(derivativeBlock).^2));
        else
            meanAbsDeltaBlock = NaN;
            variationPerHeight = NaN;
            discreteBlockRMS = NaN;
            derivativeBlockRMS = NaN;
        end

        blockRow = blockRow+1;

        blockRows(blockRow,:) = { ...
            Q,hb,indices(idx(1)),indices(idx(end)), ...
            gamma(idx(1)),gamma(idx(end)), ...
            real(blockMean),imag(blockMean),mean(abs(Fbc).^2), ...
            max(abs(Fb-cBothResidual)),max(abs(Fbc)), ...
            meanAbsDeltaBlock,variationPerHeight, ...
            complexAnisotropy(Fbc,minimumVariance), ...
            discreteBlockRMS,derivativeBlockRMS};
    end

    FblockMeanStd = sqrt(mean(abs(blockMeans-mean(blockMeans)).^2));

    % TEST 4: positividad elemental/Gram.
    normE2 = mean(abs(E).^2);
    normCarrier2 = mean(abs(carrier).^2);

    innerCarrierE = mean(conj(carrier).*E);
    innerEMinus = mean(conj(E).*Pminus);
    innerEPlus = mean(conj(E).*Pplus);
    innerMinusPlus = mean(conj(Pminus).*Pplus);

    carrierSchur = mean(abs(Rboth).^2) ...
        -abs(mean(conj(carrier).*Rboth))^2/normCarrier2;

    cauchyDetEMinus = normE2*energyMinus-abs(innerEMinus)^2;
```

```
cauchyDetEPlus = normE2*energyPlus-abs(innerEPlus)^2;
cauchyDetMinusPlus = energyMinus*energyPlus ...
    -abs(innerMinusPlus)^2;

QEplusMinus = mean(abs(E+Pminus).^2);
QEminusMinus = mean(abs(E-Pminus).^2);

polarizationErrorMinus = abs( ...
    0.5*(QEplusMinus+QEminusMinus)-normE2-energyMinus);

QEplusPlus = mean(abs(E+Pplus).^2);
QEminusPlus = mean(abs(E-Pplus).^2);

polarizationErrorPlus = abs( ...
    0.5*(QEplusPlus+QEminusPlus)-normE2-energyPlus);

positivityRows(qi,:) = { ...
    Q,normE2,carrierSchur, ...
    abs(innerCarrierE),real(innerCarrierE),imag(innerCarrierE), ...
    real(innerEMinus),real(innerEPlus), ...
    cauchyDetEMinus,cauchyDetEPlus,cauchyDetMinusPlus, ...
    QEplusMinus,QEminusMinus,polarizationErrorMinus, ...
    QEplusPlus,QEminusPlus,polarizationErrorPlus};

if numTermsUsed>0
    currentA = AQ(numTermsUsed);
    currentB = BQ(numTermsUsed);
else
    currentA = 0;
    currentB = 0;
end

summaryRows(qi,:) = { ...
    Q,numTermsUsed, ...
    real(cMinusResidual),imag(cMinusResidual), ...
    real(cPlusResidual),imag(cPlusResidual), ...
    real(cBothResidual),imag(cBothResidual),abs(cBothResidual), ...
    abs(cBothResidual-0.5), ...
    real(jointConstant),imag(jointConstant), ...
    real(cBothResidual),imag(cBothResidual),abs(cBothResidual-0.5), ...
    real(projectionMinus),imag(projectionMinus), ...
    real(projectionPlus),imag(projectionPlus), ...
    carrierCancellationIdentityError, ...
    sqrt(mean(abs(RminusOnly).^2)), ...
    sqrt(mean(abs(RplusOnly).^2)), ...
    sqrt(mean(abs(Rboth).^2)),sqrt(normE2), ...
    energyMinus,energyPlus,crossMinusPlus,energyCombined, ...
    crossEnergyIdentityError, ...
    centeredEnergyMinus,centeredEnergyPlus,centeredCross, ...
    centeredEnergyCombined, ...
    Fvariance,FmaxDeviation,FmeanAbsIncrement,FtotalVariation, ...
    FblockMeanStd,Fanisotropy, ...
    sqrt(mean(abs(discreteDefect).^2)), ...
    sqrt(mean(abs(derivativeDefect).^2)), ...
```

```
        real(currentA),imag(currentA)};

    fprintf(['Q=%-6g terms=%-5d  c^-res=%.7f%+.7fi  ' ...
        'c^+res=%.7f%+.7fi  c^both=%.7f%+.7fi  ' ...
        'Var(F)=%.4e  RMS(E)=%.4e\n'], ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual), ...
        Fvariance,sqrt(normE2));

    if Q==Qmax
        RbothAtQmax = Rboth;
        FAtQmax = F;
        EAtQmax = E;
        PminusAtQmax = Pminus;
        PplusAtQmax = Pplus;
        cBothAtQmax = cBothResidual;
    end
end

summaryVariableNames = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag','c_after_both_abs', ...
    'distance_c_after_both_to_half', ...
    'joint_constant_real','joint_constant_imag', ...
    'joint_carrier_real','joint_carrier_imag', ...
    'distance_joint_carrier_to_half', ...
    'projection_Pminus_real','projection_Pminus_imag', ...
    'projection_Pplus_real','projection_Pplus_imag', ...
    'carrier_cancellation_identity_error', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'energy_Pminus_plus_Pplus','cross_energy_identity_error', ...
    'centered_energy_Pminus','centered_energy_Pplus', ...
    'centered_cross_term','centered_energy_combined', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_mean_abs_increment','F_total_variation', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms', ...
    'A_Q_real','A_Q_imag'};

summaryTable = cell2table(summaryRows, ...
    'VariableNames',summaryVariableNames);

positivityVariableNames = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','carrier_projection_real', ...
    'carrier_projection_imag','Re_inner_E_Pminus', ...
```

```
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'Q_E_plus_Pminus','Q_E_minus_Pminus', ...
    'polarization_error_minus','Q_E_plus_Pplus', ...
    'Q_E_minus_Pplus','polarization_error_plus'};

positivityTable = cell2table(positivityRows, ...
    'VariableNames',positivityVariableNames);

blockVariableNames = { ...
    'Q','block','k_start','k_end','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

blockTable = cell2table(blockRows(1:blockRow,:), ...
    'VariableNames',blockVariableNames);

%% ------------------------------------------------------------------------
% DISPLAY PRINCIPAL: TEST 1 Y TEST 5
% ------------------------------------------------------------------------

fprintf('\n----------------------------------------------------------------\n');
fprintf('TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL\n');
fprintf('----------------------------------------------------------------\n');

mainDisplayVariables = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag', ...
    'distance_c_after_both_to_half', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(summaryTable(:,mainDisplayVariables));

fprintf('max error R_j-R_{j-1}=-P_j                  : %.6e\n', ...
    maxIncrementIdentityError);

fprintf('max error ley de energia de ramas           : %.6e\n', ...
    max(summaryTable.cross_energy_identity_error));

fprintf('max error identidad de proyeccion portadora : %.6e\n', ...
    max(summaryTable.carrier_cancellation_identity_error));
fprintf('max error identidad de constantes            : %.6e\n', ...
    maxConstantCancellationIdentityError);
```

```
%% ------------------------------------------------------------------------
% TEST 3: TELESCOPADO DENTRO DE CADA PRIMO p
% ------------------------------------------------------------------------

[groupSummaryTable,groupPartialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier);

fprintf('\n----------------------------------------------------------------\n');
fprintf('TEST 3: TELESCOPADO EN p,p^2,...,p^R\n');
fprintf('----------------------------------------------------------------\n');

if isempty(groupSummaryTable)
    fprintf('No hay primos con al menos dos potencias <=Qmax.\n');
else
    [~,orderTelescoping] = sort( ...
        groupSummaryTable.boundary_fit_relative_rms_combined,'ascend');

    numDisplay = min(20,height(groupSummaryTable));

    telescopingDisplayVariables = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
        'boundary_alpha_first_real','boundary_alpha_first_imag', ...
        'boundary_alpha_last_real','boundary_alpha_last_imag', ...
        'tail_primitive_start_abs_minus', ...
        'tail_primitive_start_abs_plus'};

    disp(groupSummaryTable( ...
        orderTelescoping(1:numDisplay), ...
        telescopingDisplayVariables));

    fprintf(['boundary_fit_relative_rms_combined cercano a cero seria ' ...
        'evidencia de colapso a los dos bordes del bloque p^r.\n']);
end

%% ------------------------------------------------------------------------
% TEST 4: FUNCIONALES POSITIVOS Y MATRICES DE GRAM
% ------------------------------------------------------------------------

Gminus = double(gramMinus);
Gplus = double(gramPlus);

GcombinedCentered = Gminus+Gplus;
GcombinedCentered = GcombinedCentered-mean(GcombinedCentered,1);

GbranchCentered = [ ...
    Gminus-mean(Gminus,1), ...
    Gplus-mean(Gplus,1)];
```

```
gramCombined = (GcombinedCentered'*GcombinedCentered)/numGramPoints;
gramBranch = (GbranchCentered'*GbranchCentered)/numGramPoints;

gramCombined = 0.5*(gramCombined+gramCombined');
gramBranch = 0.5*(gramBranch+gramBranch');

eigCombined = sort(real(eig(gramCombined)));
eigBranch = sort(real(eig(gramBranch)));

fprintf('\n------------------------------------------------------------\n');
fprintf('TEST 4: POSITIVIDAD Y GRAM\n');
fprintf('------------------------------------------------------------\n');

positivityDisplayVariables = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'polarization_error_minus','polarization_error_plus'};

disp(positivityTable(:,positivityDisplayVariables));

fprintf('Gram combinada: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigCombined),max(eigCombined),nnz(eigCombined<-1e-10));

fprintf('Gram por ramas: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigBranch),max(eigBranch),nnz(eigBranch<-1e-10));

%% ------------------------------------------------------------------------
% TEST 5 POR BLOQUES PARA Qmax
% ------------------------------------------------------------------------

blockQmax = blockTable(blockTable.Q==Qmax,:);

fprintf('\n------------------------------------------------------------\n');
fprintf('TEST 5: F_Q POR BLOQUES DE ALTURA, Q=%g\n',Qmax);
fprintf('------------------------------------------------------------\n');

blockDisplayVariables = { ...
    'block','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(blockQmax(:,blockDisplayVariables));

fprintf('Rango Re media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_real),max(blockQmax.F_mean_real));

fprintf('Rango Im media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_imag),max(blockQmax.F_mean_imag));
```

```
fprintf('Max sup global por bloque : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_global_c));

fprintf('Max sup local por bloque  : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_block_c));

%% ------------------------------------------------------------------------
% TEST 6: FRECUENCIAS OMITIDAS
% ------------------------------------------------------------------------

spectralIndices = unique(round(linspace(1,N, ...
    min(N,maxSpectralPoints)))).';

Ts = gamma(spectralIndices);
Fs = FAtQmax(spectralIndices)-cBothAtQmax;
Es = EAtQmax(spectralIndices);

testStart = Qmax+1;
testEnd = Qmax+omittedTestWidth;
nCandidates = (testStart:testEnd).';

[qTestAll,pTestAll,rTestAll,lambdaTestAll] = makePrimePowers(testEnd);

[isPrimePowerCandidate,locPP] = ismember(nCandidates,qTestAll);

candidateLambda = zeros(size(nCandidates));
candidateBasePrime = zeros(size(nCandidates));
candidateExponent = zeros(size(nCandidates));

candidateLambda(isPrimePowerCandidate) = ...
    lambdaTestAll(locPP(isPrimePowerCandidate));

candidateBasePrime(isPrimePowerCandidate) = ...
    pTestAll(locPP(isPrimePowerCandidate));

candidateExponent(isPrimePowerCandidate) = ...
    rTestAll(locPP(isPrimePowerCandidate));

omittedRows = cell(numel(nCandidates),19);

for jj = 1:numel(nCandidates)

    n = nCandidates(jj);
    ell = log(n);

    basisMinusF = exp(-1i*ell*Ts);
    basisPlusF = exp(1i*ell*Ts);

    fitMinusF = fitCarrier(Fs,basisMinusF);
    fitPlusF = fitCarrier(Fs,basisPlusF);

    basisMinusR = exp(1i*(omega-ell)*Ts);
    basisPlusR = exp(1i*(omega+ell)*Ts);
```

```
    fitMinusR = fitCarrier(Es,basisMinusR);
    fitPlusR = fitCarrier(Es,basisPlusR);

    theoryMinus = 0;
    theoryPlus = 0;

    if isPrimePowerCandidate(jj)
        lam = candidateLambda(jj);

        theoryMinus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryPlus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    omittedRows(jj,:) = { ...
        n,isPrimePowerCandidate(jj),candidateBasePrime(jj), ...
        candidateExponent(jj),candidateLambda(jj), ...
        real(fitMinusF),imag(fitMinusF),abs(fitMinusF), ...
        real(fitPlusF),imag(fitPlusF),abs(fitPlusF), ...
        real(fitMinusR),imag(fitMinusR), ...
        real(fitPlusR),imag(fitPlusR), ...
        real(theoryMinus),imag(theoryMinus), ...
        real(theoryPlus),imag(theoryPlus)};
end

omittedVariableNames = { ...
    'n','is_prime_power','base_prime','power_exponent','Lambda_n', ...
    'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
    'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
    'fit_R_minus_real','fit_R_minus_imag', ...
    'fit_R_plus_real','fit_R_plus_imag', ...
    'theory_minus_real','theory_minus_imag', ...
    'theory_plus_real','theory_plus_imag'};

omittedTable = cell2table(omittedRows, ...
    'VariableNames',omittedVariableNames);

ppMask = omittedTable.is_prime_power;
nullMask = ~ppMask;

medianPPMinus = median(omittedTable.fit_F_minus_abs(ppMask));
medianNullMinus = median(omittedTable.fit_F_minus_abs(nullMask));

medianPPPlus = median(omittedTable.fit_F_plus_abs(ppMask));
medianNullPlus = median(omittedTable.fit_F_plus_abs(nullMask));

omittedPPMask = qTestAll>Qmax & qTestAll<=testEnd;
omittedPP = qTestAll(omittedPPMask);
omittedPPLambda = lambdaTestAll(omittedPPMask);

Xomitted = complex(zeros(numel(Ts),2*numel(omittedPP)));
```

```
for jj = 1:numel(omittedPP)
    ell = log(omittedPP(jj));
    Xomitted(:,jj) = exp(-1i*ell*Ts);
    Xomitted(:,numel(omittedPP)+jj) = exp(1i*ell*Ts);
end

if isempty(Xomitted)
    fittedOmittedCoefficients = [];
    FsAfterOmittedFit = Fs;
    omittedExplainedFraction = NaN;
    fittedTheoryCorrelation = NaN;
    relativeTheoryCoefficientError = NaN;
else
    fittedOmittedCoefficients = Xomitted\Fs;
    FsAfterOmittedFit = Fs-Xomitted*fittedOmittedCoefficients;

    omittedExplainedFraction = 1 ...
        -sum(abs(FsAfterOmittedFit).^2)/sum(abs(Fs).^2);

    theoryOmittedCoefficients = complex(zeros(2*numel(omittedPP),1));

    for jj = 1:numel(omittedPP)

        n = omittedPP(jj);
        ell = log(n);
        lam = omittedPPLambda(jj);

        theoryOmittedCoefficients(jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryOmittedCoefficients(numel(omittedPP)+jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    fittedTheoryCorrelation = complexVectorCorrelation( ...
        fittedOmittedCoefficients,theoryOmittedCoefficients);

    relativeTheoryCoefficientError = norm( ...
        fittedOmittedCoefficients-theoryOmittedCoefficients) ...
        /max(norm(theoryOmittedCoefficients),eps);
end

nuMax = log(testEnd);
nuGrid = linspace(-nuMax,nuMax,numFrequencyGridPoints).';
frequencyAmplitude = zeros(size(nuGrid));

for jj = 1:numel(nuGrid)
    basis = exp(1i*nuGrid(jj)*Ts);
    frequencyAmplitude(jj) = abs(fitCarrier(Fs,basis));
end

[sortedAmplitude,sortOrder] = sort(frequencyAmplitude,'descend');
topCount = min(30,numel(sortOrder));
```

```
topFrequencyTable = table( ...
    (1:topCount).', ...
    nuGrid(sortOrder(1:topCount)), ...
    sortedAmplitude(1:topCount), ...
    'VariableNames',{'rank','demodulated_frequency','fit_amplitude'});

uniquenessTable = table( ...
    Qmax,numel(omittedPP), ...
    medianPPMinus,medianNullMinus,medianPPMinus/medianNullMinus, ...
    medianPPPlus,medianNullPlus,medianPPPlus/medianNullPlus, ...
    omittedExplainedFraction, ...
    real(fittedTheoryCorrelation),imag(fittedTheoryCorrelation), ...
    abs(fittedTheoryCorrelation),relativeTheoryCoefficientError, ...
    mean(abs(Fs).^2),mean(abs(FsAfterOmittedFit).^2), ...
    complexAnisotropy(Fs,minimumVariance), ...
    safeCorrelation(real(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(imag(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(abs(Fs),gapBefore(spectralIndices)), ...
    'VariableNames',{ ...
    'Qmax','num_omitted_prime_powers_joint_fit', ...
    'median_minus_amplitude_prime_power', ...
    'median_minus_amplitude_null', ...
    'minus_prime_power_to_null_ratio', ...
    'median_plus_amplitude_prime_power', ...
    'median_plus_amplitude_null', ...
    'plus_prime_power_to_null_ratio', ...
    'joint_omitted_explained_fraction', ...
    'fit_theory_correlation_real','fit_theory_correlation_imag', ...
    'fit_theory_correlation_abs','relative_theory_coefficient_error', ...
    'F_energy_before_omitted_fit','F_energy_after_omitted_fit', ...
    'F_anisotropy','corr_gap_realF','corr_gap_imagF','corr_gap_absF'});

fprintf('\n------------------------------------------------------------\n');
fprintf('TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS\n');
fprintf('------------------------------------------------------------\n');

disp(uniquenessTable);

omittedPrimePowerOnly = omittedTable(omittedTable.is_prime_power,:);

if ~isempty(omittedPrimePowerOnly)

    omittedPrimePowerOnly.max_branch_fit_abs = max( ...
        omittedPrimePowerOnly.fit_F_minus_abs, ...
        omittedPrimePowerOnly.fit_F_plus_abs);

    omittedPrimePowerOnly = sortrows( ...
        omittedPrimePowerOnly,'max_branch_fit_abs','descend');

    numDisplay = min(20,height(omittedPrimePowerOnly));

    fprintf('\nPotencias primas omitidas con mayor amplitud:\n');

    disp(omittedPrimePowerOnly(1:numDisplay,{ ...
```

```
        'n','base_prime','power_exponent','Lambda_n', ...
        'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
        'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
        'theory_minus_real','theory_minus_imag', ...
        'theory_plus_real','theory_plus_imag', ...
        'max_branch_fit_abs'}));
end

fprintf('\nFrecuencias no nulas dominantes de F_Q-mean(F_Q):\n');
disp(topFrequencyTable);

fprintf('\nDIAGNOSTICO FINAL PARA Q=%g\n',Qmax);
fprintf('mean(F_Q)                          = %.12g %+.12gi\n', ...
    real(cBothAtQmax),imag(cBothAtQmax));
fprintf('|mean(F_Q)-1/2|                    = %.6e\n', ...
    abs(cBothAtQmax-0.5));
fprintf('Var(F_Q)                           = %.6e\n', ...
    mean(abs(FAtQmax-cBothAtQmax).^2));
fprintf('sup |F_Q-mean(F_Q)|                = %.6e\n', ...
    max(abs(FAtQmax-cBothAtQmax)));
fprintf('RMS E_Q                            = %.6e\n', ...
    sqrt(mean(abs(EAtQmax).^2)));
fprintf('razon omitidas potencia/null (-) = %.6f\n', ...
    medianPPMinus/medianNullMinus);
fprintf('razon omitidas potencia/null (+) = %.6f\n', ...
    medianPPPlus/medianNullPlus);
fprintf('fraccion explicada ajuste omitido= %.6f\n', ...
    omittedExplainedFraction);

%% ------------------------------------------------------------------------
% FIGURAS: NO SE CIERRA NINGUNA
% ------------------------------------------------------------------------

figure('Name','Coeficientes de portadora por rama','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_real,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_real,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_real,'-o','LineWidth',1.2);
yline(0.5,':');
grid on;
xlabel('Q');
ylabel('Parte real');
legend('Y-P^-','Y-P^+','Y-P^--P^+','1/2','Location','best');
title('Coeficiente ajustado de exp(i omega T)');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_imag,'-o','LineWidth',1.2);
```

```
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_imag,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_imag,'-o','LineWidth',1.2);
yline(0,':');
grid on;
xlabel('Q');
ylabel('Parte imaginaria');
legend('Y-P^-','Y-P^+','Y-P^--P^+','0','Location','best');

figure('Name','Energia y termino cruzado de ramas','Color','w');
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pplus,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.cross_term_2Re_inner_Pminus_Pplus, ...
    '-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus_plus_Pplus,'-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Energia');
legend('||P^-||^2','||P^+||^2','2Re<P^-,P^+>', ...
    '||P^-+P^+||^2','Location','best');
title('Cancelacion entre ramas');

figure('Name','Ecuacion funcional del residuo','Color','w');
tiledlayout(2,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1),summaryTable.F_variance, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Var(F_Q)');
title('Dependencia residual de T');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.distance_c_after_both_to_half, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('|mean(F_Q)-1/2|');
title('Distancia del coeficiente a 1/2');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.carrier_discrete_defect_rms, ...
    '-o','LineWidth',1.2);
grid on;
```

```
xlabel('Q');
ylabel('RMS defecto discreto');
title('R_{k+1}-e^{i omega Delta T}R_k');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.F_block_mean_std, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('STD de medias por bloque');
title('Deriva por altura');

figure('Name','Sumas parciales A_Q y B_Q','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
plot(qVec,real(AQ),'LineWidth',1);
hold on;
plot(qVec,imag(AQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('A_Q');
legend('Re A_Q','Im A_Q','1/2','-1/2','Location','best');

nexttile;
plot(qVec,real(BQ),'LineWidth',1);
hold on;
plot(qVec,imag(BQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('B_Q');
legend('Re B_Q','Im B_Q','1/2','-1/2','Location','best');

blockSupMatrix = reshape( ...
    blockTable.sup_abs_F_minus_global_c, ...
    numHeightBlocks,numel(Qlist)).';

figure('Name','Desviacion suprema por bloques','Color','w');
imagesc(1:numHeightBlocks,1:numel(Qlist),log10(blockSupMatrix));
colorbar;
yticks(1:numel(Qlist));
yticklabels(string(Qlist));
xlabel('Bloque de altura');
ylabel('Q');
title('log10 sup_{T en I_j}|F_Q(T)-c_Q|');

figure('Name','Eigenvalores de Gram','Color','w');
```

```
tiledlayout(1,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogy(max(eigCombined,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram combinada');

nexttile;
semilogy(max(eigBranch,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram por ramas');

figure('Name','Frecuencias omitidas','Color','w');
semilogy(omittedTable.n,omittedTable.fit_F_minus_abs,'.');
hold on;
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_minus_abs(ppMask),'o');
semilogy(omittedTable.n,omittedTable.fit_F_plus_abs,'.');
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_plus_abs(ppMask),'s');
grid on;
xlabel('n');
ylabel('Amplitud ajustada');
legend('Rama - todos','Rama - potencias primas', ...
    'Rama + todos','Rama + potencias primas','Location','best');

figure('Name','Espectro no nulo de F_Q','Color','w');
plot(nuGrid,frequencyAmplitude,'LineWidth',1);
grid on;
xlabel('nu en exp(i nu T)');
ylabel('Amplitud');
title(sprintf('F_Q-mean(F_Q), Q=%g',Qmax));

%% ------------------------------------------------------------------------
% GUARDADO OPCIONAL; NUNCA ES NECESARIO PARA EJECUTAR
% ------------------------------------------------------------------------

if saveResults

    if ~isfolder(outputDirectory)
        mkdir(outputDirectory);
    end

    writetable(summaryTable, ...
        fullfile(outputDirectory,'SIX_TESTS_SUMMARY.csv'));

    writetable(coefficientCutoffDisplay, ...
        fullfile(outputDirectory,'COEFFICIENT_CUTOFF_SUMMARY.csv'));

    writetable(groupSummaryTable, ...
```

```
        fullfile(outputDirectory,'TELESCOPING_SUMMARY.csv'));

    writetable(groupPartialTable, ...
        fullfile(outputDirectory,'TELESCOPING_PARTIALS.csv'));

    writetable(positivityTable, ...
        fullfile(outputDirectory,'POSITIVITY_FUNCTIONALS.csv'));

    writetable(blockTable, ...
        fullfile(outputDirectory,'BLOCKWISE_FUNCTIONAL_EQUATION.csv'));

    writetable(omittedTable, ...
        fullfile(outputDirectory,'OMITTED_FREQUENCIES.csv'));

    writetable(uniquenessTable, ...
        fullfile(outputDirectory,'UNIQUENESS_SUMMARY.csv'));

    save(fullfile(outputDirectory,'ALL_RESULTS.mat'), ...
        'summaryTable','coefficientCutoffDisplay','radialAuditTable', ...
        'matchedNullTable', ...
        'groupSummaryTable','groupPartialTable', ...
        'positivityTable','blockTable','omittedTable', ...
        'uniquenessTable','topFrequencyTable', ...
        'gramCombined','gramBranch','eigCombined','eigBranch', ...
        'RbothAtQmax','FAtQmax','EAtQmax', ...
        'PminusAtQmax','PplusAtQmax','cBothAtQmax','-v7.3');

    fprintf('\nResultados guardados en %s\n',outputDirectory);
end

fprintf('\n------------------------------------------------------------\n');
fprintf('TERMINADO\n');
fprintf('Tablas disponibles directamente en el workspace:\n');
fprintf('  radialAuditTable\n');
fprintf('  matchedNullTable\n');
fprintf('  summaryTable\n');
fprintf('  coefficientCutoffDisplay\n');
fprintf('  groupSummaryTable\n');
fprintf('  groupPartialTable\n');
fprintf('  positivityTable\n');
fprintf('  blockQmax\n');
fprintf('  omittedTable\n');
fprintf('  uniquenessTable\n');
fprintf('  topFrequencyTable\n');
fprintf('------------------------------------------------------------\n');

%% ------------------------------------------------------------------------
% FUNCIONES LOCALES
% ------------------------------------------------------------------------

function c = fitCarrier(z,b)

    z = z(:);
    b = b(:);
```

```
    denominator = b'*b;

    if abs(denominator)<=eps
        c = NaN;
    else
        c = (b'*z)/denominator;
    end
end

function [a,c] = fitConstantAndCarrier(z,b)

    z = z(:);
    b = b(:);

    X = [ones(size(b)),b];
    coefficients = X\z;

    a = coefficients(1);
    c = coefficients(2);
end

function value = complexAnisotropy(z,minimumVariance)

    z = z(:);
    z = z-mean(z,'omitnan');

    denominator = mean(abs(z).^2,'omitnan');

    if ~isfinite(denominator) || denominator<=minimumVariance
        value = NaN;
    else
        value = abs(mean(z.^2,'omitnan'))/denominator;
    end
end

function r = safeCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(x) & isfinite(y);

    if nnz(valid)<3
        r = NaN;
        return;
    end

    x = x(valid);
    y = y(valid);

    x = x-mean(x);
    y = y-mean(y);
```

```
    denominator = sqrt(sum(abs(x).^2)*sum(abs(y).^2));

    if denominator<=eps
        r = NaN;
    else
        r = real((x'*y)/denominator);
    end
end

function r = complexVectorCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(real(x)) & isfinite(imag(x)) ...
        & isfinite(real(y)) & isfinite(imag(y));

    x = x(valid);
    y = y(valid);

    denominator = norm(x)*norm(y);

    if denominator<=eps
        r = NaN;
    else
        r = (x'*y)/denominator;
    end
end

function [qVec,basePrimeVec,powerExponentVec,lambdaVec] = ...
    makePrimePowers(Qmax)

    if Qmax<2
        qVec = zeros(0,1);
        basePrimeVec = zeros(0,1);
        powerExponentVec = zeros(0,1);
        lambdaVec = zeros(0,1);
        return;
    end

    pList = primes(Qmax);

    qList = zeros(0,1);
    baseList = zeros(0,1);
    exponentList = zeros(0,1);
    lambdaList = zeros(0,1);

    for jj = 1:numel(pList)

        p = pList(jj);
        currentPower = p;
        exponent = 1;

        while currentPower<=Qmax
```

```
            qList(end+1,1) = currentPower; %#ok<AGROW>
            baseList(end+1,1) = p; %#ok<AGROW>
            exponentList(end+1,1) = exponent; %#ok<AGROW>
            lambdaList(end+1,1) = log(p); %#ok<AGROW>

            if currentPower>floor(Qmax/p)
                break;
            end

            currentPower = currentPower*p;
            exponent = exponent+1;
        end
    end

    [qVec,order] = sort(qList);
    basePrimeVec = baseList(order);
    powerExponentVec = exponentList(order);
    lambdaVec = lambdaList(order);

    [qVec,uniqueIndex] = unique(qVec,'stable');
    basePrimeVec = basePrimeVec(uniqueIndex);
    powerExponentVec = powerExponentVec(uniqueIndex);
    lambdaVec = lambdaVec(uniqueIndex);
end

function [summaryTable,partialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier)

    uniquePrimes = unique(basePrimeVec);

    keep = false(size(uniquePrimes));

    for jj = 1:numel(uniquePrimes)
        keep(jj) = nnz(basePrimeVec==uniquePrimes(jj))>=2;
    end

    uniquePrimes = uniquePrimes(keep);

    summaryNames = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'minus_energy_ratio_sum_over_individual', ...
        'plus_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
        'boundary_alpha_first_real','boundary_alpha_first_imag', ...
        'boundary_alpha_last_real','boundary_alpha_last_imag', ...
        'tail_primitive_error_minus','tail_primitive_error_plus', ...
        'tail_primitive_start_abs_minus', ...
        'tail_primitive_start_abs_plus'};
```

```
partialNames = { ...
    'base_prime','power_exponent','n_prime_power', ...
    'term_rms_minus','term_rms_plus','term_rms_combined', ...
    'partial_rms_minus','partial_rms_plus','partial_rms_combined', ...
    'partial_coefficient_minus_real', ...
    'partial_coefficient_minus_imag', ...
    'partial_coefficient_plus_real', ...
    'partial_coefficient_plus_imag'};

if isempty(uniquePrimes)
    summaryTable = cell2table(cell(0,numel(summaryNames)), ...
        'VariableNames',summaryNames);

    partialTable = cell2table(cell(0,numel(partialNames)), ...
        'VariableNames',partialNames);

    return;
end

totalPartialRows = 0;

for jj = 1:numel(uniquePrimes)
    totalPartialRows = totalPartialRows ...
        +nnz(basePrimeVec==uniquePrimes(jj));
end

summaryRows = cell(numel(uniquePrimes),numel(summaryNames));
partialRows = cell(totalPartialRows,numel(partialNames));

partialRow = 0;

for jp = 1:numel(uniquePrimes)

    p = uniquePrimes(jp);
    idxTerms = find(basePrimeVec==p);

    [~,order] = sort(powerExponentVec(idxTerms));
    idxTerms = idxTerms(order);

    R = numel(idxTerms);

    sumMinus = complex(zeros(size(gamma)));
    sumPlus = complex(zeros(size(gamma)));

    sumEnergyMinus = 0;
    sumEnergyPlus = 0;
    sumEnergyCombined = 0;
    sumTermRMSCombined = 0;

    firstMinus = [];
    firstPlus = [];
    lastMinus = [];
    lastPlus = [];
```

```
cumulativeCoefficientMinus = 0;
cumulativeCoefficientPlus = 0;

for rr = 1:R

    j = idxTerms(rr);
    ell = log(qVec(j));
    z = exp(1i*ell*gamma);

    termMinus = cMinusVec(j).*carrier.*conj(z);
    termPlus = cPlusVec(j).*carrier.*z;
    termCombined = termMinus+termPlus;

    if rr==1
        firstMinus = termMinus;
        firstPlus = termPlus;
    end

    if rr==R
        lastMinus = termMinus;
        lastPlus = termPlus;
    end

    sumMinus = sumMinus+termMinus;
    sumPlus = sumPlus+termPlus;

    termEnergyMinus = mean(abs(termMinus).^2);
    termEnergyPlus = mean(abs(termPlus).^2);
    termEnergyCombined = mean(abs(termCombined).^2);

    sumEnergyMinus = sumEnergyMinus+termEnergyMinus;
    sumEnergyPlus = sumEnergyPlus+termEnergyPlus;
    sumEnergyCombined = sumEnergyCombined+termEnergyCombined;

    sumTermRMSCombined = sumTermRMSCombined ...
        +sqrt(termEnergyCombined);

    cumulativeCoefficientMinus = cumulativeCoefficientMinus ...
        +cMinusVec(j);

    cumulativeCoefficientPlus = cumulativeCoefficientPlus ...
        +cPlusVec(j);

    partialRow = partialRow+1;

    partialRows(partialRow,:) = { ...
        p,powerExponentVec(j),qVec(j), ...
        sqrt(termEnergyMinus),sqrt(termEnergyPlus), ...
        sqrt(termEnergyCombined), ...
        sqrt(mean(abs(sumMinus).^2)), ...
        sqrt(mean(abs(sumPlus).^2)), ...
        sqrt(mean(abs(sumMinus+sumPlus).^2)), ...
        real(cumulativeCoefficientMinus), ...
```

```
            imag(cumulativeCoefficientMinus), ...
            real(cumulativeCoefficientPlus), ...
            imag(cumulativeCoefficientPlus)};
    end

    sumCombined = sumMinus+sumPlus;

    energyRatioCombined = mean(abs(sumCombined).^2) ...
        /max(sumEnergyCombined,eps);

    energyRatioMinus = mean(abs(sumMinus).^2) ...
        /max(sumEnergyMinus,eps);

    energyRatioPlus = mean(abs(sumPlus).^2) ...
        /max(sumEnergyPlus,eps);

    rmsOverSumRMS = sqrt(mean(abs(sumCombined).^2)) ...
        /max(sumTermRMSCombined,eps);

    firstCombined = firstMinus+firstPlus;
    lastCombined = lastMinus+lastPlus;

    [relativeCombined,alphaCombined] = boundaryFit( ...
        sumCombined,firstCombined,lastCombined);

    [relativeMinus,~] = boundaryFit( ...
        sumMinus,firstMinus,lastMinus);

    [relativePlus,~] = boundaryFit( ...
        sumPlus,firstPlus,lastPlus);

    cM = cMinusVec(idxTerms);
    cP = cPlusVec(idxTerms);

    % Toda sucesion admite esta primitiva de cola. La identidad es
    % tautologica; lo informativo es su tamano y si existe forma cerrada.
    aM = [-flipud(cumsum(flipud(cM)));0];
    aP = [-flipud(cumsum(flipud(cP)));0];

    primitiveErrorM = max(abs(cM-(aM(2:end)-aM(1:end-1))));
    primitiveErrorP = max(abs(cP-(aP(2:end)-aP(1:end-1))));

    summaryRows(jp,:) = { ...
        p,R,max(qVec(idxTerms)), ...
        energyRatioCombined,energyRatioMinus,energyRatioPlus, ...
        rmsOverSumRMS, ...
        relativeCombined,relativeMinus,relativePlus, ...
        real(alphaCombined(1)),imag(alphaCombined(1)), ...
        real(alphaCombined(2)),imag(alphaCombined(2)), ...
        primitiveErrorM,primitiveErrorP, ...
        abs(aM(1)),abs(aP(1))};
end

summaryTable = cell2table(summaryRows, ...
```

```
        'VariableNames',summaryNames);

    partialTable = cell2table(partialRows(1:partialRow,:), ...
        'VariableNames',partialNames);
end

function [relativeRMS,alpha] = boundaryFit(target,firstTerm,lastTerm)

    target = target(:);
    firstTerm = firstTerm(:);
    lastTerm = lastTerm(:);

    X = [firstTerm,lastTerm];
    alpha = X\target;

    residual = target-X*alpha;

    relativeRMS = sqrt(mean(abs(residual).^2)) ...
        /max(sqrt(mean(abs(target).^2)),eps);
end


function [radius,centers,densitySmooth,peakRadii] = ...
    radialDistributionSummary(z,numBins,smoothWidth)

    z = z(:);
    z = z(isfinite(real(z)) & isfinite(imag(z)));

    center = mean(z);
    radius = abs(z-center);

    rMin = min(radius);
    rMax = max(radius);

    if rMax<=rMin
        rMax = rMin+eps(rMin+1);
    end

    edges = linspace(rMin,rMax,numBins+1);
    density = histcounts(radius,edges,'Normalization','pdf');
    centers = ((edges(1:end-1)+edges(2:end))/2).';

    densitySmooth = smoothdata(density(:),'gaussian',smoothWidth);

    candidates = find( ...
        densitySmooth(2:end-1)>densitySmooth(1:end-2) & ...
        densitySmooth(2:end-1)>=densitySmooth(3:end))+1;

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    candidates = candidates( ...
```

```
        densitySmooth(candidates)>=0.12*max(densitySmooth));

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    % Conserva maximos separados al menos seis bins, ordenados por altura.
    [~,heightOrder] = sort(densitySmooth(candidates),'descend');
    selected = zeros(0,1);

    for j = 1:numel(heightOrder)
        candidate = candidates(heightOrder(j));

        if isempty(selected) || all(abs(candidate-selected)>=6)
            selected(end+1,1) = candidate; %#ok<AGROW>
        end

        if numel(selected)>=4
            break;
        end
    end

    peakRadii = sort(centers(selected));
end

function s = joinNumericVector(v)

    v = v(:).';

    if isempty(v)
        s = "";
        return;
    end

    pieces = strings(size(v));

    for j = 1:numel(v)
        pieces(j) = sprintf('%.9g',v(j));
    end

    s = strjoin(pieces,', ');
end
```

Appendix B. Complete console transcript of the full-data run

Remark B.1. The transcript records the existing full-data run of the corrected prime-power audit. It predates the newly added matched-frequency null block, so no null-test outcome is inferred from it. All numerical claims used in the main text are reproduced in this annex.

Listing 2. Complete console transcript for the full-data run.

```
------------------------------------------------------------
ZETA PRIME-POWER STRUCTURE: AUDITORIA ANALITICA
```

```
------------------------------------------------------------
Archivo de ceros       : zeross.bin
Ceros disponibles      : 2001052
q                      : 2.7182818284590451
omega=log(q)           : 1
stride                 : 1
Qlist                  : 0 3 10 100 1000 10000
Puntos analizados       : 2001052
gamma inicial           : 14.134725142
gamma final             : 1132490.65871441

------------------------------------------------------------
TEST 2: IDENTIDADES DE COEFICIENTES
------------------------------------------------------------
Potencias primas <=Qmax                 : 1280
max |(c^-+c^+)-formula|                : 2.220446e-16
max |(c^--c^+)-formula|                : 2.220446e-16
max |c^-/c^+-formula|                  : 1.776357e-15
      Q      num_terms    Re_AQ          Im_AQ           abs_AQ_minus_half    Re_BQ          Im_BQ
     abs_BQ_minus_half    abs_BQ_plus_half
    -----    ---------    -----    ------------------    -----------------    -----    ------------------
     -----------------    -----------------

        0         0         0                       0                 0.5      0                       0
                  0.5                 0.5
        3         2         0      -0.675311086221108    0.840264876793701      0      -0.863660742917475
     0.997952844004547    0.997952844004547
       10         7         0       -1.07698676185356     1.1873923046777       0       -1.54062594080643
      1.6197309311999      1.6197309311999
      100        35         0       -1.47655654519435     1.55891604365222      0       -2.87744845271495
     2.92056665700882     2.92056665700882
     1000       193         0       -1.89626714054451     1.9610785472053       0       -5.31082516388603
     5.33431006985581     5.33431006985581
    10000      1280         0       -2.55521376172467     2.60367382137378      0       -10.6932626383477
     10.7049458593999     10.7049458593999


------------------------------------------------------------
AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD
------------------------------------------------------------
      case_name        mean_radius          std_radius        num_detected_peaks      detected_peak_radii
    -------------    ----------------    -----------------    ------------------    ------------------------

    "original"       1.92031044763071    0.726547772615351            2             "1.35731059, 2.49789956"
    "after_n3"       1.80792479214104    0.308312030736288            1             "1.68017926"
    "after_n2_n3"    1.78844590851816     0.22140483373138            1             "1.81300495"

Coeficiente teorico total n=3:
  c3^- = 0 -1.02370050729i
  c3^+ = 0 +0.0481029537851i
Q=0      terms=0      c^-res=0.4999989+0.1381673i  c^+res=0.4999989+0.1381673i  c^both=0.4999989+0.1381673i
     Var(F)=1.2420e+00  RMS(E)=1.1145e+00
Q=3      terms=2      c^-res=0.4999999+0.0909043i  c^+res=0.4999989+0.1429495i  c^both=0.4999999+0.0956865i
     Var(F)=1.3393e-01  RMS(E)=3.6597e-01
Q=10     terms=7      c^-res=0.5000000+0.0636593i  c^+res=0.4999989+0.1500304i  c^both=0.5000000+0.0755224i
     Var(F)=5.9610e-02  RMS(E)=2.4415e-01
Q=100    terms=35     c^-res=0.5000000+0.0191267i  c^+res=0.4999989+0.1736326i  c^both=0.5000000+0.0545920i
     Var(F)=2.2040e-02  RMS(E)=1.4846e-01
```

```
Q=1000   terms=193    c^-res=0.5000000-0.0206343i  c^+res=0.4999990+0.2014465i  c^both=0.5000001+0.0426449i
     Var(F)=1.1282e-02  RMS(E)=1.0622e-01
Q=10000  terms=1280   c^-res=0.4999999-0.0581400i  c^+res=0.4999989+0.2306199i  c^both=0.4999999+0.0343125i
     Var(F)=8.2559e-03  RMS(E)=9.0862e-02

------------------------------------------------------------
TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL
------------------------------------------------------------
      Q      num_prime_power_terms    c_after_minus_real    c_after_minus_imag     c_after_plus_real
     c_after_plus_imag    c_after_both_real    c_after_both_imag     distance_c_after_both_to_half
     rms_centered_Y_minus_Pminus    rms_centered_Y_minus_Pplus    rms_centered_Y_minus_both
     rms_after_both_fitted_carrier     energy_Pminus         energy_Pplus
     cross_term_2Re_inner_Pminus_Pplus        F_variance         F_max_abs_deviation_from_mean
     F_block_mean_std        F_anisotropy       carrier_discrete_defect_rms    derivative_defect_rms
    -----    ---------------------    ------------------    -------------------    -----------------
     -----------------    ------------------    -------------------    ------------------------------
     ----------------------------    --------------------------    -------------------------
     ------------------------------    -----------------    --------------------
     ----------------------------------    --------------------    ------------------------------
     --------------------    -----------------    ----------------------------    ---------------------

        0               0             0.499998912312768       0.138167317526428     0.499998912312768
     0.138167317526428     0.499998912312768      0.138167317526428          0.138167317530709
     1.22927492233399              1.22927492233399              1.22927492233399
     1.11446297144985                          0                     0                                0
         1.24202771473284           2.53363749717284            0.0101613448739479     0.164704934429685
       0.828969921883291           2.13681541452671
        3               2              0.49999988687813       0.0909043288314701     0.499998893284938
     0.142949512755117     0.499999867850236     0.0956865240601598          0.095686524060251
     0.630307565358949              1.22653992517659             0.626967498686433
     0.365967847866614          1.11258876067543     0.00443650350346841             0.00251726408927867
         0.133932465672121            1.3998447731532           0.00612569360136624     0.670875082815371
        0.476587597924875            1.7823665130755
       10               7              0.49999998247139       0.0636593469198101     0.499998916032037
     0.150030381454066     0.499999986190665      0.075522410847451          0.0755224108474523
     0.568023295911874              1.22456854935429             0.561527861557534
     0.244151425501691          1.18492844347144     0.00871311004796696             0.00421177723983395
        0.0596099185745075           1.05195331714233             0.0042120343511029     0.886135136342853
        0.398840669251055           1.69391674505877
      100              35             0.500000017346496       0.0191267412460307      0.49999893653376
     0.173632564206007      0.50000004156745      0.054591987925598          0.0545919879256138
     0.536555599882361              1.22197814824027             0.524423432995709
     0.148457436196022          1.21916077552125      0.0180717973642778             0.00501426487870665
        0.0220396103618959           0.799206246858361            0.0022297937207615     0.958121438757457
          0.2682174006908           1.45913395069404
     1000             193             0.500000036693381      -0.0206342853236017     0.499998951118377
     0.201446524072374     0.500000075498899     0.0426449212223482          0.042644921222415
     0.526364711348993              1.22166555561873              0.51293359872215
     0.10621775695736           1.23384826802886      0.0252917244776332             0.00469116700005195
       0.0112822118930529           0.668993854159613            0.0011112380638497     0.970189423496525
       0.179876374959657           1.21197809635573
    10000            1280             0.499999922217885      -0.0581399985655881     0.499998861049832
     0.230619861962331     0.499999870954991     0.0343125458703112          0.0343125458705538
     0.523988825848724              1.22334869951959             0.509345817216466
     0.0908622020249825          1.24333446127337       0.030996201483447          -0.000596344803520444
        0.00825593975682874           0.61202824502835          0.000381258918696766     0.981211983063741
         0.150986530997687           1.0471736199134

max error R_j-R_{j-1}=-P_j                 : 8.962228e-16
```

```
max error ley de energia de ramas           : 2.220446e-16
max error identidad de proyeccion portadora : 3.559044e-14
max error identidad de constantes           : 5.446388e-14

------------------------------------------------------------
TEST 3: TELESCOPADO EN p,p^2,...,p^R
------------------------------------------------------------
    base_prime    num_powers    largest_power    combined_energy_ratio_sum_over_individual
     combined_rms_over_sum_term_rms    boundary_fit_relative_rms_combined    boundary_fit_relative_rms_minus
       boundary_fit_relative_rms_plus    boundary_alpha_first_real    boundary_alpha_first_imag
     boundary_alpha_last_real    boundary_alpha_last_imag    tail_primitive_start_abs_minus
     tail_primitive_start_abs_plus
    ----------    ----------    -------------    ----------------------------------------
     ------------------------------    ----------------------------------    ------------------------------
       ------------------------------    -------------------------    -------------------------
     -------------------------    -------------------------    ------------------------------
     ----------------------------

        89            2             7921                       0.99578526600946
    0.951568779802395                 3.25731221427626e-15               1.66529459300868e-14
     5.68490958536155e-15             0.999999999999997           5.38298239788951e-18
    0.999999999999988        -6.13165759929884e-17             0.022712415828171
    0.0146012082752434
        37            2             1369                       0.992325345063945
    0.929272152959258                 3.32342530365488e-15               6.90975846079265e-15
     2.57414422667853e-14             0.999999999999997          -8.74885970690982e-18
    1.00000000000001         -8.01331294276757e-17             0.0386826763307904
    0.0223795089575229
        41            2             1681                       0.992798804843908
    0.932179807015267                 4.01764974378634e-15               5.71123669041389e-15
     6.00240878492108e-15                             1          -4.41766278752356e-18
    0.999999999999997         6.86611361451242e-17             0.0362585450928221
    0.0212931769572583
        23            2              529                       0.98981553486963
    0.914722655159467                 5.29829056019486e-15               3.01909112408219e-14
     5.06477325930078e-14             0.999999999999995           1.5256443527521e-17
    1.00000000000002         -4.12721429521546e-17             0.052842635596283
    0.0281454412919937
        67            2             4489                       0.994867301216485
    0.945073421083208                 6.62713306705871e-15               2.92396921729543e-14
     3.16762417963749e-14             0.999999999999993           4.06385013591999e-18
    0.999999999999977         -6.6991869985725e-17             0.0268592065905553
    0.0167695322528614
        73            2             5329                       0.995124247078512
    0.947082293993756                 8.02253098923419e-15               4.21876931539434e-14
     2.84327775551962e-14             0.999999999999992           2.29352846836488e-17
    0.999999999999967         1.25402372809613e-16             0.0255226923919741
    0.0160830067737586
        29            2              841                       0.991065442073751
    0.921970578232658                 1.69958093184349e-14               3.92022883362443e-15
     4.85298540260265e-14             1.00000000000002            5.31348409790742e-18
    0.999999999999986        -1.67495027840586e-16             0.0452613167826162
    0.0251764129412383
        59            2             3481                       0.994356231992412
    0.941904651888199                 1.82233584563793e-14               4.44612460135524e-14
      3.3145884657823e-14             1.00000000000002            3.31967944946604e-17
    0.999999999999965         2.13895102136555e-16             0.0289902084009457
    0.0178408950865232
```

```
   61            2            3721                    0.994484921893845
0.942760057482987                1.89061413493395e-14              4.06345157893362e-14
 1.56552838293448e-14            0.999999999999981        -1.56580841679002e-17
0.999999999999966       -2.22783722808597e-17           0.0284133680209016
0.0175536598492679
   43            2            1849                    0.993018430417279
0.933512335891677                1.91767251608662e-14              9.19221287192386e-15
 2.67694821919958e-14            1.00000000000002         -5.47518533793213e-18
0.999999999999988        1.24241990600724e-16           0.0351952587338588
0.0208067465987886
   97            2            9409                    0.996034159322401
0.953412551798485                2.19836316719303e-14              2.48682193279047e-15
 4.67252549818149e-15            0.999999999999978        -1.96285882336457e-17
0.999999999999975       -2.20623599182162e-16           0.0216013618259659
0.0140005339242615
   71            2            5041                    0.995055469368263
0.94645716703403                 2.45606582770506e-14              1.40537772877289e-14
1.35601376253121e-14             1.00000000000002         -2.17658103478551e-17
1.00000000000007        -3.81439794611675e-16           0.0259471363764858
0.0163022731943209
   83            2            6889                    0.995556221672419
0.950034257670606                2.80380400407119e-14              7.38122722146823e-15
 1.69133036510308e-14            1.00000000000003         -1.68047878446504e-17
1.00000000000006        -3.60217187758929e-16           0.0236603972161033
0.0151069817398424
   53            2            2809                    0.993923584391844
0.939166861275507                3.64856201837887e-14              5.33595501442798e-14
 6.78833363346336e-15            1.00000000000004         -1.38986297087105e-17
0.999999999999991       -4.48286028808063e-16            0.030939494184714
0.0187966802009105
   47            2            2209                     0.99342415440416
0.935964632967202                4.02588932254384e-14              4.69976317545872e-15
 3.70119612870561e-14            1.00000000000004          3.64557105503385e-17
0.999999999999993       -5.00636376597972e-16           0.0333062074556399
0.0199271852036153
   79            2            6241                    0.995407211972277
0.948922007921855                4.26585420767673e-14              6.84664346328644e-16
 1.95720970351145e-14            0.999999999999957         4.32727341964049e-17
0.999999999999994        6.93817715660186e-16           0.0243581235750551
0.015475353217867
   31            2             961                     0.99142834108407
0.924010046881153                4.48560611982839e-14              1.73461849505888e-14
 4.95966705898557e-15            1.00000000000004          1.34960651075476e-18
1.00000000000001        -1.47682556498033e-16           0.0433336826232223
0.0243790159002065
    3            8            6561                    0.994136424160185
0.925379114250609                 0.0526293055865214                0.0488679460726364
    0.38539545977793            0.997125200778133          1.04928001769985e-08
-0.373319182377488          0.00313372076449291             1.09725325128044
0.080998072339286
   19            3            6859                    0.988180958947934
0.896291448276131                 0.099571908742451                 0.0910436270892582
    0.130813323815928           0.994234206127615          8.21480982437346e-07
0.587653567802991       -6.66788850988072e-05           0.0610579674917809
0.0314116469586121
```

```
        17            3             4913                      0.987462564808847
     0.891547371679716                      0.104055647033926                  0.0950571818522644
         0.138802345385611              0.993904379532631            7.82474768756128e-08
     0.603159942163849         -1.05608473149603e-05             0.0661990286058359
     0.0331867205015699

boundary_fit_relative_rms_combined cercano a cero seria evidencia de colapso a los dos bordes del bloque p^r.

------------------------------------------------------------
TEST 4: POSITIVIDAD Y GRAM
------------------------------------------------------------
      Q            norm_E2          carrier_schur_complement    carrier_projection_abs     Re_inner_E_Pminus
           Re_inner_E_Pplus       cauchy_det_E_Pminus     cauchy_det_E_Pplus     cauchy_det_Pminus_Pplus
     polarization_error_minus    polarization_error_plus
     -----    -------------------    ------------------------    ---------------------    --------------------
         ---------------------    --------------------    --------------------    ------------------------
     ------------------------    ------------------------

        0       1.24202771473284         1.24202771473284         4.1753563397406e-15                         0
                              0                        0                          0                          0
                         0                        0
        3      0.133932465672121        0.133932465672121        3.19449329162395e-14     -0.00516035229366167
          0.000339021642957032     0.148985126760466     0.000594076917452501      0.00493441978003228
     2.22044604925031e-16       8.67361737988404e-19
       10      0.0596099185745075       0.0596099185745074        7.25002166702646e-15     -0.00596341179707172
          0.000207836720997232    0.0705979257513303     0.000519344584387362       0.0103199771600532
     2.22044604925031e-16       3.46944695195361e-18
      100      0.0220396103618959       0.0220396103618958        1.12998339769851e-14     -0.00697316806144494
         -0.000663780178893137     0.026821203386263     0.000397854767628292        0.022026140776441
                         0                        0
     1000      0.0112822118930529       0.0112822118930528        3.76207946660896e-14      -0.0090127255333809
          -0.00296783860935025     0.0138393083795104     0.000276538526093234       0.0312006486796008
     2.22044604925031e-16       1.38777878078145e-17
     10000     0.00825593975682874      0.00825593975682881        2.86661244710573e-14     -0.00982808606142384
          -0.00476027747414303     0.010168303134064     0.000233242529391743       0.0385385565639377
     2.22044604925031e-16       3.46944695195361e-18

Gram combinada: min=2.291421e-03, max=1.119809e+00, negativos<-1e-10=0
Gram por ramas: min=4.858227e-04, max=1.112574e+00, negativos<-1e-10=0

------------------------------------------------------------
TEST 5: F_Q POR BLOQUES DE ALTURA, Q=10000
------------------------------------------------------------
     block      gamma_start          gamma_end           F_mean_real          F_mean_imag            F_variance
              sup_abs_F_minus_global_c     sup_abs_F_minus_block_c     mean_abs_delta_F
      variation_per_height       anisotropy        carrier_discrete_defect_rms    derivative_defect_rms
     -----    ----------------    ----------------    -----------------    ------------------
      -------------------    ------------------------    ------------------------    ------------------
      --------------------    -----------------    --------------------------    --------------------

       1          14.134725142      63699.04754739    0.499994991361003     0.0350523539924386
      0.0054744556993458        0.542532351404617           0.542510726459913        0.0963670633500536
      0.126163323741996       0.979858632599489          0.121147135176878            0.538564667956422
       2       63699.700151835     118462.736278749     0.500001451274137      0.035105891285452
      0.00701927614984981        0.530144044699919           0.530116241500027        0.110272510195026
      0.167888441917521       0.980474241382446          0.137953328199234            0.726396110177662
       3       118463.232066064     170651.787459452     0.499998562254379     0.0349210363561646
      0.00745311407766993        0.549893284088027           0.549871555991103        0.113755262822731
      0.181736637024686       0.980873421275848          0.142306171066992            0.7574604344515
```

```
 4        170652.38272511     221291.519378741    0.500000256946116     0.0348042049768492
0.00767564743723714         0.581090952553066          0.581063883641327         0.116011219222976
0.191009406027093        0.980878273487911          0.144913294713216            0.880192519706131
 5        221292.295838797     270836.328162977    0.500000955646856     0.0347080955882823
0.007831121437243          0.57962624512367           0.579600207778524         0.117381041928519
0.197536641502951        0.980905060378457          0.146681991054392            0.923600652953341
 6        270836.931684803     319539.773103185    0.499999624306421     0.0346475825508576
0.00808884143363537         0.534809090007345          0.53479827792399          0.118850165982451
0.203463517740728        0.981360633102241          0.14914505740182             1.16913063854399
 7        319540.570678831     367564.435949345    0.49999957921803      0.0345578903969988
0.00818709113446676          0.52294466209729           0.52292439742989          0.119910420246151
0.208180893856153        0.981406498730984          0.150029861940657            0.880453279774671
 8        367565.006156693     415022.283240227    0.500000487984706     0.0344519123654481
0.00814253081563704         0.600282121372596          0.600276208543475         0.119491391046747
0.209930169452775        0.981010897479751          0.149551484403089            1.00807618824186
 9         415023.15758963     461995.409615676    0.500002125859323     0.0343892982332912
0.00818298949111592         0.572654690283211          0.572652885861422           0.1200026729604
0.213007944730247        0.980937348210029          0.150172616772631            1.08111614297039
10        461995.761482843     508544.337297334    0.499999128775315     0.0343361215632085
0.00835629092630199         0.542768812320236          0.542768199088721           0.12137168582189
0.217396246824282        0.981335496066421          0.152037594340513            0.934750513408792
11        508544.765796632     554719.129189251     0.49999902683957     0.0342841671661181
0.00838655885140866         0.555269896680713           0.55526995649336          0.121722656912353
0.219791838956823        0.981261093718569          0.152335939237595            1.12844994151658
12        554719.634077261     600557.928334761    0.499999052950246     0.0342521074641879
0.00848384933678091         0.606087165112338          0.606091467206371           0.12213899474892
0.222160553553316        0.981322097240396          0.153080139635091            1.07065793214649
13        600558.535755246     646093.102175532    0.500000235514038     0.0342171059277988
0.00852631056710716         0.524846602477766          0.524849827921532           0.122203119197641
0.223759839331275        0.981196827744373          0.153515460380432            1.14290197565882
14         646094.01946502     691351.826509571    0.499999459797555     0.0341425285063966
0.00852459159575678          0.61202824502835           0.612034526321802          0.122717190800213
0.226075215930959         0.98130473863714          0.153711972752659            1.52515702182085
15        691352.526827545     736357.304575928     0.50000063261387     0.0340996079052541
0.00863390467854271         0.603869638677834          0.603880275368038           0.123435919580137
0.228680535306117          0.9813459888377           0.154802894954523            1.13198994875413
16        736358.196419715     781127.536440619    0.500000507064813     0.0340763607134163
0.00863949315690915         0.551934114882366          0.551943379429299           0.123479710718261
0.229961941722585        0.981411615889858          0.154800438591316            1.02064882439406
17        781128.075976077     825679.648395047    0.499999867045727     0.0340594494960409
0.00867051311819898         0.586536403755823          0.586546166766736           0.123341062550961
0.230826520207627        0.981193765149418          0.155214716157765            1.15057126782006
18        825680.493113046     870029.136314775    0.499999157278646     0.0340083355422251
0.00880860286374675         0.564338258958265          0.564345300438399           0.124310351932475
0.233705005485217        0.981676906683931          0.156532439348918            1.09637171176575
19        870029.689029286     914187.909332815    0.499999527336173     0.0339700206865135
0.00876154865120847          0.56850486610471           0.568522179523045           0.124185981373753
0.234477981944182        0.981561079878874           0.15588197711697             0.995676199295695
20        914188.612117913     958168.018537517     0.50000099473625     0.0339345816929237
0.00878369871741152         0.599171890474147          0.599196831871485           0.124030304858425
0.235136204413765          0.98142509043805          0.156206512272264            1.06669516660454
21         958168.60666595     1001979.98016229    0.499998363067321     0.0338951222311845
0.0088529271020219          0.597221450310013           0.597233477557632           0.125000771021225
0.237887755021112        0.981656183204604          0.157057387478477            1.16995875593502
22        1001980.68512795     1045632.21311476    0.500003656384172     0.0338638859862856
0.00874571864686109         0.562103795049353          0.562114143907445           0.124273605310392
0.237367088718873          0.98131312028741          0.156023921934697            1.01328485216897
```

```
 23       1045632.45556665     1089133.21595891     0.499998860220289     0.0338981896499074
 0.00898518332990823        0.571629074187526            0.571636102485886           0.1254657058582
 0.240474616933205       0.981670300817732          0.158067171395785             1.12663969590805
 24       1089133.56438868     1132490.65871441     0.500000398441743     0.0338252499519786
 0.00892480514817264        0.590816935924078            0.590833242133321         0.125308756813391
 0.240969628397672       0.981703647581976          0.157535630788163             1.18442415402658

Rango Re media por bloque : [0.499994991361, 0.500003656384]
Rango Im media por bloque : [0.033825249952, 0.0351058912855]
Max sup global por bloque : 6.120282e-01
Max sup local por bloque  : 6.120345e-01

------------------------------------------------------------
TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS
------------------------------------------------------------
    Qmax     num_omitted_prime_powers_joint_fit    median_minus_amplitude_prime_power
     median_minus_amplitude_null    minus_prime_power_to_null_ratio    median_plus_amplitude_prime_power
     median_plus_amplitude_null    plus_prime_power_to_null_ratio    joint_omitted_explained_fraction
     fit_theory_correlation_real    fit_theory_correlation_imag    fit_theory_correlation_abs
     relative_theory_coefficient_error    F_energy_before_omitted_fit    F_energy_after_omitted_fit
     F_anisotropy        corr_gap_realF        corr_gap_imagF        corr_gap_absF
    -----    ----------------------------------    ----------------------------------
     ----------------------------    ----------------------------------    ----------------------------------
     ----------------------------    ----------------------------------    ----------------------------------
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    10000                    56                              0.0011038260045886
     0.000329157644250013               3.35348737564237                  0.000885575009662317
     0.000330982677195599              2.6755932279168                   0.0144445968386994
     0.916473061428038             0.00108815005327049            0.916473707420905
     0.480753428806987                0.00822729788430414           0.00810845788329348
     0.981063724279348    0.187253690050176    0.0933065219921283    -0.139580596671858


Potencias primas omitidas con mayor amplitud:
      n      base_prime    power_exponent        Lambda_n          fit_F_minus_real         fit_F_minus_imag
           fit_F_minus_abs         fit_F_plus_real         fit_F_plus_imag          fit_F_plus_abs
     theory_minus_real     theory_minus_imag      theory_plus_real     theory_plus_imag
     max_branch_fit_abs
    -----    ----------    --------------    -----------------    ----------------------    --------------------
        --------------------    ----------------------    --------------------    --------------------
     -----------------    --------------------    -----------------    --------------------
     --------------------

    10301      10301             1           9.23999625688369     0.000371892244369111    -0.00161813748988026
        0.00166032309433371      0.00033865779123921      0.00138200597601097      0.00142289480190808
       0            -0.00175843054430363           0             0.0014149869530774     0.00166032309433371
    10169      10169             1           9.22709915579114     -0.000115283050974615    -0.00156964135711816
        0.00157386917239577    -5.07240436672432e-05      0.00133670966276089       0.0013376717277136
       0            -0.00177010677368571           0            0.00142394668533194     0.00157386917239577
    10181      10181             1            9.2282785171072     0.000278847600421771     -0.00152895592997706
        0.00155417573654751      0.000286682906630887      0.00132354059079306      0.00135423291365672
       0            -0.00176903580369307           0            0.00142312504251575     0.00155417573654751
    10039      10039             1           9.21423278669153      0.00041374673843581     -0.00144753771339835
        0.00150550715550501      0.000421812244560221      0.00124369263450663      0.00131327717515647
       0            -0.00178183332924528           0            0.00143294107929247     0.00150550715550501
```

```
    10333      10333             1           9.24309793621435     6.14996145285231e-05     -0.00149251220327071
         0.00149377872507917     5.12963409860825e-05      0.00123167205653074      0.00123273978172087
       0            -0.00175563411722169           0            0.00141284053501554     0.00149377872507917
    10331      10331             1           9.24290436284945     -0.000230890640791369     -0.00147441795772886
          0.0014923869478384    -0.000347632129299537      0.00129628666277216      0.00134209061147238
       0            -0.00175580850836064           0            0.00141297439683093      0.0014923869478384
    10211      10211             1           9.23122084955554     -0.000633103531925687     -0.00134320284045877
         0.00148492893862073    -0.000631071689296745      0.00106646662668294      0.00123919431198676
       0            -0.00176636674804379           0            0.00142107721241217     0.00148492893862073
    10369      10369             1           9.24657586455828      4.67547186174683e-05     -0.00147559974971223
         0.00147634028091894     7.99910522234896e-05      0.00124576018232505      0.00124832567878033
       0            -0.00175250381884864           0            0.00141043758286627     0.00147634028091894
    10453      10453             1           9.25464429753271     -0.000304478868200946     -0.00141117850295559
         0.00144365236410464    -0.000223525782210025        0.00124767153432      0.00126753620574129
       0            -0.00174526357211357           0             0.0014048785677242     0.00144365236410464
    10271      10271             1           9.23707966916581      0.000151926677381929     -0.00139593924366338
         0.00140418235542959     0.000199437478543235      0.00117395228955498      0.00119077255846742
       0            -0.00176106420552525           0            0.00141700823205664     0.00140418235542959
    10141      10141             1            9.224341891612     -2.69378011998342e-05     -0.00139213634163686
         0.00139239694011425     1.72266295677679e-05      0.00116438745087312      0.00116451487432195
       0             -0.001772613186228           0            0.00142586946325406     0.00139239694011425
    10177      10177             1           9.22788555119193      0.000562556014238549     -0.00123881678217888
         0.00136056469488371     0.000525626337530265      0.00105890848795867      0.00118218874659523
       0            -0.00176939258053915           0            0.00142339876359955     0.00136056469488371
    10061      10061             1           9.21642184229205     -0.000271184796372421      -0.0012905239456047
         0.00131870893223739    -0.000199026354140983      0.00104720828811089      0.00106595341752386
       0            -0.00177983266568738           0            0.00143140682870411     0.00131870893223739
    10247      10247             1           9.23474025879972      0.000296226420166288     -0.00127916398853374
         0.00131301584208496     0.000225798114775531      0.00100579696158174      0.00103083088650042
       0            -0.00176317956235639           0             0.0014186315781825     0.00131301584208496
    10457      10457             1           9.25502688959859     -0.00038302950951308     -0.00124755231657541
         0.00130502811760916    -0.000336465701900575       0.0010635359416545      0.00111548996756868
       0            -0.00174492100357133           0            0.00140461550806041     0.00130502811760916
    10223      10223             1           9.23239536275701      4.37551338344118e-05     -0.00126550404521048
         0.00126626024188591     6.46513618466203e-05      0.00112056448509538      0.00112242797713069
       0            -0.00176530245864271           0            0.00142026058016567     0.00126626024188591
    10099      10099             1           9.22019168802656      3.77615682350026e-05      -0.0012349486588066
         0.00123552584996178     0.000122474628046823      0.00095475772183488     0.000962581083295603
       0            -0.00177639258075131           0            0.00142876845882169     0.00123552584996178
    10429      10429             1           9.25234566612121      0.000104275707715879     -0.00122341220360067
         0.00122784805376671     0.000135594574873729     0.000939541687326359     0.000949275761282911
       0            -0.00174732317276288           0            0.00140646007086086     0.00122784805376671
    10343      10343             1            9.2440652413778      8.65573758937093e-05     -0.00120755464889939
         0.00121065288559526     0.000151903149928664     0.000998017665518561      0.00100951167781525
       0            -0.00175476293000098           0            0.00141217180262058     0.00121065288559526
    10111      10111             1            9.2213792190914       9.1842043748573e-05     -0.00120508425512231
         0.00120857892706419     5.50085507262679e-05     0.000972297490544918      0.00097385232493071
       0            -0.00177531032055336           0            0.00142793835000022     0.00120857892706419


Frecuencias no nulas dominantes de F_Q-mean(F_Q):
    rank    demodulated_frequency       fit_amplitude
    ----    ---------------------    --------------------

      1         7.22212181819358      0.00102998542892402
      2        -0.509252179488009     0.000972670281121882
      3         -7.22212181819358     0.000962787079088038
      4         0.509252179488009     0.000944393206974707
      5         -7.26841747087431     0.000923900818886456
```

```
      6        0.694434790210921     0.000912309613268334
      7       -0.694434790210921     0.000907278710300437
      8         7.26841747087431     0.000828631152433364
      9        -2.82403481352441     0.000809343213775063
     10        -6.89805224942848     0.000799201794454691
     11         6.89805224942848     0.000798947722046012
     12         8.42580878789251     0.000771802707211477
     13        0.879617400933833     0.000761874424348131
     14       0.0462956526807281     0.000760539159031102
     15       -0.879617400933833     0.000742831699098273
     16        -5.32400005828373     0.000727744418239324
     17        -8.42580878789251     0.000703624825291971
     18        -8.56469574593469     0.000703431607127999
     19        -8.61099139861542     0.000697831699989827
     20         8.61099139861542     0.000697444147794987
     21         8.93506096738052     0.000693242426624136
     22        -9.02765227274197     0.000691823571238767
     23      -0.0462956526807281     0.000690811206825879
     24         8.56469574593469     0.000687798036575012
     25         9.02765227274197     0.000680600662464099
     26                        0      0.00067560438002473
     27        -8.33321748253105     0.000673228670039579
     28         3.84253917250043     0.000672369683136021
     29        -3.84253917250043       0.0006713463801668
     30        -6.24991311189829     0.000667528070597226


DIAGNOSTICO FINAL PARA Q=10000
mean(F_Q)                          = 0.499999870955 +0.03431254587703i
|mean(F_Q)-1/2|                   = 3.431255e-02
Var(F_Q)                           = 8.255940e-03
sup |F_Q-mean(F_Q)|                = 6.120282e-01
RMS E_Q                            = 9.086220e-02
razon omitidas potencia/null (-) = 3.353487
razon omitidas potencia/null (+) = 2.675593
fraccion explicada ajuste omitido= 0.014445

------------------------------------------------------------
TERMINADO
Tablas disponibles directamente en el workspace:
  radialAuditTable
  summaryTable
  coefficientCutoffDisplay
  groupSummaryTable
  groupPartialTable
  positivityTable
  blockQmax
  omittedTable
  uniquenessTable
  topFrequencyTable
------------------------------------------------------------
>>
```

Appendix C. Matched-frequency null-test report

Listing 3. Text report produced by the matched-amplitude frequency scan.

```
LOG(3) MATCHED-AMPLITUDE DETERMINISTIC FREQUENCY SCAN
```

```
Data source: archived gammaZeta array in ZETA_DYNAMIC_RADIAL_DIAGNOSTICS.mat
Number of available ordinates: 2001052
First ordinate: 14.134725142000001
Last ordinate: 1132490.658714411081746
Audit subsample size: 100000
Carrier frequency omega: 1.000000000000000
Exact target frequency log(3): 1.098612288668110
Number of admissible null frequencies: 199
Null interval: [0.848612288668110, 1.348612288668110]
Prime-power log exclusion radius: 0.0005

Observed radial-variance reduction after subtracting the exact n=3 mode:
0.820094348913472

Matched-amplitude non-arithmetic frequency scan:
minimum reduction: -0.608944143730128
median reduction: -0.595772672531983
maximum reduction: -0.583195205316322
number of null frequencies with reduction >= observed: 0

Interpretation: on this deterministic 100000-point audit, every matched-amplitude
non-arithmetic replacement increased rather than decreased the radial variance,
whereas the exact log(3) frequency reduced it by about 82 percent. This is finite
numerical evidence for frequency specificity. It is not an analytic theorem that
the full radial density must be bimodal.
```

Appendix D. Matched-frequency null-test data

Listing 4. Complete CSV output of the matched-amplitude frequency scan.

```
index,ell_null,radial_variance_reduction
1,0.8486122886681098,-0.5898231459921301
2,0.8511122886681097,-0.5964731867226032
3,0.8536122886681098,-0.5957763395441549
4,0.8561122886681097,-0.5931695179103056
5,0.8586122886681098,-0.5963882785577987
6,0.8611122886681097,-0.5908138552254003
7,0.8636122886681098,-0.594411386106116
8,0.8661122886681097,-0.5942588746407349
9,0.8686122886681098,-0.5959029568334553
10,0.8711122886681097,-0.5960259706742361
11,0.8736122886681098,-0.5940433562846092
12,0.8761122886681098,-0.5973335753679774
13,0.8786122886681098,-0.5917617409063023
14,0.8811122886681098,-0.5931656560977243
15,0.8838622886681098,-0.5933016115075245
16,0.8863622886681097,-0.5911331502212909
17,0.8888622886681098,-0.5955300262442895
18,0.8913622886681097,-0.5895678309512211
19,0.8938622886681098,-0.5938294084115763
20,0.8963622886681097,-0.6036847052605221
21,0.8988622886681098,-0.5941968353014626
22,0.9013622886681097,-0.5954549002972771
23,0.9038622886681098,-0.5971917603783188
24,0.9063622886681098,-0.5913960257240796
25,0.9088622886681098,-0.592406551908826
26,0.9113622886681098,-0.6006926883222456
27,0.9138622886681098,-0.6041652200607102
```

```
28,0.9166122886681098,-0.5942709872483887
29,0.9191122886681098,-0.6010741034480371
30,0.9216122886681097,-0.5938215927767858
31,0.9241122886681098,-0.6014264269398479
32,0.9266122886681097,-0.6000977936063119
33,0.9291122886681098,-0.5986590817063933
34,0.9316122886681097,-0.5933460239061525
35,0.9341122886681098,-0.5939522785819968
36,0.9366122886681097,-0.5937668436203267
37,0.9391122886681098,-0.5928487966623421
38,0.9416122886681098,-0.5960897255176403
39,0.9441122886681098,-0.5999989443577753
40,0.9466122886681098,-0.5972805617649581
41,0.9493622886681098,-0.5920702386444918
42,0.9518622886681098,-0.5900733489006347
43,0.9543622886681098,-0.5977363720264928
44,0.9568622886681097,-0.5986763169766829
45,0.9593622886681098,-0.5936326528582041
46,0.9618622886681097,-0.5956795128341212
47,0.9643622886681098,-0.6000707844692603
48,0.9668622886681097,-0.5889970967112432
49,0.9693622886681098,-0.5942656056744011
50,0.9718622886681098,-0.6026153846648397
51,0.9743622886681098,-0.5990894760772068
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53,0.9793622886681098,-0.5993537374196101
54,0.9821122886681097,-0.5915411009419629
55,0.9846122886681098,-0.599522539452715
56,0.9871122886681098,-0.5874961037158515
57,0.9896122886681098,-0.583461605815911
58,0.9921122886681097,-0.5970345280676785
59,0.9946122886681098,-0.5933806774243919
60,0.9971122886681097,-0.5947893562698121
61,0.9996122886681098,-0.5941976347259101
62,1.0021122886681098,-0.5904062130829175
63,1.0046122886681097,-0.5946243436640437
64,1.0071122886681099,-0.596657377748762
65,1.0096122886681098,-0.5963508771812935
66,1.0121122886681098,-0.596209691196204
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69,1.0198622886681097,-0.5985885114396072
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74,1.0323622886681099,-0.5949978535849914
75,1.0348622886681098,-0.5950431976337405
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79,1.0448622886681098,-0.6007609262077918
80,1.0473622886681098,-0.5986368879593436
81,1.0501122886681098,-0.5929361512948881
82,1.0526122886681097,-0.5872356037546511
83,1.0551122886681097,-0.591744486468406
84,1.0576122886681099,-0.5976847786985864
85,1.0601122886681098,-0.5890834252547483
86,1.0626122886681098,-0.5982786339322246
```

```
87,1.0651122886681097,-0.5894233377692719
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89,1.0701122886681098,-0.5987454317631775
90,1.0726122886681098,-0.5964860009873283
91,1.0751122886681097,-0.5887924566330356
92,1.0776122886681099,-0.5945817431914575
93,1.0801122886681098,-0.5923817367762954
94,1.0828622886681099,-0.6033054603974601
95,1.0853622886681098,-0.5974050632037073
96,1.0878622886681097,-0.5910743340358569
97,1.0903622886681097,-0.5990090883118928
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106,1.1141122886681099,-0.5932206691125488
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162,1.2553622886681097,-0.5868989133123979
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186,1.3158622886681097,-0.5890281693738901
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188,1.3208622886681098,-0.6015939814558708
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191,1.3283622886681097,-0.5983539293288906
192,1.3308622886681098,-0.5977487866152462
193,1.3333622886681098,-0.5932795249622664
194,1.3358622886681097,-0.597455528236549
195,1.33836228866811,-0.5978126729190543
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197,1.3433622886681098,-0.598107716362853
198,1.3458622886681098,-0.6019204476572204
199,1.3486122886681098,-0.5976193743286629